

The Wiryadana's Notebook

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Table of contents

Preface	10
Acknowledgements	11
1 Introduction	12
I Internal Medicine	13
2 Emergency & Intensive Care Medicine	14
2.1 Approach to Critical Ill Patients	14
2.1.1 Shock	15
2.1.2 Initial Evaluation	15
2.1.3 Respiratory Failure	19
2.1.4 Multiorgan system failure	31
2.2 Specific Critical illness	31
2.2.1 Acute Respiratory Distress Syndrome (ARDS)	31
2.2.2 Diagnostic Criteria for ARDS	31
2.2.3 Epidemiology	31
2.2.4 Approach to Circulatory Shock	35
2.2.5 Sepsis	40
2.2.6 Cardiogenic Shock and Pulmonary edema	40
2.2.7 Cardiac Arrest	40
2.2.8 Neurological Critical illness	40
2.3 Important Calculation	40
2.3.1 Airway & Ventilation	40
2.3.2 Circulation	40
3 Hematology & Medical Oncology	42
3.1 Oncological Emergency	42
3.2 Hematology	42
3.2.1 Transfusion Medicine	42
3.3 Medical Oncology	42
3.4 Procedures in Hematology and Medical Oncology	42
3.4.1 Examination of Blood Cells & Marrow Cells	42

4 Allergy & Immunology	45
5 Rheumatology	46
5.1 Autoimmune	46
5.2 Joint Disease	46
5.2.1 Crystal Deposition Disease	46
5.2.2 Arthrocentesis Technique	53
6 Infectious Disease	54
6.1 Bacterial Infection	54
6.2 Viral Infection	54
6.2.1 HIV	54
6.2.2 Dengue	54
6.3 Fungal Infection	54
7 Cardiology & Vascular Medicine	55
8 Respiratory Medicine	56
9 Nefrology & urology	57
9.1 Nefrology	57
9.1.1 Hypertension	57
9.1.2 FLuid and Electrolyte Disorders	57
9.1.3 Infectious Disease in Kidney	59
9.1.4 Acute Kidney Injury	59
9.1.5 GLomerular Disease	59
9.1.6 Tubulointerstitial Disease	59
9.1.7 Diabetic Kidney Disease	59
9.1.8 Renovascular Disease	59
9.1.9 Hepatorenal Syndrome	59
9.1.10 Cardiorenal Syndrome	59
9.1.11 Pregnancy and Renal Disease	59
9.1.12 Congenital Disease of the Kidney	59
9.1.13 Chronic Kidney Disease	59
9.1.14 Dialytic Therapy	59
9.1.15 Transplantation	59
9.1.16 Edema and Diuretics use	59
9.2 Urology	65
10 Gastroenterology & Hepatology	67
10.1 Gastroenterology	67
10.1.1 Anorectal Disease	67
10.2 Hepatology	68

11 Endocrine & Metabolism	69
11.1 Emergency	69
11.1.1 Diabetes	69
11.2 Non-emergency	69
12 Psychosomatic Medicine	70
13 Geriatrics	71
13.1 Emergency	71
13.2 Non-emergency	71
II The Rest of Medicine	72
14 Clinical Nutrition	73
15 Surgery	74
16 neurology	75
16.1 Coma & Decreased Consciousness	75
16.2 Pain Management	75
17 Perioperative Consultation & Anesthesia	76
17.1 Perioperative Consultation	76
17.1.1 Common Risk Score	76
17.2 Anesthesia	78
17.2.1 Standard Anesthesia (just to know)	78
17.2.2 Standard Postoperative Analgesia Protocol	78
18 Obstetric & Gynecology	80
19 Dermato-venereology	81
20 Poison, Envenomation and Drug Overdose	82
21 Disease due to Environmental Exposure	83
22 Travel Medicine	84
23 Radiology	85
23.1 Konvensional	85
23.2 CT Scan	85
23.3 Ultrasonography	85
23.3.1 POCUS in Emergency/Critical Care Setting	85
23.4 MRI	85

23.5 Nuclear Medicine	85
24 Anatomical Pathology	86
25 Clinical Pathology	87
25.1 Hematology Laboratory	87
25.1.1 Hematologic Malignancy	87
25.2 Clinical Chemistry	87
25.2.1 Acid-Base Analysis	87
25.2.2 Liver Function Test	89
25.2.3 Disorder of Lipids	102
25.2.4 Cardiac markers	102
25.2.5 Coagulation Study	102
25.2.6 Renal Function Test	102
25.2.7 Urine Analysis	110
26 Clinical Microbiology	111
27 Ear, Nose & Throat	112
28 Ophtalmology	113
29 Rehabilitation & Physical Medicine	114
30 Disaster Medicine, Wilderness medicine, Transport Medicine	115
31 Pharmacology	116
32 Common Medical Procedures for Internist	117
32.1 Diagnostics	117
32.2 Therapeutics	117
32.2.1 Blood Transfusion	117
III Bioinformatics & Genomic Medicine	118
33 Genomic Medicine	119
34 Linux & Bioinformatic Programs	120
IV Statistics and R language	121
Dataframe Operation	122
Central Tendency	128
Variance	128

Single-Samples	130
Two-Samples	130
Regression	130
Analysis of Variance	130
Analysis of Covariance (ANCOVA)	130
Multiple Regression	135
Contrast	135
Other Response Variables	135
Count Data	135
Proportion Data	135
Binary Response Variable	135
Death and Failure	135
35 Basic R and Base Plot	136
36 Tidyverse & ggplot2	137
V Mata Kuliah Dasar Umum	138
37 Biologi Molekular	140
37.1 Cell Structure & Organel	140
37.1.1 Organel Sel	140
37.2 Genome Organization	141
37.2.1 Protein coding genes	141
37.2.2 Codon	141
37.3 DNA & RNA Structure	143
37.4 Signal Transduction	143
37.5 Cell Cycle	144
37.5.1 G1	146
37.5.2 S	146
37.5.3 G2	146
37.5.4 M	146
37.6 Mutastion & Free Radical & Antioksidan	146
37.6.1 SUBstitusi biasanya terjadi 1 basa saja, sehingga sering juga disebut point mutation.	147
37.6.2 Delesi dan Inseri	147
37.6.4 Molecular effect of Mutation	147
37.6.5 Clinical Effect of Mutation	147
37.6.6 DNA Repair	147
37.6.7 Free Radical	148
37.7 DNA Replication	148
37.8 Mutation & DNA Repair	149

37.9 Gene Expression	149
37.9.1 Transkripsi	149
37.9.2 Translasi	150
37.9.3 Regulasi Transkripsi	150
37.10 Control of Gene Expression	150
37.11 Genetic Variation	150
37.12 Genotyping	150
37.12.1 Polimorfisme vs Mutasi	150
37.13 Stem Cell	151
37.14 Molecular Biology Technique	151
37.15 Application of Molecular Biology in Medicine	151
38 Metodologi Penelitian	152
38.1 Overview of Clinical Research	152
38.2 Association & Causation	152
38.3 Research Measurement	154
38.4 Variables and Relationship Between Variables	154
38.5 Hypothesis and Hypothesis Testing & Effect Size, Confident Interval, and P Value	154
38.6 Sample Size and Power	154
38.7 Research Design	156
38.7.1 Cross sectional	156
38.7.2 Kasus-Kontrol	156
38.7.3 kohort	156
38.7.4 Nested Case-Cohort Study atau Nested Control Study	156
38.7.5 Uji Diagnostik	156
38.8 Cross Sectional & Case-Control Study	157
38.8.1 Cross-sectional Study	157
38.8.2 Case-Control Study	157
38.9 Cohort Study	157
38.10 Experiments	157
38.11 Diagnostic Test	157
38.12 Data Management and Analysis	157
38.13 Research Proposal	157
38.14 Ethical Issue	157
38.15 Novelty Aspect	157
39 Filsafat Ilmu	158
39.1 Pengantar Filsafat Ilmu	158
39.2 Ilmu Sebagai Produk Berpikir	159
39.3 Ilmu dan Teknologi Kedokteran	159
39.4 Ontologi, Epistemologi, dan Aksiologi Kedokteran	159
39.5 Intelektualitas dan Aktivitas Intelektual	159
39.6 Pengetahuan dan Ilmu Pengetahuan	159

39.7 Teori Kebenaran Ilmiah	159
39.8 Perkembangan Ilmu Kedokteran	159
39.9 Aksiologi, Sikap Ilmiah dan Etika Ilmiah	159
39.10 Complementary & Alternative Medicine	159
39.11 Komunikasi Ilmiah	159
39.12 Bioetika Penelitian	159
39.13 Publikasi dan Hak Paten	159
39.14 Filsafat Ketuhanan dan Kemanusiaan	159
39.15 Metode Penalaran Ilmiah	159
40 Statistik Kedokteran Dasar	160
40.1 Dasar-dasar Penulisan Ilmiah Kedokteran dan Plagiarisme	160
40.1.1 Methods of acquiring knowledge	160
40.1.2 Benefits of Conducting Research	160
40.1.3 Qualities of a researcher	160
40.1.4 Bioethical	161
40.1.5 Induction and Deduction in Research	161
40.1.6 Scientific Method	161
40.1.7 First Draft	162
40.1.8 Structure of a Research Paper	162
40.1.9 Seven Categories of plagiarism	162
40.1.10 Characteristics of Good Research	162
40.2 Kiat-kiat Publikasi Majalah Terakreditasi dengan OJS	163
40.2.1 Kendala Publikasi di Indonesia:	163
40.2.2 Kenapa publikasi penting?	163
40.2.3 Jurnal	163
40.3 Effect Size, COntidence Interval	165
40.4 Uji Hipotesis	165
40.4.1 Clinician and Statistician	166
40.5 Korelasi Linier dan multipel	166
40.6 Regresi Linier Sederhana	169
40.7 Distribusi normal	170
40.8 Uji T berpasangan dan tak berpasangan	171
40.9 One Way ANOVA	171
40.10 Ancova dan Repeated Measurements ANOVA	171
40.11 Tabel 2 x 2, Odd ratio, Prevalence Ratio dan Relative Risk	172
40.12 Prinsip Analisis Data Biner	172
40.13 Regresi Logistik Sederhana	172
40.14 Regresi Logistik Berganda	172
41 Evidence-based Medicine	173
41.1 Introduction to EBM and Critical Appraisal	173

41.2 Clinical Question berdasarkan PICO	174
41.2.1 Step 1	174
41.2.2 Prinsip Dasar Searching	174
41.3 Effect Size, Hypothesis, and Confidence Interval	175
41.4 Study About Risk Factor/Case Control Study	176
41.5 Study About Diagnosis	176
41.6 Study about Therapy	177
41.7 Study about Prognosis	177
41.8 Summary Capita Selekta	177
42 Tugas MKDU	178
42.1 Biologi Molekuler	178
42.2 Statistik	178
42.3 Metode Penelitian	178
42.4 Filsafat Ilmu Kedokteran	180
42.5 Evidence-based Medicine	180
References	182

Preface

In the middle of 2023, I was so fortunate that I passed the admission for Internal Medicine residency program. I believe my knowledge has been barely enough just to start the program. To compensate my lack in memorizing and attention to detail, I build this books. I published it online but not intended to disseminate widely, at least not yet. It is only to help me doing my work.

By the way, this is a Quarto book. Quarto is amazing innovation from Rstudio.org that makes it possible for a lay person creating online books like this one. To learn more about Quarto books visit <https://quarto.org/docs/books>.

Acknowledgements

This book made possible by years of clinical education given by all of my teachers.

1 Introduction

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

Part I

Internal Medicine

2 Emergency & Intensive Care Medicine

2.1 Approach to Critical Ill Patients

2.1.0.1 Assessment of illness severity

Important for:

1. resource allocation
2. Hospital administrative policies
3. asses quality of care

Two most commonly used scoring systems:

1. SOFA (Sequential Organ Failure Assessment)
2. APACHE (Acute Physiology and Chronic Health Evaluation)

2.1.0.1.1 SOFA Score

Includes 6 organ systems, with each graded 0 to 4 according to the degree of dysfunction. Increased score correlate with mortality and can be evaluated repeatedly (Loscalzo et al. 2022).

A derivation, the qSOFA or quick SOFA intended to screen patients for srisk of poor outcomes from sepsis. It is not intended for sepsis diagnostic screening tool, but in it often used as such in resources poor area. qSOFA used for bedside evaluation that may identify patient with suspected infection who are at greater risk for a poor outcome outside the ICU. The score range from 0 to 3 points in each three category including blood preasure, respiratory rate and mental status. Poor outcimes predicted if there at least 2 clinical criteria: (1) respiratory rate $\geq 22/\text{min}$, altered mental status, or systolic BP ≤ 100 mmhg.

2.1.0.1.2 APACAHE Score

SLightly more complicated than SOFA Score. (updated later)

Calculation of SOFA Score

System	0	1	2
Respiration (PaO ₂ / FiO ₂)	≥ 400	<400	<300
Coagulation (plt)	≥ 150	<150	<100
Liver (bilirubin, mg/dl)	<1,2	1,2 - 1,9	2,0 - 5,9
Cardiovascular	MAP ≥ 70	MAP < 70	Dopamin <5 or dobutar
CNS (GCS)	15	13-14	2023-12-10
Renal (Creatinin mg/dl or urine output ml/day)	1.2	1,2 - 1,9	2,0 - 3,4

2.1.1 Shock

2.1.2 Initial Evaluation

shock is a multisystem end-organ hypoperfusion. The resulting hypoperfusion followed by tissue hypoxia with accompanying lactic acidosis. Clinical Indicator:

1. Reduced MAP
2. Tachycardia
3. Tachypnea
4. Cool Skin and Extremities
5. Acute Altered mental Status
6. Oliguria

Reduced MAP could be the product of decreased cardiac output and/or systemic vascular resistance (SVR). Thus every shock patients should be evaluated for adequacy of cardiac output. Sign of diminished cardiac output includes (Cold Shock):

- A narrow pulse pressure (SBP - DBP), marker of stroke volume
- Cool extremities and delayed capillary refill time (COLD SHOCK). Palpate proximal extremities (eg Thigh) rather than distal extremities to determine relative “coolness” as peripheral artery disease may always have cool distal extremities.

Contrary, there are sign of increased cardiac output (Warm Shock), that may results from decreased SVR:

- A widened pulse pressure (particularly reduced DBP)
- Warm extremities with bounding pulse,
- Rapid capillary refill time

If reduced cardiac output found, then conduct assesment of volume status.

- History suggesting fluid loss or hemorrhage
- Reduced JVP
- Straight leg raise or fluid challenge
- USG Marker: inferior vena cava collapse, left ventricular stroke volume

reduced cardiac function with increased intravascular volume - S3 or S4 gallop - JVP increased
 - Extremity Edema - Crackles on lung auscultation - Chest Xray show cardiomegaly, widening vascular pedicle, Kerley B lines, pulmonary edema. - ECG: ischemic with or without chest pain.

If sign of increased cardiac output found, conduct search for cause of reduced SVR.

- Sepsis
- Liver Failure
- Severe Pancreatitis
- Adrenal Insufficiency
- Burns
- Trauma
- Anaphylaxis
- Thyrotoxicosis
- Peripheral AV shunts

The need for arterial line and CVC - if shock prolonged and does not resolve with proper fluid resuscitation and vasoactive agent.

Initial evaluation followed with early aggressive targeted resuscitation improve survival. If initial bedside evaluation yield confounding data, objective assessment with USG/Echo needed (look Figure 2.2).

2.1.2.1 The need for Mechanical Ventilation

Always assess the ability of a patient to protect his or her airway and to maintain adequate gas exchange. Early intubation or mechanical ventilation often required for two main reasons:

1. Acute hypoxemic respiratory failure.

It may occur in cardiogenic shock and pulmonary edema, septic shock with pneumonia or acute respiratory distress syndrome (ARDS).

2. Ventilatory failure

Often occurs as a consequence of an increased load on the respiratory system in the form of acute metabolic acidosis or decreased lung compliance.

Also in shock, a large percentage of CO need for respiratory muscle (10 folds), lactic acid production from inefficient respiratory muscle activity presents an additional ventilatory load.

Ventilatory supports relieve work of breathing and allow redistribution of limited CO to other vital organ.

Sign of respiratory distress:

- inability to speak full sentences,
- accessory muscle activity
- paradoxical abdominal muscle activity
- extreme tachypnea (>40 breath/min)

After intubation and mechanical ventilation, declines in MAP frequently seen. The reasons:

1. Impended Venous Return from positive pressure ventilation (PPV)
2. reduced endogenous catecholamine secretion once stress associated with respiratory failures abates
3. Actions of drugs used in endotracheal intubation.
4. Right heart failure patients or preexisting pulmonary hypertension, due to increased right ventricula afterload due to PPV.

Many patients may be fluid responsive. Therefore, fluid administration and vasopressor support might needed before intubation.

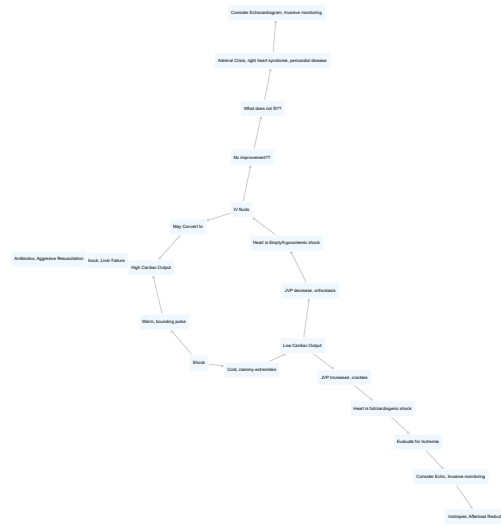


Figure 2.1: Approach to the patient in shock

2.1.3 Respiratory Failure

Respiratory failure mechanistically on the basis of pathophysiology can be categorized onto:

2.1.3.0.1 Type 1: Acute Hypoxemic Respiratory Failure

It occurs due to alveolar flooding, subsequent Ventilation-perfusion (VP) mismatch and intra-pulmonary shunt physiology.

Alveolar flooding occurs due:

- pulmonary edema, pulmonary edema further caused by:
 - elevated pulmonary microvascular pressures (heart failure or fluid overload) or
 - ARDS (low pressure pulmonary edema), defined as: acute onset (≤ 1 week) of bilateral opacities on chest imaging that are not fully explained by cardiac failure or fluid overload and ventilation perfusion mismatch, and shunt physiology that require positive end-expiratory pressure (PEEP)
- lung injury
- pneumonia
- alveolar hemorrhage

Type 1 RF occurs in sepsis, gastric aspiration, pneumonia, covid-19, near drowning, multiple blood transfusion & pancreatitis.

2.1.3.0.1.1 General Treatment

- Mechanical ventilation. Mechanical ventilation with high tidal volume (12 ml/kg ideal body weight) traditionally further injured the lung due to repeated alveoli overdistention and stretching. This is called ventilator induced volutrauma and even induce ARDS in patients without ARDS initially. many studies point toward low tidal volume (6 ml/kg ideal body weight) improve survival.

reproduce Figure of Pressure volume relationship!!!

- Prone positioning
- Low CVP or low PCWP or fluid conservative approach fluid conservative management to maintain low CVP or PCWP associated with fewer days with mechanical ventilation than fluid liberal approach in patients that who have been resuscitated from shock.

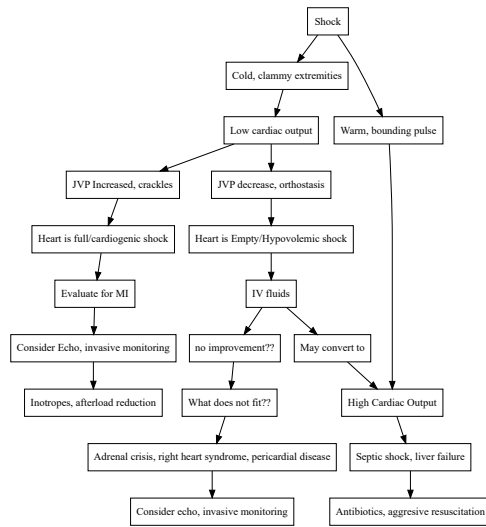


Figure 2.2: Approach to Shock

2.1.3.0.2 Type 2 Respiratory Failure: Hypercapnic Respiratory Failure

It is a consequence of alveolar hypoventilation and the results from inability to eliminate CO₂ effectively. The cause categorized into:

- impaired CNS drive to breathe (“wont breathe”) Cause: drug overdose, brainstem injury, sleep-disordered breathing, severe hypothyroidism
- reduced strength or impaired neuromuscular function in respiratory systems (“cant breathe”) cause: impaired neuromuscular transmission (eg. myasthenia gravis, Guillain-Barre syndrome, amyotrophic lateral sclerosis) or respiratory muscle weakness (eg. myopathy, electrolyte problem, fatigue)
- increased load on respiratory system (“cant breathe”) cause:
 - increased resistive load eg. bronchospasm
 - reduced lung compliance eg. alveolar edema, atelectasis, intrinsic PEEP
 - reduced chest wall compliance eg. pneumothorax, pleural effusion, abdominal distention
 - increased minute ventilation requirements eg. pulmonary embolus with increased dead space fraction, sepsis

2.1.3.0.2.1 general Treatment

Principal: reversing the underlying cause(s). Non-invasive ventilation (NIV) with positive pressure ventilation with tight fitting mask or nasal mask (avoidance of intubation) may stabilize these patients. It tested extensively in exacerbation of COPD. Contraindication for NIV include hemodynamic instability, inability to protect airway, respiratory arrests, significant airway secretions, significant aspiration risk.

2.1.3.0.3 Type 3 Respiratory Failure: Lung Atelectasis

The primary cause is lung atelectasis. It is common in perioperative period, this form is also called perioperative respiratory failure. Post general anesthesia, there are decrease in functional residual capacity lead to collapse of dependent lung units.

2.1.3.0.3.1 general treatment

treatment include: frequent positional changes, chest physiotherapy, upright positioning, control of incisional/or abdominal pain.

2.1.3.0.4 Type 4 Respiratory Failure: Metabolic Demands

Most often due to hypoperfusion of respiratory muscles in patients in shock. Normally, respiratory muscle consume < 5% total cardiac output (CO) and oxygen delivery. Patients in shock often experience respiratory distress due to pulmonary edema (eg. in cardiogenic shock), lactic acidosis and anemia. In these conditions, up to 40% CO distributed to respiratory muscle.

2.1.3.0.4.1 General Treatment

Endotrachea intubation & mechanical ventilation allow blood redistributed to vital organ and reverse the underlying cause.

2.1.3.1 Mechanical Ventilation

mechanical ventilation is used to assist or replace spontaneous breathing. Its application of high oxygen content + positive pressure. Primary indication is **acute respiratory** failure which there are two basic types: 1. hypoxemic and 2. hypercarbic. *Hypoxemic* present when arterial O₂ saturation <90% despite increased inspired O₂ fraction due to ventilation perfusion mismatch or shunting. *Hypercarbic* characterized by elevated arterial CO₂ (usually >50 mmHg) resulting from conditions that decrease minute ventilation or increased physiologic dead space.

2.1.3.1.1 Types of Mechanical Ventilation

there are two types of MV: noninvasive ventilation (NIV) and invasive (conventional MV)

2.1.3.1.1.1 Noninvasive ventilation

NIV: effective in acute and chronic respiratory failure with fewer complications (pneumonia & tracheolaringeal trauma). NIV provided by tight fitting mask, a nasal mask similar to that used for sleep apnea. It is especially effective in COPD. Modes used are bilevel positive airway pressure (BiPAP) or pressure support ventilation (PSV). Both used positive pressure during inspiration and expiration, although lower in expiration. Trial showed: COPD exacerbation pH 7.25 - 7.35 associated with good outcomes, for pH <7.25 the failure increase with decreasing pH. Patient with pH >7.35 NIV is not better than conventional treatment (controlled O₂, systemic glucocorticoid, bronchodilator and antibiotics).

The use of NIV in other respiratory failure is limited, and in some conditions contraindicated (cardiac arrest, respiratory arrest, decreased consciousness, GI bleeding, hemodynamic instability, MI, facial trauma, upper airway obstruction, inability to clear secretion), and may delay life saving measure. Reduced respiratory rate, decreased the use of accessory muscle (scalene,

sternomastoid and intercostal) are good clinical indicator of improvement. ABG should be evaluated hours after NIV initiation, if no improvement it may need a conventional MV

2.1.3.1.1.2 Invasive Ventilation (Conventional MV)

Intubation (cuffed tube inserted to trachea) to allow conditioned (warmed, oxygenated, humidified) gas delivered to the airways.

Principles of MV: optimize oxygenation while avoiding ventilator-induced lung injury due to overstretch and collapse/re-recruitment (protective ventilatory strategy)

2.1.3.1.2 Modes of Ventilation

modes are the manner in which breaths are triggered, cycled and limited. - trigger: inspiratory effort or time based signal - cycle : refers to the factors that determine the end of inspiration (volume, pressure and time cycling) - limiting factors are operator-specified values: airway pressure.

Most patient ventilated with assist-control, intermittent mandatory ventilation, PSV.

2.1.3.1.2.1 Assist control ventilation

Assist control (ACMV) is the most widely used. Used in initiation of MV. trigger: initiated by patient inspiratory effort, if none detected, by specified time signal. cycle: pressure or volume cycle. limit: specified tidal-volume, RR defined by patients rate or backup rate.

problem: auto-peep might occurs if dynamic hyperinflation due to high RR happens.

2.1.3.1.2.2 Intermittent Mandatory Ventilation

Most common modes: SIMV trigger: set mandatory breath, between each mandatory breath patient can breath spontaneously. in SIMV, mandatory breath delivered in synchrony with patients breath. if patient fail to initiate a breath, the machine give fixed volume and reset its timer for next breath. cycle: defined volume, limit:

difference with ACMV: only the a preset number of breath are ventilator-assisted. SIMV allowd patients with intact respiratory drive to exercise inspiratory muscles between assisted breath, useful for supporting and weaning intubation. problem: difficult to use in tachypnea, due to expiration attemp during ventilator inspiratory cycle. The pressure increase, and may abort the inspiration and thus lower tidal volume. In contrast, for tachypnea represent of metabolic or respiratory acidosis, change in ACMV will increase minute ventilation and help normalize pH

2.1.3.1.2.3 Pressure-support Ventilation

trigger: patient breath cycle: flow, inspiration is terminated when airflow fall below a certain level. limit: pressure

provides graded assistance and differs from the other two in that the operator set the pressure level to augment every spontaneous breath. The level of pressure is adjusted by observing the patient's RR.

2.1.3.1.2.4 Pressure-Control Ventilation (PCV)

useful to limit peak pressure, in patients with preexisting barotrauma or post thoracic surgery.

trigger: time cycle: time, limit: pressure

minute ventilation is altered through RR and pressure control.

2.1.3.1.2.5 Continuous Positive Airway Pressure (CPAP)

All ventilation occurs through the patient's spontaneous efforts. The machine provides fresh gas to the circuit and set constant pressure.

2.1.3.1.3 Protective Ventilatory Strategy

Whatever the modes of MV, the principles:

1. Set target tidal volume close to 6 ml/kg of ideal body weight
2. Prevent plateau pressure exceed 30 cm H₂O
3. Use the lowest possible FIO₂ to keep SaO₂ ≥ 90%
4. Adjust the PEEP to maintain alveolar patency while preventing overdistention and closure/reopening.

2.1.3.1.4 Maintenance for mechanical Ventilated patients

Respiratory System Mechanics

Peak airway pressure determined by two variables: airway resistance and respiratory system compliance. At the end of inspiration, there is a transient pause of inspiratory flow, the pressure called plateau pressure. Plateau pressure determined by respiratory system compliance. During volume controlled ventilation, the difference peak airway pressure (airway + respiratory system) and plateau pressure (respiratory system) provides a quantitative assessment of airway resistance. Patients with increased airway resistance typically have increased peak airway pressures and abnormally high gradients between peak and plateau airway pressure (> 10 - 15 cmH₂O) at a constant inspiratory flow rate of 1 L/s (shown in figure Figure 2.3). Normally, respiratory system

compliance is $\sim 100 \text{ ml/cmH}_2\text{O}$, divided by the lungs and chest wall. Pathophysiologic process decrease wall compliance such as pleural effusion, pneumothorax & increase abdominal girth. Lung compliance decreased by pneumonia, pulmonary edema, alveolar hemorrhage, interstitial lung disease, or auto PEEP. Auto-PEEP occurs when there is insufficient time for emptying of alveoli before the next inspiratory cycle, thus alveoli failed to decompress fully and remains positive pressure at all phase of respiration. This phenomenon results most commonly from obstruction of distal airways such as Asthma or COPD.

Patients frequently require sedatives & analgetics.

2.1.3.1.4.1 Analgesics

Opiates is the mainstay of mechanically ventilated patients

2.1.3.1.4.2 Sedatives

After pain adequately controlled, sedation needed for anxiolysis, treatment of subjective dyspnea, reduction of autonomic hyperactivity, reduction of total O₂ consumption. Nonbenzodiazepine are preferred because benzodiazepine associated with increased delirium and worse patients outcome.

Amnesia could be achieved with propofol or benzodiazepine such lorazepam or midazolam.

2.1.3.1.4.3 Neuromuscular Blocking Agents

Patients with ventilatory dyssynchrony despite optimal sedation may need paralytic agents. Without prolonged weakness, a myopathy known as the postparalytic syndrome. Thus, neuromuscular blocking is the last agent, only in patients which fail to achieve ventilator synchrony with aggressive sedation. Remember, neuromuscular blocking agent do not result in pharmacological paralysis without altering mental status, thus sedative is almost always required for amnesia.

2.1.3.1.5 Ventilator Weaning and Extubation

Recognition of patients readiness to be liberated from mechanical ventilation is important. Screening conducted daily, criteria:

1. Stable oxygenation ($\text{PaO}_2 / \text{FiO}_2 > 200$ and $\text{PEEP} \leq 5 \text{ cm H}_2\text{O}$)
2. Cough and Airway Reflexes Intact
3. No Vasopressor administered
4. No Sedatives administered

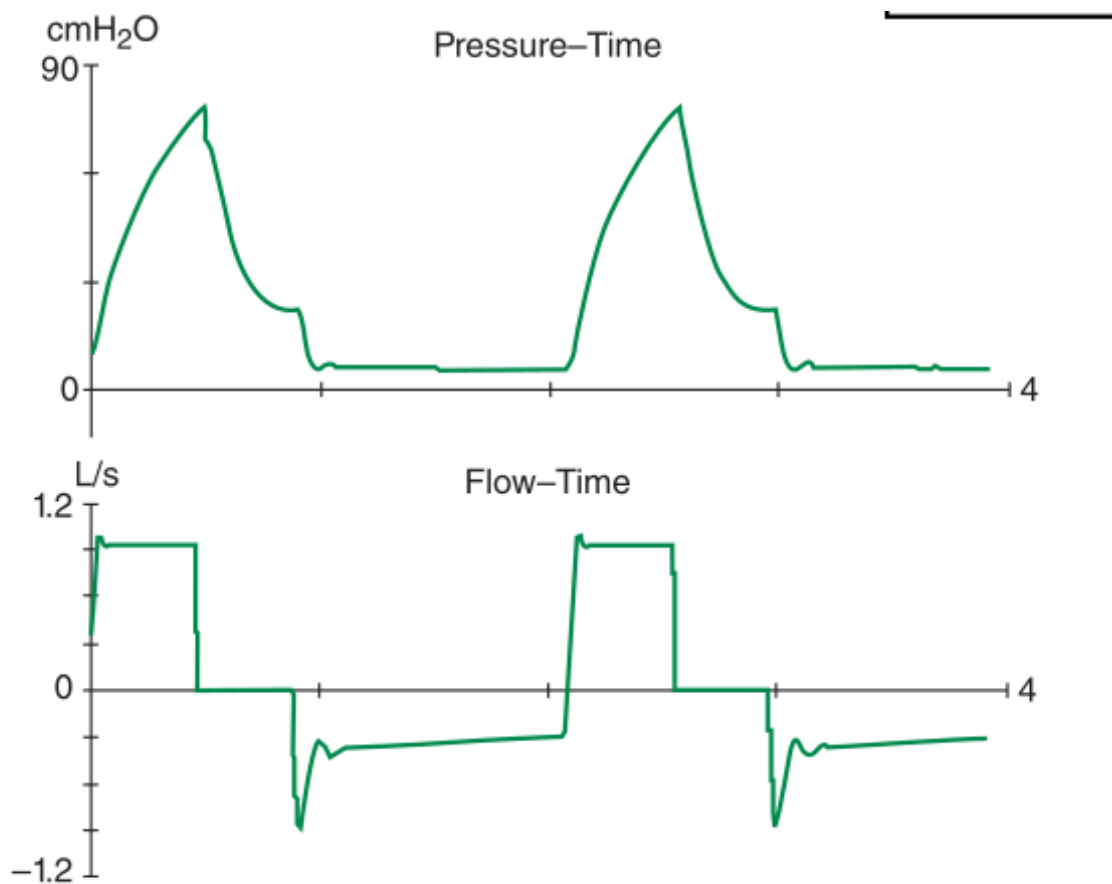


FIGURE 300-6 Increased airway resistance with auto-PEEP. The top waveform (airway pressure vs time) shows a large difference between the peak airway pressure (80 cmH₂O) and the plateau airway pressure (20 cmH₂O). The bottom waveform (flow vs time) demonstrates airflow throughout expiration (reflected by the flow tracing on the negative portion of the abscissa) that persists up to the next inspiratory effort.

Figure 2.3: Graph Illustration taken from harrison

if all requirements passed, patient should undergo a spontaneous breathing trial (SBT), if sedative still administered, patients could also undergo spontaneous awakening trial (SAT).

SAT/SBT trial consists of a period of breathing through endotracheal tube with little or without ventilator support (CPAP 1-5 cmH₂O with or without low-level pressure support (PSV) and open T-piece breathing system) for 30 - 120 minutes. SBT Failure or should be stopped if:

1. RR >35/min for >5 min
2. O₂ saturation <90%
3. HR > 140x/min or a 20% increase or decrease from baseline
4. SBP <90 mmHg or >180 mmHg
5. Increased anxiety or diaphoresis

if none of these events occurred, patients should be considered for an extubation trial. Despite these careful criteria, 10% of patients develop respiratory distress after extubation and may require resumption of mechanical ventilation and reintubation.

Once patients determined that able to breathe spontaneously, then proceed to remove the artificial airway which should be undertaken only when it is patients showed the ability to protect airway, able to cough and clear secretions, alert enough to follow commands. If reintubation is deemed difficult if required, evaluation with cuff-leak test is supported by evidence. If possibility of post-extubation stridor, administration of systemic steroid should be given.

2.1.3.2 Prolonged MV and Tracheostomy

5-13 % of patients with MV will go on to require prolonged MV (>21 days). It is important to decide whether and when to perform a tracheostomy. Tracheostomy is thought to be more comfortable, require less sedation, and provide a more secure airway and reduce weaning time.

In general, if a need of MV for >10-14 days, a tracheostomy would be indicated.

2.1.3.2.1 Maintenance for mechanical Ventilated patients

Respiratory System Mechanics

Peak airway pressure determined by two variables: airway resistance and respiratory system compliance. At the end of inspiration, there is a transient pause of inspiratory flow, the pressure called plateau pressure. Plateau pressure determined by respiratory system compliance. During volume controlled ventilation, the difference peak airway pressure (airway + respiratory system) and plateau pressure (respiratory system) provides a quantitative assessment of airway resistance. Patients with increased airway resistance typically have increased peak airway pressures and abnormally high gradients between peak and plateau airway pressure (> 10 - 15 cmH₂O) at a

constant inspiratory flow rate of 1 L/s (shown in figure Figure 2.3). Normally, respiratory system compliance is $\sim 100 \text{ ml/cmH}_2\text{O}$, divided by the lungs and chest wall. Pathophysiologic processes decrease wall compliance such as pleural effusion, pneumothorax & increase abdominal girth. Lung compliance decreased by pneumonia, pulmonary edema, alveolar hemorrhage, interstitial lung disease, or auto PEEP. Auto-PEEP occurs when there is insufficient time for emptying of alveoli before the next inspiratory cycle, thus alveoli failed to decompress fully and remains positive pressure at all phase of respiration. This phenomenon results most commonly from obstruction of distal airways such as Asthma or COPD.

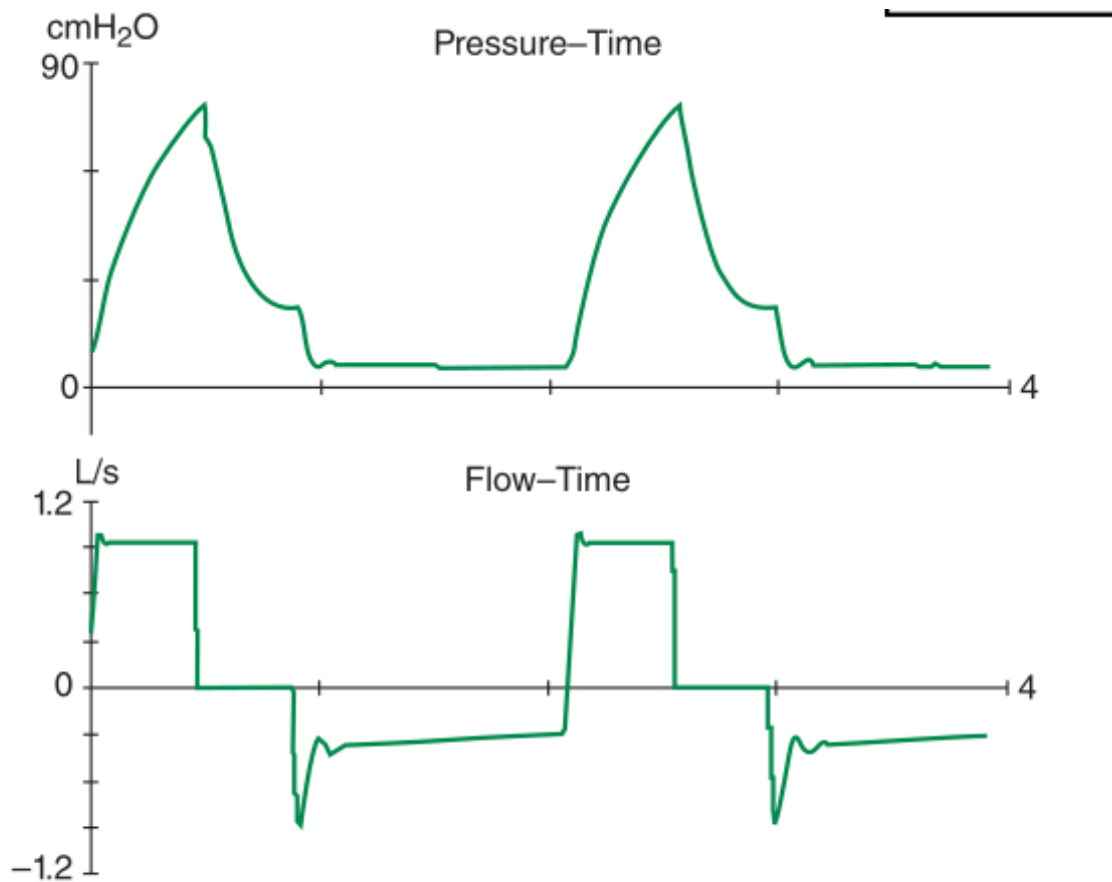


FIGURE 300-6 Increased airway resistance with auto-PEEP. The top waveform (airway pressure vs time) shows a large difference between the peak airway pressure (80 cmH₂O) and the plateau airway pressure (20 cmH₂O). The bottom waveform (flow vs time) demonstrates airflow throughout expiration (reflected by the flow tracing on the negative portion of the abscissa) that persists up to the next inspiratory effort.

Figure 2.4: Graph Illustration taken from harrison

Patients frequently require sedatives & analgetics.

2.1.3.2.1.1 Analgesics

Opieates is the mainstay of mechanically ventilated patients

2.1.3.2.1.2 Sedatives

After pain adequately controlled, sedation needed for anxiolysis, treatment of subjective dyspnea, reduction of autonomic hyperactivity, reduction of total O₂ consumption. Nonbenzodiazepine are preferred because benzodiazepine associated with increased delirium and worse patient outcome.

Amnesia could be achieved with propofol or benzodiazepine such as lorazepam or midazolam.

2.1.3.2.1.3 Neuromuscular Blocking Agents

Patients with ventilatory dyssynchrony despite optimal sedation may need paralytic agents. Without prolonged weakness, a myopathy known as the postparalytic syndrome. Thus, neuromuscular blocking is the last agent, only in patients which fail to achieve ventilator synchrony with aggressive sedation. Remember, neuromuscular blocking agent does not result in pharmacological paralysis without altering mental status, thus sedative is almost always required for amnesia.

2.1.3.2.2 Ventilator Weaning and Extubation

Recognition of patient readiness to be liberated from mechanical ventilation is important. Screening conducted daily, criteria:

1. Stable oxygenation ($P_{aO_2} / F_{iO_2} > 200$ and $PEEP \leq 5$ cm H₂O)
2. Cough and Airway Reflexes Intact
3. No Vasopressor administered
4. No Sedatives administered

If all requirements passed, patient should undergo a spontaneous breathing trial (SBT), if sedative still administered, patients could also undergo spontaneous awakening trial (SAT).

SAT/SBT trial consists of a period of breathing through endotracheal tube without ventilator support (CPAP 5 cmH₂O with or without low-level pressure support and open T-piece breathing system) for 30 - 120 minutes. SBT Failure or should be stopped if:

1. RR >35/min for >5 min
2. O₂ saturation <90%
3. HR > 140/min or a 20% increase or decrease from baseline
4. SBP <90 mmHg or >180 mmHg
5. Increased anxiety or diaphoresis

If none of these events occurred, patients should be considered for an extubation trial. Despite these careful criteria, 10% of patients develop respiratory distress after extubation and may require resumption of mechanical ventilation and reintubation.

Severity (PaO₂/FiO₂)

mild: 200 mmHg < X ≤ 300 mmHg; moderate: 100 mmHg < X ≤ 200 mmHg; Severe: ≤ 100 mmHg

2.1.4 Multiorgan system failure

Multiorgan system failure, defined by the simultaneous presence of physiologic dysfunction and/or failure of two or more organs. Organ failure, no matter how it is defined must persist beyond 24 hours.

2.2 Specific Critical illness

2.2.1 Acute Respiratory Distress Syndrome (ARDS)

ARDS is clinical syndrome of **severe dyspnea** of rapid onset, **hypoxemia**, and **diffuse pulmonary infiltrates** leading to respiratory failures.

2.2.2 Diagnostic Criteria for ARDS

ddx: - cardiogenic pulmonary edema - bilateral pneumonia - alveolar hemorrhage

2.2.3 Epidemiology

Estimated incidence: 60/100.000 population.

2.2.3.1 Etiology

ARDS can be caused by diffuse lung injury from many medical and surgical disorders. It may be direct (on the lung) or indirect (extrapulmoner including thoracic structure). The most case (>80%) are caused by pneumonia and sepsis. Then, gastric aspiration, trauma, multiple transfusiion, pancreatitis, and drug overdose. Traumatic cause include pulmonary contusion, multiple bone fractures, chest trauma/flail chest. The risk increase with predisposing medical condition.

2.2.3.2 Pathophysiology

Natural history of ARDS divided into 3 phases - exudative, proliferative, and fibrotic.

2.2.3.2.1 Exudative Phase

This phase encompasses the first 7 days after precipitating ARDS risk factor. The alveolar capillary endothelial cells and type 1 pneumocytes injured, loss of tight alveolar barrier to fluid and macromolecules. Edema fluid containing protein accumulates in the interstitial and alveolar spaces. Proinflammatory cytokines increase followed by leukocyte recruitment. The leaked plasma condenses and aggregates in the air spaces + cellular debris + dysfunctional surfactant to form hyaline membrane whorls.

Alveolar edema predominantly involves dependent portions of the lung with diminished aeration. Large lung diminished aeration causes increase in lung compliance. Alteration in alveolar spaces are exacerbated by microvascular occlusion leading to reduced perfusion and pulmonary hypertension. It also causes intrapulmonary shunting and hypoxemia, leading to dyspnea.

sign: - dyspnea, may lead to respiratory fatigue - chest radiograph reveals opacities resembling pulmonary edema. Distinguishing features from cardiogenic pulmonary edema are no cardiomegaly, no pleural effusion, or no pulmonary vascular redistribution. - laboratory finding indicative of the underlying disorder.

2.2.3.2.2 Proliferative Phase

Phase day 7 to 21. May show an improvement, including weaning of mechanical ventilation. It may show some degree of lung repair, reduced alveolar exudates, and shift to lymphocyte predominant infiltrate, increased surfactant.

sign: - improvement in symptoms, but still dyspnea - tachypnea - hypoxemia

2.2.3.2.3 Fibrotic Phase

Phase 3-4 weeks after lung injury. Many patients may recover, but some enter a fibrotic phase. The exudate converts to extensive alveolar duct and interstitial fibrosis. Destruction of acinar architecture leads to emphysema-like changes (bullae), and may also develop microvascular occlusion and pulmonary hypertension.

2.2.3.3 Treatment

2.2.3.3.1 General Principles

Emphasize on treatment of the Critical ill

1. treatment of underlying cause
2. minimization of unnecessary procedures
3. standardized bundle care
4. prompt recognition of nosocomial infections

5. adequate nutrition

2.2.3.3.2 Mechanical Ventilation

Patient meeting criteria ARDS frequently become respiratory fatigues, so it requires mechanical ventilation.

2.2.3.3.2.1 Minimize ventilator induced lung injury

Mechanical ventilatio (MV) can save lifes, but also can aggravate lung injury due to volutrauma and atelectrauma. Volutrauma can occurs due to overdistention of alveoli due to effort to inflate the consolited lungs thus damaging the still functioning alveoli.

- **Low tidal volume 6ml/kg predicted body weight are better than coenventional 12 ml/kg bodyweight.**
- lower airway pressure, plateau pressure ≤ 30 cm H₂O.
- RR ≤ 35

2.2.3.3.2.2 Minimize atelectrauma

In presence of fluid and loss of surfactant, PEEP is required to prevent atelectasis. PEEP is adjusted to minimize FIO₂ while able to provide adequate PaO₂ without causing alveolar overdistention. Follow table of PEEP-FiO₂ combinations from ARDS Network trial.

other technique includes **recruitment maneuver**, that transiently increase PEEP to high levels to recruit ateletatic lung can increase oxygenation, but a mortality benefit has not been established.

2.2.3.3.2.3 Prone positioning

It improves survival.

2.2.3.3.3 Fluid Management

increased pulmonary vascular permeability leading to interstitial and alveolar edema fluid rich in protein is a centra features of ARDS. impaired vascular integrity augments the normal increase in extravascular lung water. Thus keeping low left atrial filling pressure could minimize pulmonary edema. Fluid restriction and diuretics should be an important aspect of ARDS management, limited only by hypotension and hypoperfusion.

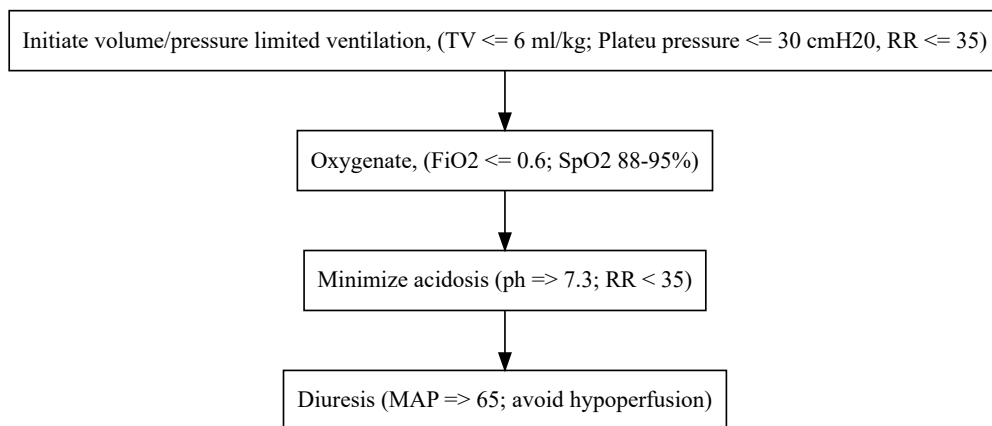


Figure 2.5: ARDS Algorithm

2.2.3.3.4 Neuromuscular blockage

Sedation alone may not adequate for ventilator synchrony. Selective use of neuromuscular blockage for ventilator synchrony could improves survival.

2.2.3.3.5 Glucocorticoids

Current evidence does not support the routine use of glucocorticoids.

2.2.3.4 Prognosis

extrapulmonary organ failures and sepsis counts > 80% of deaths. Risk factor for reduced survival are age (>60 yo), preexistin organ dysfunction. Patients who survives despite prolonged respiratory failures and dependent on MV may recover within 6 months.

2.2.4 Approach to Circulatory Shock

shock is the clinical condition of organ dysfuntion resulting from imbalance between celular oxygen supply and demand. Shock is reversible in early state, but if left untreated leads to multisystem organ dysfunction and death.

2.2.4.1 Pathophysiology of Shock

The cellular oxygen imbalance most commonly related to impaired oxygen delivery in the setting circulatory failure. - shock may also occurs in the setting of failure oxygen utilization (cyanide poisoning) - and state of increased oxygen demand.

In the setting of inadequate oxygen supply, the cell cannot maintain aerobic metabolism, thus forced into anaerobic metabolism and produce lactate and less ATP. Less ATP supply impaired cell ability to maintain hemeostasis such as maintaining N/K pump. Celular pump failure cause influx of calcium and leads to activation of phospolipases and protease, causing cellular swelling and death. In addition, hypoxia cause leakage of intracellular contents to the extracellular space, activating inflammatory cascades.

Oxygen delivery influenced by two major components, namely Cardiac Output (CO) and arterial oxygen content (CaO₂).

$DO_2 = CO \times CaO_2$, major determinants of CO are HR and SV

$DO_2 = (HR \times SV) \times CaO_2$, major determinants of SV are preload, afterload (SVR), and cardiac contractility. thus

$SV \sim (Preload \times Contractility) / SVR$

Preload refers to myocardial fibers length before contraction (LV-EDV). Contractility refers to the ability of the ventricle to contract independent of preload and afterload. SVR represents afterload, force that ventricle must overcome to send the blood out of the cardiac chambers.

The CaO_2 is composed of oxygen carried by convection with hemoglobin and oxygen dissolved in blood,

$$\text{CaO}_2 = (\text{Kg} \times 1.39 \times \text{SaO}_2) + (\text{PaO}_2 \times 0.03)$$

A disease process that affects these variables (HR, preload, contractility, SVR, SaO_2 , or Hb) has potential to reduce DO_2 .

2.2.4.2 Classification of Shock

Four major shock types are distributive, cardiogenic, hypovolemic, and obstructive. Each has unique hemodynamics profile.

2.2.4.2.1 Distributive Shock

Distributive shock is condition of reduced O_2 delivery where the primary physiologic disturbance is **reduction in SVR**. There are **increase of CO** as a compensatory effort. CVP and PCWP reduced. The most common etiology is sepsis. Sepsis is life-threatening organ dysfunction due to dysregulated host response to infection. If accompanied by persistent hypotension despite adequate fluid resuscitation and needed vasopressor support, it is called septic shock. Other medical cause : Anaphylaxis (IgE mediated vasodilatation after exposure to allergen), pancreatitis, severe burns, liver failure. There almost 35% extravasation of circulating blood volume within 10 minutes. Patients with severe brain or spinal cord injury may have a reduction in SVR due to disruption of autonomic pathways and blood pooling in venous system. Other cause is adrenal insufficiency due to chronic steroid use, medication , malignancy, adrenal hemorrhage, infection or autoimmune. Acute stress condition, created deficit in steroid, and cause vasodilatation as well as aldosterone deficiency-mediated hypovolemia.

2.2.4.2.2 Cardiogenic Shock

Cardiogenic shock is characterized by reduced O_2 delivery related to a reduction in CO owing to a primary cardiac problem. There are compensatory response as an increase in SVR. If left ventricle is dysfunction, there will be increased PCWP, and if right ventricle is will present as increased CVP. SV may reduced due to reduced contractility (MI, Cardiomyopathies, myocarditis), valvular disease, or due to alteration in HR (bradycardia and tachycardia)

2.2.4.2.3 Hypovolemic Shock

Hypovolemic shock encompasses disease that reduce CO and DO_2 via a reduction in preload. There is elevation in SVR and low CVP-PCWP. Most common cause is hemorrhage followed by fluid loss from diarrhea or vomiting, diuresis, skin damage.

2.2.4.2.4 Obstructive Shock

Obstructive shock is reduced O_2 delivery related to reduced CO because impairment in blood flow, both inflow and outflow, in extracardiac pulmonary vascular or other mechanical process. Inflow obstruction Etiology: tension pneumothorax, cardiac tamponade, restrictive pericarditis. Outflow obstruction etiology: pulmonary embolism, venous air embolism, fat embolism or aortic dissection.

UGD $\Rightarrow$ hypovolemic (30,8%) > septic (27,2%) > distributive non septic (23,4%) > cardiogenic (14%) $\gg$ obstructive shock (0,9%). ICU $\Rightarrow$ Shock predominates (62%) $\gg$ Hypovolemic (16%) $\sim$ cardiogenic shock (15%) $\gg$ obstructive shock (2%).

Mortality rate: cardiogenic $\sim$ septic > hypovolemic

2.2.4.3 Stages of Shock

There are 3, preshock, shock and irreversible shock. Body utilizes variety of physiologic response to counter initial insult. There are no overt signs of organ dysfunction. Lab: slight elevation of creatine or troponin or mild elevation of lactate. If host compensatory mechanisms are not enough, organ dysfunction occurs and irreversible shock ensues.

2.2.4.4 Evaluation of the patient with Shock

Early recognition is paramount important to reverse the stages of shock. Anamnesis to find the cause. Physical examination to identify circulatory failure (hypotension SBP <90 mmHg, or MAP < 65 mmHg) and signs of organ dysfunction. Check urine output. Finding: High CO shock: warm peripherals, CRT <2 s, large pulse pressure (LOW DBP). Low CO shock: delayed CRT, cool extremities, weak pulses.

Increased intravascular filling pressure and JVP (cardiogenic, obstructive shock)

Shock Index : defined as the HR/SBP with normal SI 0,5 - 0,7

Lab test: - Lactate - Renal function - liver function - cardiac enzymes - CBC - Hemostatic index - U/L - PP test if needed - ECG - CXR - POCUS (RUSH, ACES, SESAME protocol)

2.2.4.5 Initial Treatment of Shock

2.2.4.5.1 Initiate treatment for circulatory shock.

Find venous access and invasive monitoring if needed

2.2.4.5.2 Volume Resuscitation

The physiologic goal of volume resuscitation is to move the patient to the **nonpreload-dependent portion of the Starling curve**. Most patients with any of the four shock types will benefit from an increase in intra-vascular volume. suspected septic shock, a minimum of 30 mL/kg is recommended by the Surviving Sepsis Campaign. While the need for volume resuscitation is most apparent for patients with distributive or hypovolemic shock, even patients with cardiogenic shock may benefit from cautious volume replacement. In these patients, there should be a careful assessment of volume status prior to volume administration.

Volume resuscitation will begin with **crystalloid**. In patients with hypovolemic shock due to ongoing hemorrhage, volume replacement with packed red blood cells is warranted. In cases of massive transfusion, platelets and fresh frozen plasma should be provided to offset the dilution of these components during volume replacement. Because hemoglobin is a key determinant of CaCO₂, **red cell administration may be a part of volume replacement even without hemorrhage in order to optimize oxygen delivery if hemoglobin content is <7 g/dL**.

Assessment of intravascular volume status (and the adequacy of volume resuscitation) begins with the physical examination. The **passive leg raise (PLR) test can predict responsiveness to additional intravenous fluid (IVF) by providing the patient with an endogenous volume bolus**. While the patient is resting in a semirecumbent position at a 45° angle, the bed is placed in Trendelenburg position such that the patient's head becomes horizontal and the legs are extended at a 45° angle. There is then an immediate (within 1 min) assessment of changes in CO (or pulse pressure variation as a surrogate). It is important to emphasize that one does not merely look for changes in blood pressure; if the shock patient is mechanically ventilated there is the option of looking at changes in **SV variation (or pulse pressure variation)** during the respiratory cycle to assess volume responsiveness. A >12% SV variation suggests a volume-responsive state. This measurement requires that the patient be in a volume cycle mode of ventilation, without breath-to-breath variations in intrathoracic pressure and without arrhythmias.

There is also increased use of echocardiography to assist in determination of intravascular fluid status, with a variety of static and dynamic variables. The most commonly used parameters to assess adequacy of volume resuscitation are inferior vena cava (IVC) diameter and IVC collapse. Alternatively, serial assessments of LV function can be performed while volume is being administered. Placement of a pulmonary artery catheter (PAC) is another tool for assessment of volume status. This more invasive measure involves placement of the PAC into

the central venous circulation and through the right heart. Ports in the PAC (Swan-Ganz catheter) allow for direct measurement of CVP, pulmonary artery (PA), and PCWPs. The PCWP is used as a surrogate for LA pressure.

2.2.4.5.3 Vasopressor and Inotropic Support

If hypotension and inadequate tissue perfusion persist, then vasopressor and inotropic support should be initiated. **The use of vasopressors and inotropes must be tailored to the primary physiologic disturbance.**

In distributive shock, the aim is to increase the SVR. **Norepinephrine is the first-choice vasopressor**, with potent α_1 and α_2 adrenergic effects. The α_1 causes vasoconstriction while α_2 has positive inotropic and chronotropic effects. At high doses, epinephrine has a similar profile (at lower doses the β effects predominate) but is associated with tachyarrhythmia, myocardial ischemia, decreased splanchnic blood flow, pulmonary hypertension, and acidosis. In distributive shock, vasopressin deficiency may be present. Vasopressin acts on the vasopressin receptor to reverse vasodilation and redistribute flow to the splanchnic circulation. **Dopamine does not have a role as a first-line agent in distributive shock.**

For patients with **cardiogenic shock**, **dobutamine is the first-line agent**; it is a synthetic catecholamine with primarily β_1 -mediated effects and minimal α adrenergic effects. The β_1 effect is manifest in increased inotropy and the β_2 effect leads to vasodilation with decreased afterload; it can be used with norepinephrine in patients with mixed distributive and cardiogenic shock.

2.2.4.5.4 Oxygenation and Ventilation Support

Supplemental oxygen should be initiated and titrated to maintain SpO₂ of 92–95%. patients may require intubation and MV for two reason:

1. primary pulmonary process or related to LV dysfunction and elevations of PCWP. For patients with all types of shock, there can be development of ARDS and subsequent V/Q mismatch and shunt.
2. high minute ventilatory needs to compensate for metabolic acidosis. As shock progresses, they may not be able to maintain adequate respiratory compensation.

It is important to provide ventilation with lung-protective strategies focused on low tidal volume ventilation and optimization of positive end-expiratory pressure to minimize ventilator-induced lung injury.

2.2.4.5.5 Antibiotics

For patients presenting with undifferentiated shock, if the diagnosis of septic shock is being entertained, then broad-spectrum antibiotics should be administered after obtaining appropriate cultures. While it is ideal to initiate antibiotics after appropriate cultures, the inability to obtain cultures should not delay the start of treatment

2.2.4.5.6 Other specific treatment for specific etiology

The initial evaluation (history, physical examination, and diagnostic testing) may have identified an etiology of shock that requires urgent lifesaving intervention in addition to the initial treatment steps outlined above.

2.2.5 Sepsis

2.2.6 Cardiogenic Shock and Pulmonary edema

2.2.7 Cardiac Arrest

2.2.8 Neurological Critical illness

2.3 Important Calculation

2.3.1 Airway & Ventilation

2.3.2 Circulation

2.3.2.1 Oxygen Delivery

Oxygen delivery (DO_2 or QO_2) is a function of cardiac output and the content of O_2 in the arterial blood (CaO_2). CaO_2 is determined by HB concentration, arterial HB saturation, and dissolved O_2 dissolved in plasma (not bound to HB). Equation provided at Equation 2.1. Nearly all oxygen delivered to the tissue is bound to HB. Dissolved O_2 are negligible.

Arterial oxygen content (CaO_2) normally around 21.16 mL/dL, while the mixed venous blood (VO_2) is 15.76 mL/dL, thus normal tissue extraction is around 25%. Reduced VO_2 may result from inadequate CO, reduced HB or reduced SaO_2 . Abnormally high VO_2 may be caused by fever, agitation, shivering and thyrotoxicosis.

$$\text{QO}_2 = \text{Cardiac Output (dL/min)} \times (1.39 \times \text{HB concentration} \times \text{HB\% Saturation} + 0.0031 \times \text{PaO}_2) \quad (2.1)$$

$$QO_2 = 50 \text{ dL/min} \times (1.39 \times 15\text{g/dL} \times 100\% + 0.0031 \times 100)$$

$$QO_2 = 50 \text{ dL/min} \times 21.6\text{mL O}_2 \text{ per dL blood or } CaO_2 = 1058 \text{ mL O}_2 \text{ per min}$$

3 Hematology & Medical Oncology

3.1 Oncological Emergency

3.2 Hematology

3.2.1 Transfusion Medicine

3.3 Medical Oncology

3.4 Procedures in Hematology and Medical Oncology

3.4.1 Examination of Blood Cells & Marrow Cells

3.4.1.1 Examination of Blood Cells

3.4.1.1.1 Automated Quantitative Parameter

Red cell parameter that measured directly: MCV, Red cell number, hemoglobin concentration, and red cell distribution width. Parameter derived from calculation: HCT, MCH, MCHC

3.4.1.1.1.1 Measurement of RBC count and hematocrite

HCT are measured from $RBC \times MCV / 10$, falsely elevated MCV and decreased RBC can be observed when red cell autoantibodies are presents and binding capability at room temperature, thus creatig red cell clumping.

3.4.1.1.1.2 Measurement of Hemoglobin

RBC contain mixture of HB, including: oxyhemoglobin, carboxyhemoglobin, methemoglobin, and other minor hemoglobin. HB quantification done lysing the cell and converted the all HB variants to cyanmethemoglobin for quantification by absorptio at 540 mm. Major interference with this method is the present of chylomicronemia.

3.4.1.1.1.3 Standard Red Cell Indices

Mean Cell Volume

Mean cell volume (MCV) are measured directly by electrical impedance or light scattering.

Mean Cell Hemoglobin

Mean cell hemoglobin (MCH) is the amount of HB per red cell. It is derived from Hemoglobin content and red cell count, two most accurate measurement, thus could provide consistent result among many type of analyzer. Reduced cell hemoglobin is paralel to the reduced MCV. Mean cell hemoglobin concentration is not used much for diagnostic, it is primarily ised for quality control purpose.

Red Cell Distribution Width

Red cell distribution widht or RDW is an estimate of volume variability in red cells population. RDW expressed as 1 SD of red cell volume measurements divided by the MCV. RDW has developed as biomarker for various clinical condition and may be a surrogate of systemic inflammation

Reticulocyte Count & RNA Content

Reticulocyte is a newly released anucleate red cell that enters bloodstream with residual but detectable RNA. The Reticulocyte counts permits an estimate of marrow erythrocyte production, thus provide tools for evaluation anemia due to inadequate production or accelerated destruction. Reticulocytes are now measured by exposig the samples with Fluoresced, RNA binding Dyes. Measurement are reported in abosolute counts, reticulosytes per microliter of Blood, obviating the need to correct for reduced red cell counts in anemia.

The measuremet of Reticulocytes specific hemoglobin content (CHr and RET-He) are closely related to iron availability withiin 24-48 hours, and is very usefull in detecting functional iron deficiency in complex clinical situation such as chronic inflamation or chronic kidney disease. Reticulocytes specific hemoglobin are superior to traditional iron parameter due to feritin also acts as a acute phase reactants.

Other Indices

Nucleated red cells: are presents in newborns, physiologically stress individual and variety disorders such as hypoxic states, severe hemolytics anemia, primary myelofibrosis, infiltrative disease of the marrow.

3.4.1.1.1.4 Leukocyte Counts and Differetial

3.4.1.1.2 Qualitative Parameter

3.4.1.2 Examination of Marrow Cells

Bone marrow aspiration was devised by Arinkin in 1923, then the regular use posterior iliac crest for aspiration and biopsy occurred in 1970.

4 Allergy & Immunology

5 Rheumatology

5.1 Autoimmune

5.2 Joint Disease

5.2.1 Crystal Deposition Disease

5.2.1.1 Gout and Hyperuricemia

Gout is a metabolic disorder resulting from the tissue deposition of monosodium urate crystals in or around joints and/or the crystallization of uric acid in the renal collecting system. The metabolic derangement responsible for gout is the supersaturation of blood and body fluids with the urate ion to the point that crystal formation is possible. At physiologic pH and at normal body temperature, urate is considered to be supersaturated at concentrations of 6.8 mg/dL or greater. Therefore, from a biologic perspective, hyperuricemia is any serum urate level greater than 6.8 mg/dL in both men and women. Although hyperuricemia is a necessary prerequisite for developing gout, only 20% of all hyperuricemic subjects will ultimately develop gout.

5.2.1.1.1 Epidemiologi

A number of factors have been proposed to explain this dramatic rise. These include the overall increase in longevity; the increased prevalence of hypertension, metabolic syndrome, and obesity; the ubiquitous use of thiazide diuretics and low-dose aspirin; changes in dietary trends, including the greater use of high-fructose corn syrup as a sweetener; and finally, the increase in survival of patients with end-stage renal disease and organ transplantation. There is a direct correlation between the degree of serum urate elevation and the likelihood of developing gout. The reported annual incidence of gout in subjects with baseline serum urate levels greater than or equal to 9 mg/dL is 4.9%, compared with only 0.5% in people with serum urate levels of 7.0 to 8.9 mg/dL.

5.2.1.1.2 pathobiology

Uric acid is the end product of purine metabolism in humans. In most mammals, purine catabolism is taken one step further through the enzyme uric acid oxidase or uricase, with the purine end product in these species being the very soluble allantoin. Humans and most other hominoids lost the ability to produce the enzyme uricase nearly 18 million years ago. As a result, uric acid accumulation is possible. Whether caused by overproduction of uric acid or its underexcretion by the kidneys, this accumulation leads to supersaturation of urate ion in blood and the precipitation of monosodium urate crystals in synovial fluid, soft tissues, and organs.

Urate is produced by the conversion of a very soluble molecule, hypoxanthine, to the less soluble xanthine, which, in turn, is converted to the very insoluble uric acid by progressive purine ring oxidations catalyzed by the enzyme xanthine oxidase. Xanthine oxidase is present in several organs, but most activity in the body is found in the liver and intestines. Because of its potential for causing disease, urate elimination is very important. The total daily accumulation of uric acid from de novo synthesis, nucleotide degradation, and dietary consumption is normally balanced by renal excretion of approximately two thirds of the total urate turnover and intestinal elimination by the remaining one third.

Simply put, hyperuricemia occurs when urate production is not balanced by renal excretion. In 90% of all gout patients, the cause of this imbalance is renal underexcretion. The remaining 10% of gout cases are caused by purine overproduction or a combination of overproduction and underexcretion.

uric acid is small and not protein bound, it is completely filtered by the glomerulus. In normal persons, approximately 8 to 10% of the filtered load is ultimately cleared in the urine. Both reabsorption and secretion occur in this segment through the actions of several organic acid transporters, with the net effect being the reabsorption of nearly 90% of the uric acid filtered at the glomerulus. These organic acid transporters are also responsible for eliminating organic acids other than uric acid as well as many commonly used medications. The gouty subjects excrete only 70% as much uric acid as normal individuals at any given serum urate concentration.

10% of gouty subjects, hyperuricemia is caused by uric acid overproduction rather than reduced renal excretion. In most of these people, the hyperuricemia reflects accelerated cell turnover (e.g., lymphoproliferative and myeloproliferative diseases, psoriasis, chronic hemolytic states, polycythemia vera, and certain muscle glycogenoses) or by enhanced purine nucleotide breakdown (e.g., alcohol abuse or fructose ingestion). Regardless of the cause, urate overproduction is documented by a 24-hour urine collection showing greater than 1000 mg of uric acid while eating a standard Western diet.

Monosodium urate crystals in joints, soft tissues, and organs are the cause of pain and destruction in gout. Urate crystals will form only when physiologic conditions permit. A reduction in pH or temperature will lower the solubility threshold even further. Not all people who are hyperuricemic will form crystals, however. There appears to be an additional requirement of a “nucleation factor” that is still poorly defined. Monosodium urate crystals form in joints

and soft tissues of individuals long before they cause any symptoms of gout. They deposit in small lattice structures called microtophi on the surface of cartilage and synovial lining. These microtophi slowly grow but are generally stable as long as the environment surrounding them does not change drastically with regard to pH, urate concentration, or temperature. At the time of the first and subsequent flares, something does change in the joint environment to cause these crystal lattice structures to break apart and shed a massive number of crystals into the joint space. These newly released and nonopsonized crystals activate receptors on synovial macrophages and are then phagocytized by monocytes and macrophages, leading to interaction with the NLRP3 inflammasome. This results in the rapid production of interleukin-1 (IL-1), which is responsible for all the cardinal features of the severe inflammation associated with acute gout.

Although monocytes and macrophages are the major cellular sources of IL-1, neutrophils predominate at the site of inflammation (e.g., joint), where monosodium urate crystals become extremely pro-inflammatory by activating neutrophils that then generate reactive oxygen species. However, neutrophils also have a major role in the resolution of acute gout through the formation of neutrophil extracellular traps (NETs). Neutrophil proteases are released into the NETs and cause the formation of cell aggregates that contain cellular debris and DNA. These cellular aggregates within NETs can rapidly degrade pro-inflammatory cytokines, thereby allowing for the spontaneous resolution of joint inflammation after 3 days (even in the absence of treatment). Therefore, neutrophils play a dual role in gouty inflammation, amplifying it initially but then mediating its resolution.

5.2.1.1.3 Clinical Manifestation

Classic gout passes through three stages: asymptomatic hyperuricemia, acute intermittent gout symptoms, and chronic persistent gout symptoms. Asymptomatic hyperuricemia refers to a state in which serum urate exceeds the level of solubility (6.8 mg/dL) but symptoms of crystalline deposition have not yet occurred. Only 15 to 20% of all hyperuricemic people are prone to develop monosodium urate crystals, and for this group, the period of asymptomatic hyperuricemia begins a stage of subclinical structural changes. In men, asymptomatic hyperuricemia frequently begins at puberty, whereas in women, it is usually delayed until menopause.

the first flare usually occurs between the fourth and sixth decades of life. In women, the age of onset is older and varies with several factors, most importantly the age of menopause. The classic gout flare is hallmarked by the rapid development of warmth, swelling, erythema, and exquisite pain in one or occasionally two joints. The characteristically severe pain evolves from its faintest twinge to its most intense level over an 8- to 12-hour period. Initial episodes are usually monoarticular and involve lower extremity joints. The most common joint involved is the first metatarsal phalangeal joint (termed podagra), followed by the ankle, midfoot, and knee. After years of recurrent gouty flares, upper extremity joints, including the wrists, elbows, and small joints of the hands, can also become involved. Systemic symptoms of fever, chills,

and malaise may accompany gouty flares, along with an intense erythema extending beyond the area of the involved joint. This may lead to confusion with an infectious process. Factors capable of provoking episodes of flare are those that cause fluctuations in serum urate levels, including trauma, surgery, starvation, overindulgence in certain high-purine foods, and the ingestion of any medication that raises or lowers serum urate.

Another characteristic of classic gout flares is their self-remitting nature for reasons discussed previously. For the first several acute flares, the duration of the attack is 5 to 8 days. The resolution of symptoms is gradual but complete, even if no anti-inflammatory therapy is administered. The periods between the flares are devoid of articular pain, although synovial fluid aspirates during this stage continue to show low-grade inflammation and the presence of monosodium urate crystals.

the untreated patient will progress to persistent gouty arthritis, also referred to as advanced gout. This stage usually develops after 10 or more years of acute intermittent gout and is evident when the pain-free intercritical periods have disappeared. The subcutaneous tophus is the most characteristic lesion of advanced gout. The development of tophaceous deposits of monosodium urate is a function of the duration and severity of hyperuricemia as well as the presence of proteinuria. Subcutaneous tophi may occur anywhere over the body but most commonly in the fingers, wrists, ears, knees, and olecranon bursa and at pressure points under the ulnar aspect of the forearm and Achilles tendon.

Approximately 5% of patients with gout exhibit the onset of symptoms before age 25 years. Early-onset gout represents a special subset of patients who generally have a genetic component (see Table 257-2), with a more accelerated clinical course requiring more aggressive antihyperuricemic therapy.

5.2.1.1.4 Diagnosis

Hyperuricemia is an essential risk factor for developing gout, but it is not a reliable diagnostic test because many people with elevated serum urate levels will never develop gout. Serum urate levels during an acute flare of gout are also unreliable because they may be suppressed by as much as 1.5 to 2.0 mg/dL from baseline values. The definitive diagnosis of gout is made by polarized compensated microscopy of a synovial fluid aspirate from the affected joint. The presence of intracellular needle-shaped crystals with strong negative birefringence is the diagnostic “gold standard.” Similar microscopic results from a tophus aspirate or spontaneously draining fluid would also confirm the diagnosis of gout.

The presumptive diagnosis of gout is based on a pattern of acute joint symptoms coupled with the patient’s own medical history or a family history. Key features include joint erythema, difficulty walking, time to maximal pain less than 24 hours, resolution by 2 weeks, presence of a tophus, and history of involvement of the first metatarsophalangeal joint.⁷ The patient’s medical history may reveal comorbidities frequently associated with gout or the use of medications associated with urate retention. The characteristic clinical presentation is the rapid onset (over

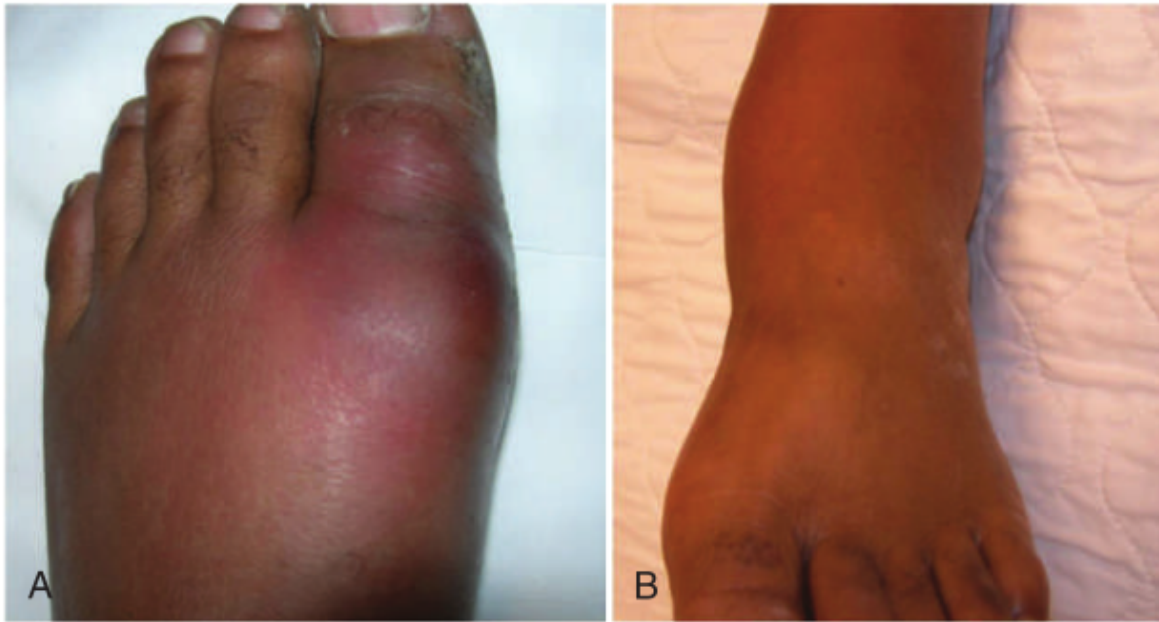


Figure 5.1: A, The marked swelling of the first metatarsophalangeal joint (podagra) is demonstrated. A dusky blue hue over an intense erythema is characteristic. B, Ankle swelling is shown with erythema extending beyond the area of the tibiotalar joint.

8 to 12 hours) of severe pain in one or several lower extremity joints (especially the great toe, midfoot, and ankle).

Radiographic evaluation is not helpful in the early phase of gout except to rule out fracture. In advanced gout, affected joints may show **punched-out periarticular erosions with classic overhanging edges**. Ultrasonography can detect monosodium urate crystals layered over articular cartilage in early disease and may be diagnostic

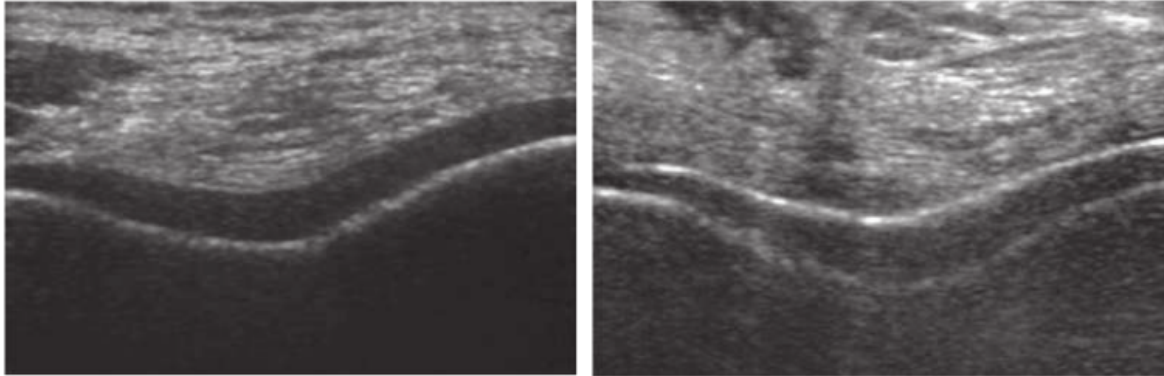


Figure 5.2: Comparison of ultrasonographic appearance of normal (left) and gouty (right) knee with the probe in the suprapatellar, transverse orientation and the knee in flexion. The normal knee shows soft tissues overlying an anechoic (black) layer that represents hyaline cartilage. The single bright stripe represents the bony cortex of the anterior distal femur. The gouty knee demonstrated the “double-contour” sign composed of a layer crystals covering the hyaline cartilage and the lower bright contour being the femoral cortex.

The differential diagnosis of acute gout includes bacterial infection, trauma, sarcoidosis, and calcium pyrophosphate arthropathy (pseudogout). Diseases occasionally confused with advanced gout include rheumatoid arthritis, reactive arthritis, and calcium pyrophosphate arthropathy.

5.2.1.1.5 Treatment

5.2.1.1.5.1 Education

Patients be informed about dietary and other lifestyle changes, such as weight loss, that will lower their serum uric acid and lessen flares. Diet education include *Avoid* Organ meats High fructose corn syrup—sweetened drinks, and Alcohol overuse. *Limit* Beef, pork, lamb, shellfish, Beer. *Encourage* Low-fat dairy, smoking cessation and well hydrated.

5.2.1.1.5.2 Gout Flare

The treatment goal for gouty flare is to relieve pain and terminate the flare as quickly as possible. The therapeutic options include nonsteroidal anti-inflammatory drugs (NSAIDs), oral colchicine, and corticosteroids. Ensuring adequate dosing of anti-inflammatory therapy, and continuing the treatment until the flare has completely resolved (usually 6 to 10 days). During the acute flare, subjects already taking urate lowering therapy should continue the drug, whereas those not receiving this therapy should not be started.

- NSAID: may be inappropriate for patients with renal insufficiency or peptic ulcer disease
- Oral colchicine administered as 1.2 mg (two tablets) at the time of flare onset followed in 1 hour by a third 0.6-mg tablet is the recommended dosing for the first 24 hours. This is followed by 7 to 10 days of once-daily or twice-daily colchicine depending on renal function.
- Corticosteroid administered if poor renal function or intolerance to colchicine, Safer and similar efficacy than NSAIDs.

5.2.1.1.5.3 Maintenance of Urate Lowering Therapy

The principal goal of treating gout is to lower the serum uric acid below its saturation point so that the process of crystallization will cease and the accumulated urate burden will be gradually diminished. Urate-lowering therapy is recommended for all patients with two or more gouty flares per year, patients with advanced disease, and those with kidney stones.

Both allopurinol and febuxostat are xanthine oxidase inhibitors that are considered first-line urate-lowering therapy.

5.2.1.1.5.4 Treat to Target Approach

The American College of Rheumatology recommends a “treat-to-target” approach, using urate-lowering medications to lower serum uric acid levels to less than 6 mg/dL in patients with a history of two or more clinical flares, with close monitoring of uric acid levels. The 2012 American College of Rheumatology guidelines recommend a target serum urate of less than 6 mg/dL in all subjects, with an even lower target (<5 mg/dL) for patients with more advanced gout. The American College of Rheumatology guidelines recommend that the allopurinol starting dose be no greater than 100 mg/day. The dose is gradually escalated by 100 mg daily every 2 to 5 weeks, with serum urate monitoring until the target serum urate is achieved. The maximal U.S. Food and Drug Administration (FDA) approved dose of allopurinol is 800 mg daily.

In subjects with advanced chronic kidney disease, the initial dose should be reduced to 50 mg daily, with incremental dose escalations of 50 mg. Febuxostat is an alternative urate-lowering therapy and should be used in patients who have failed allopurinol treatment or have

demonstrated sensitivity or intolerance to allopurinol. The initial febuxostat dose of 40 mg daily can be increased to 80 mg daily after 2 weeks of therapy if the serum urate target is not achieved.

Patient who have not attained the target serum urate despite maximal doses of either allopurinol or febuxostat, uricosuric agents (probenecid, up to 2 g daily). Pegloticase (8 mg by IV infusion every 2 weeks) is an intravenously administered monomethoxypoly(ethylene glycol)-conjugated recombinant uricase that dramatically lowers serum urate levels. It is approved by the FDA for the treatment of gout in patients for whom conventional therapy has been ineffective.

Before the initiation of any form of urate-lowering therapy, the patient should be placed on anti-inflammatory therapy to prevent or minimize the anticipated increase in flare activity that is associated with starting urate-lowering therapy. Background anti-inflammatory prophylaxis is in the form of colchicine once or twice daily or low-dose NSAIDs. Anti-inflammatory prophylaxis should be continued until the subject has been free of gout flares for 6 months or longer.

5.2.1.1.5.5 Treat to symptoms Approach

the 2016 American College of Physicians guideline advocates a “treat-to-symptoms” approach, emphasizing the use of anti-inflammatory medications to control flares and reserving urate-lowering medications for patients with frequent flares. This later approach is not evidence-based.

5.2.1.2 Calcium Pyrophosphate

5.2.1.3 Basic Calcium Arthropathy

5.2.2 Arthrocentesis Technique

Read from Robert hedge Procedures

6 Infectious Disease

6.1 Bacterial Infection

6.2 Viral Infection

6.2.1 HIV

6.2.1.1 Immunopathogenesis HIV Infection

6.2.1.2

6.2.2 Dengue

6.3 Fungal Infection

7 Cardiology & Vascular Medicine

8 Respiratory Medicine

9 Nefrology & urology

9.1 Nefrology

9.1.1 Hypertension

9.1.2 FLuid and Electrolyte Disorders

9.1.2.1 Disorders of Extracelular Volume

9.1.2.2 Disorders of Water Metabolism

9.1.2.3 Disorders of Potassium Metabolism

The most frequent electrolyte abnormalities in clinical medicine. Patients with disorders of potassium metabolism may be asymptomatic, or they may have symptoms ranging from mild weakness to sudden death.

9.1.2.3.1 Normal Physiology

9.1.2.3.1.1 Potassium Intake

Potassium is necessary for essentially all cellular functions, is present in most foods, and is excreted primarily by the kidney. Typical diet 70 mmol of potassium, while the recommended intake is 120 mmol per day. Gastrointestinal tract efficiently absorbs potassium.

9.1.2.3.1.2 Potassium Distribution

After absorption from the GI tract, potassium distributes rapidly into the extracellular fluid (ECF) and intracellular fluid (ICF) compartments. Cellular potassium uptake is rapid and limits the magnitude of changes in serum potassium concentration. During potassium deficiency, shift of potassium from intracellular to extracellular compartments limits the change in extracellular potassium concentration.

Most potassium is in the ICF. Most body potassium is intracellular, with only 1% to 2% in the ECF. Potassium is the major intracellular cation, with cytosolic K⁺ concentrations about 100

to 120 mmol/L. Total intracellular K⁺ content is 3000 to 3500 mmol in healthy adults and is found primarily in muscle (70%), with a lesser amount in bone, red blood cells, liver, and skin.

9.1.2.3.1.3 Renal Handling of Potassium in Normal Kidney

9.1.2.3.1.4 Renal Potassium Handling in Chronic Kidney Disease

9.1.2.3.2 Hypokalemia

9.1.2.3.3 Hyperkalemia

9.1.2.4 Disorders of Calcium, Phosphate and Magnesium Metabolism

9.1.2.5 Acid-Base Balance

9.1.2.5.1 Metabolic Acidosis

9.1.2.5.2 Metabolic Alkalosis

9.1.2.5.3 Respiratory Acidosis

9.1.2.5.4 Respiratory Alkalosis

9.1.2.5.5 Mixed Acid Base Balance

9.1.3 Infectious Disease in Kidney

9.1.4 Acute Kidney Injury

9.1.5 GLomerular Disease

9.1.6 Tubulointerstitial Disease

9.1.7 Diabetic Kidney Disease

9.1.8 Renovascular Disease

9.1.9 Hepatorenal Syndrome

9.1.10 Cardiorenal Syndrome

9.1.11 Pregnancy and Renal Disease

9.1.12 Congenital Disease of the Kidney

9.1.13 Chronic Kidney Disease

9.1.14 Dialytic Therapy

9.1.15 Transplantation

9.1.16 Edema and Diuretics use

9.1.16.0.1 Diuretics Type

Diuretics are based on the sites of action. These are as follows:

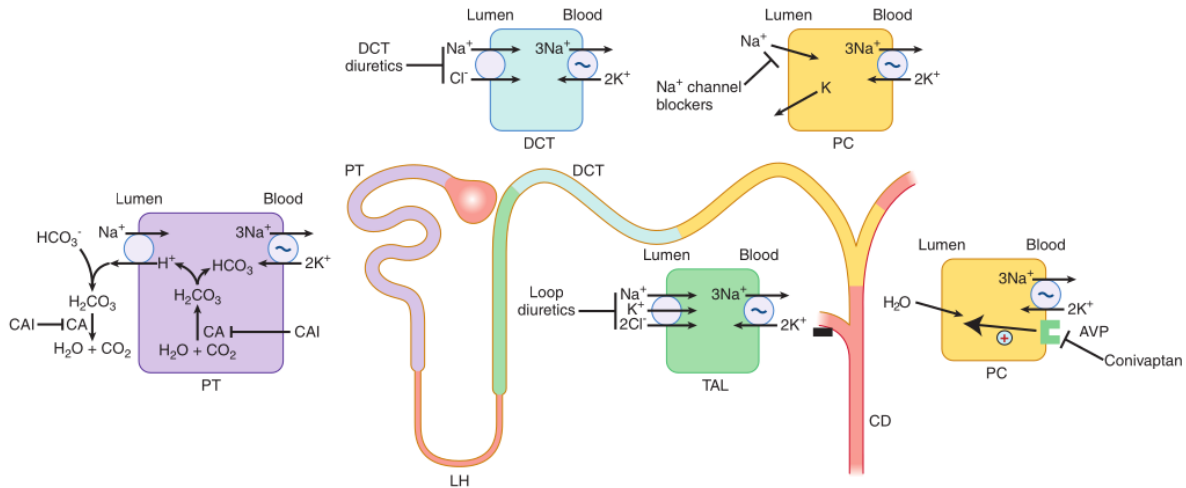


Figure 9.1: The proximal tubule (PT, purple segment) is represented by a typical PT cell. The loop of Henle (LH) includes a thick ascending limb (TAL, green segment), and a typical TAL cell is shown in green. The distal convoluted tubule (DCT, blue segment) is represented by a typical DCT cell in blue. The collecting duct (CD, yellow and orange segments) includes principal cells (PC), shown in yellow.

9.1.16.0.2 Proximal Tubular Diuretics

Carbonic Anhydrase inhibitor (CAI) work by inhibiting generation of intracellular Hydrogen ion, which is a requirement for absorption of sodium. This is the main action for its diuretics property. Most of the work in proximal tubule where bulk of sodium absorption, although Na absorption in more distal tubule will reduce CAI effects. Acetazolamide is the only diuretics used in practice. Acetazolamide has short action, due to readily absorbed and easily eliminated by tubular action. Prolonged use will cause metabolic acidosis, thus **acetazolamide 250 to 500 mg** daily used to correct metabolic alkalosis as side effects of thiazide or loop diuretics. Its use should be cautiously in advanced kidney disease as it will systemically accumulate with repeat dosing.

9.1.16.0.3 Distal Convoluted Tubule Diuretics

The major site of action of thiazide diuretics is the early DCT where they inhibit the coupled reabsorption of Na^+ and chloride (Cl^-). Thiazides also :

1. inhibit NaCl /fluid reabsorption in the medullary-collecting duct.
2. The water-soluble thiazides such as hydrochlorothiazide (HCTZ) also inhibit CA when given at high doses, and therein further increase Na^+ excretion.
3. thiazide diuretics impair urinary diluting capacity without affecting urinary concentrating mechanisms,

4. reduce calcium (Ca^{++}) and urate excretion,
5. increase magnesium (Mg^{++}) excretion
6. thiazide diuretic to minimize the development of loop diuretic-related DCT cell hypertrophy that might otherwise increase Na^{+} absorption

The onset of diuresis with hydrochlorothiazid (HCTZ) occurs within 2 hours, peaks between 3 and 6 hours, and dose-dependently continues for upward to 12 hours. The half-life ($t_{1/2}$) of HCTZ is prolonged both in decompensated HF and kidney failure. Doses of thiazide diuretics in the **100 to 200 mg/day** range are required to initiate a diuresis in patients with chronic **kidney disease (CKD)**, although the overall natriuretic response is arbitrated by the GFR/filtered Na^{+} load.

A newer thiazid diuretic is chlorthalidone. chlorthalidone and HCTZ are structurally similar, they differ pharmacokinetically in that chlorthalidone has a longer $t_{1/2}$ of 40 to 60 hours versus 3.2 to 13.1 hours for HCTZ, as well as a larger volume of distribution by virtue of its extensive partitioning into red blood cells. This latter feature creates a depot that promotes continuous streaming of chlorthalidone (red cell $\rightarrow$ plasma $\rightarrow$ tubular secretion). This plasma half-life difference for chlorthalidone correlates with a more extended diuretic effect and is a likely explanation for the observation that chlorthalidone is a better mg-for-mg antihypertensive than HCTZ.

9.1.16.0.4 Loop Diuretics

Loop diuretics act predominantly at the luminal membrane in the thick ascending limb (TAL) of the loop of Henle where they compete with Cl^{-} for binding to the $\text{Na}^{+}/\text{K}^{+}/2\text{Cl}^{-}$ cotransporter, thereby inhibiting Na^{+} and Cl^{-} reabsorption. Loop diuretics also:

1. a decrease in both free water (H_2O) excretion and absorption during H_2O loading and dehydration, respectively,
2. a 30% increase in fractional Ca^{++} excretion, a significant increase in Mg^{++} excretion,
3. brief increase followed by a more long-lived decrease in uric acid excretion.
4. Stimulate renal prostaglandin synthesis, particularly that of the vasodilatory prostaglandin E_2 (PGE_2).

Angiotensin-II, generated following the administration of intravenous loop diuretics, coupled with an increased synthesis of PGE_2 , are the likely reasons for the shift in renal blood flow (RBF) from the inner to the outer renal cortex with these drugs; however, both total RBF and GFR are maintained when loop diuretics are administered to normal subjects. Loop diuretics are highly protein bound, therefore to gain access to their tubular site of action they must be secreted as is the case for thiazide diuretics. Tubular secretion of loop diuretics may be interfered by elevated levels of endogenous organic acids, as arise in CKD, and by drugs that share the same transporter site, such as salicylates and nonsteroidal antiinflammatory drugs

(NSAIDs). Uremic toxins and fatty acids can decrease loop diuretic protein binding and further alter diuretic pharmacokinetics.

Furosemide is the most widely used diuretic in this class; however, its use is complicated by variable absorption with a bioavailability ranging from 12% to 112%. The coefficient of variation for absorption varies from 25% to 43% for different furosemide products, and exchanging one furosemide formulation for another will not standardize patient absorption and/or response to oral furosemide. Loop diuretics are commonly used in the patient with CKD, with the renal clearance of these drugs reduced in parallel with the degree of reduction in kidney function. In general, furosemide pharmacokinetics are more significantly changed in CKD than the other loop diuretics, with both its renal metabolism and intact clearance reduced in CKD.

9.1.16.0.5 Distal Potassium-Sparing Diuretics

There are two classes of K⁺-sparing diuretics: competitive antagonists of aldosterone, such as spironolactone or eplerenone, and compounds that work independent of aldosterone, such as amiloride and triamterene. Drugs in this class reduce active Na⁺ absorption in the late DCT/collecting duct (CD). Basolateral Na⁺/K⁺-ATPase activity reduced, intracellular K⁺ concentration decreases, and the electrochemical gradient for K⁺ is lowered, thereby reducing K⁺ secretion. K⁺-sparing diuretics also reduce Ca⁺⁺ and Mg⁺⁺ excretion, which is a useful feature in the HF patient. Because K⁺-sparing diuretics are only modestly natriuretic, their clinical utility resides more in their K⁺-sparing capacity, particularly when more proximally acting diuretics increase distal Na⁺ delivery or in the instance of either primary or secondary aldosteronism.

Spironolactone is a well-absorbed, highly protein-bound, lipid-soluble K⁺-sparing diuretic with a 20-hour half-life. The onset of action for spironolactone is characteristically slow, with a peak response at times 48-hours or more after the initial dose. Spironolactone, unlike amiloride and triamterene, remains active (as a diuretic and antihypertensive agent) in advanced kidney failure in that its locus of action is basolateral; thus it does not require glomerular filtration or tubular secretion to gain access to its effect site. Eplerenone is a mineralocorticoid receptor antagonist that is highly selective for the aldosterone receptor; accordingly, its much lower affinity for androgen and progesterone receptors results in considerably less gynecomastia than is the case with spironolactone.

9.1.16.0.6 Osmotic Diuretics

Mannitol is a polysaccharide diuretic given intravenously that is freely eliminated by glomerular filtration. Mannitol is poorly reabsorbed along the length of the nephron and thereby exerts a dose-dependent osmotic effect. This osmotic effect traps water and solutes in the tubular fluid, thus increasing Na⁺, K⁺, Cl[−], and HCO₃[−] excretion. The plasma half life mannitol depends on the level of kidney function but usually is between 30 and 60 minutes; thus its diuretic properties are quite transient. Mannitol has been used to reduce the incidence of acute

kidney injury (AKI) in patients undergoing cardiopulmonary bypass, having rhabdomyolysis, or following exposure to contrast media. mannitol also expands extracellular fluid (ECF) volume and can precipitate pulmonary edema in patients with HF, it should be used cautiously, if at all, in these patients. excessive mannitol administration, particularly in the setting of a reduced GFR, can cause dilutional hyponatremia, hyperkalemia, and/or AKI. The latter is dose dependent, relates to afferent arteriolar vasoconstriction, and commonly corrects with the elimination of excess mannitol as may be accomplished with hemodialysis.

9.1.16.1 Edema Treatment

The pathophysiology of Na⁺ and H₂O retention in the patient with edema is typically characterized by a complex interchange of hemodynamic and neurohumoral factors. For example, systemically perceived arterial underfilling sets into motion related Na⁺ and H₂O retention in the patient with HF. In other instances, such as in the patient with a reduced GFR and/or nephrotic syndrome, Na⁺ and H₂O retention derives from a more primary set of renal processes. There are two important considerations that should be brought to bear before and/or concurrent with initiation of diuretic therapy: the need to restrict dietary Na⁺ intake (2 to 4 g Na⁺ day) and dose reduction/elimination of drugs that foster Na⁺ retention, such as NSAIDs, nonspecific vasodilators like hydralazine and minoxidil, and high-dose α -blockers or central α -agonists.

A compound, such as chlorthalidone, can be quite effective in the setting of mild to moderate edema given once or twice daily in the 25- to 50-mg/day range. When the underlying disease state worsens and/or dietary Na⁺ restriction cannot be adequately maintained, then conversion to a loop diuretic-based regimen becomes the more practiced approach. Combination diuretic therapy can be considered thereafter, either because the severity of the edema requires “sequential nephron blockade” or because the underlying disease state is particularly sensitive to medications other than loop diuretics, as is the case for spironolactone (50 → 400-mg/day range) use in the patient with advanced cases of cirrhosis and ascites.

Determining the minimally effective dose for a diuretic effect is a necessary clinical exercise, particularly in the patient with a reduced GFR. Gradually increasing a diuretic dose until a response is seen will establish the threshold/ effect dose, and thereafter the frequency of dosing can be established based on clinical circumstances. The beginning dose from which dose titration proceeds is influenced by both the level of kidney function and the severity of the edematous state. If kidney function is reduced, the diuretic dose-response curve shifts to the right and the maximal effectiveness, based on absolute Na⁺ excretion rate, can be significantly reduced, therein making dietary Na⁺ restriction of considerable importance. In CKD patients with edema furosemid 40 mg twice daily would be considered adequate starting doses that could then be gradually incremented until the desired effect is achieved. Often, the dose that elicits an increase in urine output can be continued indefinitely unless the underlying disease state worsens and/or dietary Na⁺ intake becomes inordinately high. Conversely, a diuretic dose that establishes “euvolemia” can occasionally be lowered and restriction of dietary Na⁺ intake given added opportunity to minimize edema redevelopment. Diuretic dose reduction

should always be a sought after consideration, as it minimizes loss of K^+ , Mg^{++} , and, in the case of loop diuretics, Ca^{++} .

9.1.16.1.1 Adaptation to Diuretic Therapy

Diuretic-induced inhibition of Na^+ reabsorption in one nephron segment elicits important adaptations in other nephron segments, which not only limits their antihypertensive and fluid-depleting actions but also contributes to the development of side effects. The initial dose of a diuretic normally produces a brisk diuresis, which is quickly followed by a new equilibrium state in which daily fluid and electrolyte excretion either matches or is less than intake with body weight stabilizing. In nonedematous patients given either a thiazide or a loop diuretic, this adaptation, or **braking phenomenon**, occurs within 1 to 2 days and limits net weight loss to 1 to 2 kg.

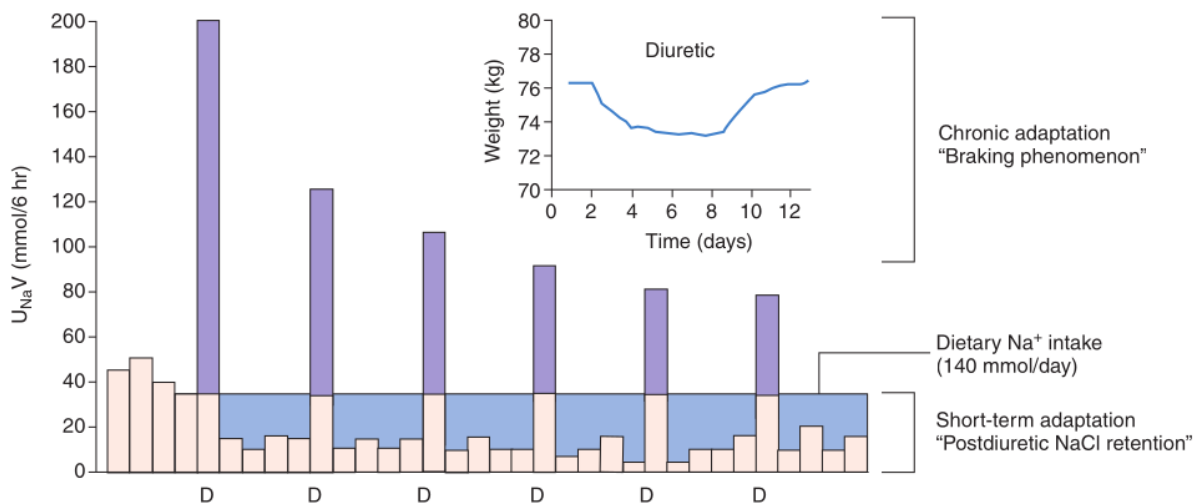


Figure 9.2: Effect of a loop diuretic on urinary sodium (Na^+) excretion. Each bar represents a 6-hour time interval. Purple bars indicate periods during which urinary Na^+ excretion ($U_{Na}V$) exceeds that of dietary intake. Blue areas indicate periods of postdiuretic Na^+ retention, during which dietary Na^+ intake exceeds urinary Na^+ excretion. The horizontal black line indicates dietary Na^+ intake per 24-hour period. Changes in the magnitude of the natriuretic response over several days are reflective of the “braking phenomenon.” Inset shows the effect of diuretics on weight (and extracellular fluid volume) during several days of diuretic administration.

For example, furosemide administered orally to subjects ingesting a high- Na^+ diet (270 mmol/24 hours) produced an initial brisk natriuresis, which resulted in a negative Na^+ balance over the ensuing 6 hours. This was followed by an 18-hour period when Na^+ excretion was reduced to levels well below the prescribed Na^+ intake, resulting in a positive Na^+ balance. This postdiuresis Na^+ retention matched the initial natriuresis with the result at the end of the day

being a neutral Na⁺ balance state and no weight loss. After 3 successive days of furosemide administration, a similar pattern of Na⁺ loss and retention was demonstrated each day. if Na⁺ intake is kept very low, balance can remain negative after a single dose of furosemide, even though there is some blunting of the initial natriuretic response.

The mechanistic basis for the braking phenomenon is complex. The relationship between natriuresis and the rate of furosemide excretion is shifted to the right in subjects receiving a low-salt diet, which denotes a blunting of tubular response. It appears to be unrelated to aldosterone, as spironolactone therapy has little effect on the Na⁺ retention. Structural hypertrophy in the distal nephron also occurs in rats receiving prolonged infusions of loop diuretics. These structural changes are marked by increased distal nephron Na⁺ and Cl⁻ absorption and K⁺ secretion, phenomena which are aldosterone independent. These structural adaptations may contribute to postdiuretic Na⁺ retention and to diuretic tolerance in humans, and could explain the **Na⁺ retention persisting for up to 2 weeks after loop diuretic therapy is discontinued.**

Plasma renin activity (PRA) and plasma aldosterone concentrations rise within minutes of receiving an intravenous diuretic, a transient process that is independent of volume loss and/or sympathetic nervous system (SNS) activation. This rise in PRA is caused by inhibition of NaCl reabsorption at the macula densa in conjunction with loop-diuretic stimulation of renal prostaglandin release. Shortly after this initial rise in PRA, diuretics cause a more sustained increase in PRA and aldosterone arising from an increase in SNS activity (α-agonism) and a fall in ECF volume. The increase in renal prostaglandin production is the likely explanation for the preload reduction and decrease in ventricular filling pressures that occur within 15 minutes of loop diuretic administration.

9.2 Urology

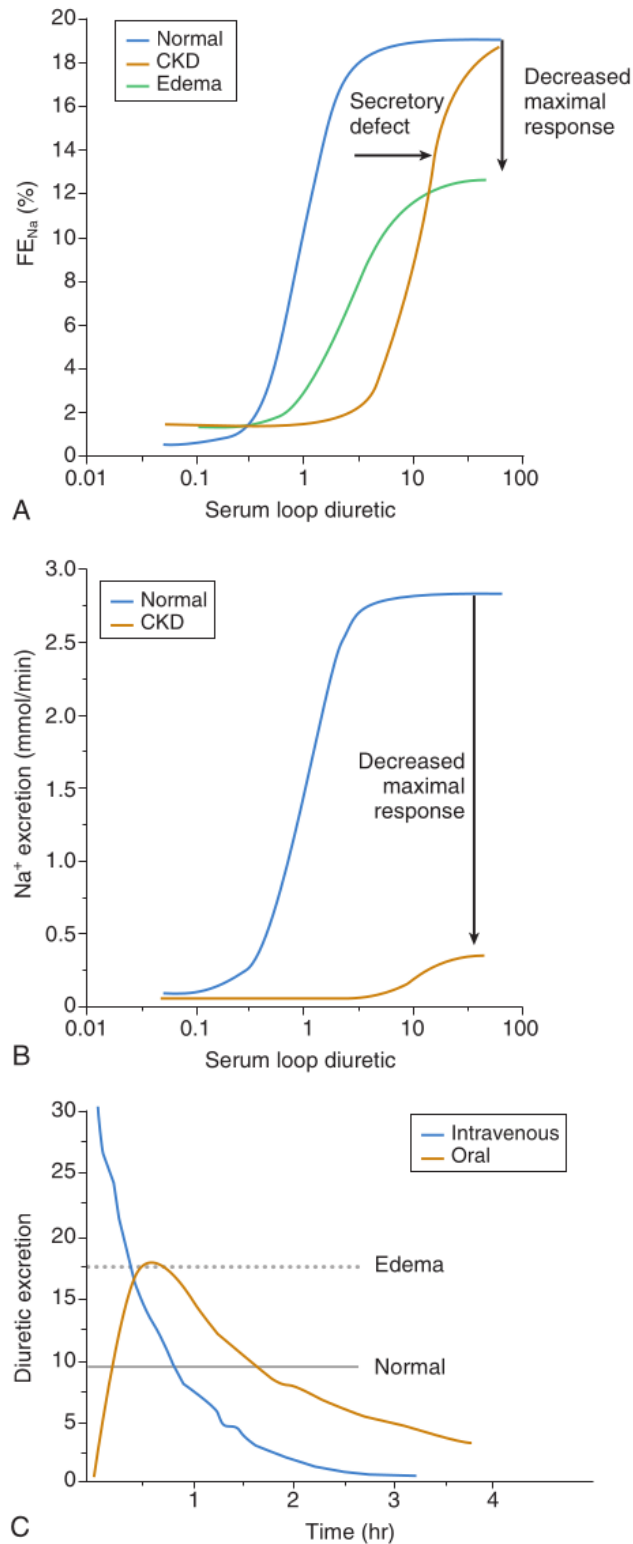


Figure 9.3: (A) Comparison of effects of chronic kidney disease (CKD) and conditions marked by edema on the loop diuretic dose response, expressed as fractional Na^+ excretion (FE_{Na}). Diuretic delivery via secretion into the lumen is impaired in CKD (pharmacokinetic abnormality), whereas the response to delivered drug is diminished with edema (pharmacodynamic defect). (B) Effect of CKD on the absolute response to a loop diuretic. Compare with panel A. (C) Pharmacokinetics of intravenous and oral loop diuretics. The diuretic thresholds for normal and edematous individuals are shown as horizontal lines; whereas a normal individual responds appropriately to either an intravenous or oral diuretic, some edematous individuals can only reach threshold excretion with intravenous diuretic administration.

10 Gastroenterology & Hepatology

10.1 Gastroenterology

10.1.1 Anorectal Disease

composed of blood vessels, smooth muscle fibers, and supporting connective tissue, normally present in the anal canal. The term “hemorrhoids” is also applied to situations when these cushions become symptomatic.

Hemorrhoids are the result of slippage of the anal cushions due to weakening of the supporting tissues. The prolapse hinders the venous flow leading to dilatation and engorgement of the anal cushions, exposing them to complications such as edema, thrombosis, and bleeding.

Hemorrhoids are classified as internal, external, or combined. Internal hemorrhoids are located proximal to the dentate line and are covered by insensate columnar mucosa or transitional epithelium. External hemorrhoids are located below the dentate line, covered with squamous epithelium rich in nerve endings.

10.1.1.1 Clinical Presentation

The most common clinical presentation are;

1. Bleeding per rectum.
2. Prolapsing tissue.

Bleeding per rectum is The bleeding is bright red and painless, and it usually follows evacuation.

Prolapse occurs during straining or heavy lifting causing discomfort, feeling of incomplete evacuation, mucus discharge, and pruritus. Prolapsing hemorrhoids can be complicated by incarceration and strangulation secondary to thrombosis, causing severe pain and requiring urgent surgery.

Internal hemorrhoids are classified according to the degree of prolapse:

1. Grade I – bleeding without prolapse;
2. Grade II – prolapse on straining that spontaneously reduces;

3. Grade III – prolapse that requires manual reduction; and
4. Grade IV – irreducible prolapse

External hemorrhoids are usually asymptomatic unless thrombosis develops, which is frequently due to trauma and straining. They present as a painful blue lump at the anal verge.

Patients older than 40 years or with atypical symptoms should undergo complete colonic investigation to exclude cancer. Inflammatory bowel disease is marked by bloody diarrhea

10.1.1.2 Treatment

Medical treatment aims to promote soft, formed, regular bowel movements: 1. with a high-fiber diet (20–30 g/day). bulking agents (psyllium husk: e.g., Metamucil™, 1 packet 1–3 times/day)

2. generous water intake (six 8-ounce glasses of water/day).
3. Warm sitz baths are helpful to relax the anal sphincter.

10.2 Hepatology

11 Endocrine & Metabolism

11.1 Emergency

11.1.1 Diabetes

11.1.1.1 Diabetic ketoacidosis

11.1.1.2 Hyperosmolar-hyperglycemic Syndrome

11.2 Non-emergency

12 Psychosomatic Medicine

13 Geriatrics

13.1 Emergency

13.2 Non-emergency

Part II

The Rest of Medicine

14 Clinical Nutrition

15 Surgery

16 neurology

16.1 Coma & Decreased Consciousness

16.2 Pain Management

17 Perioperative Consultation & Anesthesia

17.1 Perioperative Consultation

17.1.1 Common Risk Score

17.1.1.1 Assess Respiratory Risk in Surgical Patients in Catalonia (ARISCAT) Score

It used to predicts risk of pulmonary complications after surgery, including respiratory failure. Postoperative pulmonary complications (PPCs) include complications affecting the respiratory system after anesthesia and surgery. These complications are common, difficult to predict, and have significant effects on patient. The ARISCAT risk index is mostly used in operations other than thoracic surgery.

17.1.1.1.1 Applicable to:

Patients undergoing surgery under general, neuraxial, or regional anesthesia.

17.1.1.1.2 Scoring

17.1.1.1.3 Advice

Interpretation:

ARISCAT Score

17.1.1.2 Revised Lee Cardiac Risk Index for Pre-Operative Risk (RCRI)

It used to estimates risk of cardiac complications after non cardiac surgery. The RCRI score identifies a risk class based on the presence or absence of six factors (predictors) associated with preoperative cardiac complications.

17.1.1.2.1 Applicable to:

- Patients 45 years old (or 18-44 years old with significant cardiovascular disease)
- Undergoing elective non-cardiac surgery or urgent/semi-urgent (non-emergent) non-cardiac surgery.
- Use with caution in emergency surgery patients, as the score is not as well validated in this population.

17.1.1.2.2 Scoring

Baseline risk: 3.9% of 30-day risk of death, MI, or cardiac arrest Scoring Points Includes:

- **Elevated-risk surgery** [YES / NO] Intraperitoneal; intrathoracic; suprainguinal vascular
- **History of ischemic heart disease** [YES / NO] History of myocardial infarction (MI); history of positive exercise test; current chest pain considered due to myocardial ischemia; use of nitrate therapy or ECG with pathological Q waves
- **History of congestive heart failure** [YES / NO] Pulmonary edema, bilateral rales or S3 gallop; paroxysmal nocturnal dyspnea; chest x-ray (CXR) showing pulmonary vascular redistribution
- **History of cerebrovascular disease** [YES / NO] Prior transient ischemic attack (TIA) or stroke
- **Pre-operative treatment with insulin** [YES / NO] Pre-operative treatment with insulin
- **pre-operative creatinine** [YES / NO] Pre-operative creatinine >2 mg/dL / 176.8 µmol/L

Subsequently, it assigns a class (a risk index) from I-IV, listed below.

- Class I [0 predictors] correlates with a 3.9% 30-day risk of death, myocardial ischemia (MI), or cardiac arrest (CA).
- Class II [1 predictors] correlates with a 6% 30-day risk of death, MI, or CA.
- Class III [2 predictors] correlates with a 10.1% 30-day risk of death, MI, or CA.
- Class IV [greater than or equal to 3 predictors] correlates with a more than 15% 30-day risk of death, MI, or CA.

17.1.1.2.3 Advice

- If the RCRI is 1, the patient's age is 65, or they are between 45-64 with significant cardiac disease*, the next step is to measure the patient's NT-ProBNP or BNP if this is available at your institution.
- If the NT-ProBNP is 300 ng/L or BNP is 92 ng/L, then there should be an EKG ordered in the PACU and troponins should be measured daily for 48-72 hours.

- If, after risk stratification, the NT-ProBNP is <300 ng/L or BNP <92 ng/L, no routine postoperative cardiac monitoring is warranted.
- If the institution does not have these assays available, then all patients should be monitored with an EKG in the PACU and troponin measurements daily for 48-72 hours if they meet one of the following: RCRI 1, age 65, or age 45-64 with the aforementioned cardiac disease including:
 - known history of coronary artery disease,
 - cerebral vascular disease,
 - peripheral artery disease,
 - congestive heart failure,
 - severe PHTN or a severe obstructive intracardiac abnormality (e.g. severe aortic stenosis,
 - severe mitral stenosis, or severe hypertrophic obstructive cardiomyopathy).

17.1.1.3 CAPRINI VTE Score

17.1.1.4 IMPROVE Bleeding Score

17.2 Anesthesia

17.2.1 Standard Anesthesia (just to know)

Patients were monitored in the operating room under the ASA standards and 0.03 mg/kg midazolam was administered intravenously to the patients for premedication. Following preoxygenation; anesthesia was induced with 2–2.5 mg/kg propofol, 1–1.5 mcg/kg fentanyl, and 0.1 mg/kg vecuronium intravenously. After the intubation with a left-sided double-lumen endobronchial tube, anesthesia was maintained by administering 2–3% sevoflurane in oxygen and air mixture. Intraoperative lung-protective ventilation (tidal volume 5–7 mL/kg, positive end-expiratory pressure 5–7 cmH₂O, and ventilatory plateau pressures below 30 cmH₂O whenever possible) was administered in all patients.

17.2.2 Standard Postoperative Analgesia Protocol

Before the end of the surgical procedure; metoclopramide, to prevent PONV; dexketoprofen (50 mg), and tramadol (100 mg) were given to the patients intravenously. Intravenous morphine was administered via patient-controlled analgesia (PCA) pump for 24 hours in the postoperative surgical intensive care unit. The PCA pump's dose delivery was limited to administering a bolus dose of 1 mg morphine and delivering a maximum dose of 12 mg morphine in total within 4 hours with lockout intervals of 15 minutes. Paracetamol 1 g every 8 hours and dexketoprofen

50 mg twice daily were administered intravenously for multimodal analgesia. As a rescue analgesic agent, 0.5 mg/kg tramadol was given to patients intravenously when a score of VAS at coughing 4.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9333546/>

18 Obstetric & Gynecology

19 Dermato-venereology

20 Poison, Envenomation and Drug Overdose

21 Disease due to Enviromental Exposure

22 Travel Medicine

23 Radiology

23.1 Konvensional

23.2 CT Scan

23.3 Ultrasonography

23.3.1 POCUS in Emergency/Critical Care Setting

23.3.1.1 Blue Protocol

23.3.1.2 RUSH/ACES Protocol

23.4 MRI

23.5 Nuclear Medicine

24 Anatomical Pathology

25 Clinical Pathology

25.1 Hematology Laboratory

25.1.1 Hematologic Malignancy

The latest classification of hematopoietic and lymphoid malignancy was WHO 2016. This classification combines clinical, morphological, immunophenotype, cytogenetics and molecular features.

25.2 Clinical Chemistry

25.2.1 Acid-Base Analysis

Understanding Acid-base analysis is essential for emergency/critical care. (Marino 2014)

25.2.1.1 Basic Concepts of Acid-Base

hydrogen ion concentration is expressed as pH:

$$pH = \text{Log}(1/[H^+]) = -\log[H^+]$$

The physiological range is 7.40, which corresponds to a $[H^+] = 40 \text{ nEq/L}$

The balance of hydrogen ions in ECF is determined by partial pressure of CO_2 and concentration of bicarbonate. The equation is as follows:

$$[H^+] = 24x(\text{PCO}_2/\text{HCO}_3)$$

The dynamics of $\text{PCO}_2/\text{HCO}_3$ identifies the primary acid base disorders and the secondary responses. If the primary change is in the PCO_2 then the disturbance is called respiratory acidosis or alkalosis, if the primary force is the change in HCO_3 then it is metabolic acidosis or alkalosis. If there are primary forces, the body will react with secondary responses to limit the changes in H^+ . This is accomplished by changing the other component of $\text{PCO}_2/\text{HCO}_3$ ratio

in the same direction. if the primary problem is an increase in PaCO₂ (respiratory acidosis), the secondary response will involve an increase in HCO₃, and this will limit the change in [H⁺] produced by the increase in PaCO₂. Secondary responses should not be called “compensatory responses” because they do not completely correct the change in [H⁺] produced by the primary acid-base disorder.(Marino 2014)

25.2.1.1.1 Metabolic Acidosis

The secondary response to metabolic acidosis is an increase in minute ventilation (tidak volume and respiratory rate) thus resulting in decreased of PCO_2 . This response appears in 30 - 120 minutes, and can take 12 - 24 hours to complete. The magnitude of the response is defined by the equation:

$$\Delta PaCO_2 = 1.2 \times \Delta HCO_3$$

Using the normal PaCO₂ of 40 mmHg, a normal HCO₃ of 24 mEq/L

$$Expected PaCO_2 = 40 - [1.2 \times (24 - current HCO_3)]$$

For a metabolic acidosis with a plasma HCO₃ of 14 mEq/L, the ΔHCO_3 is $24 - 14 = 10$ mEq/L, the $\Delta PaCO_2$ is $1.2 \times 10 = 12$ mm Hg, and the expected PaCO₂ is $40 - 12 = 28$ mm Hg. If the PaCO₂ is > 28 mm Hg, there is a secondary respiratory acidosis. If the PaCO₂ is < 28 mm Hg, there is a secondary respiratory alkalosis.

25.2.1.1.2 Metabolic Alkalosis

The secondary response to metabolic alkalosis is a decrease in minute ventilation and a subsequent increase in PaCO₂.

$$\Delta PaCO_2 = 0.7 \times \Delta HCO_3$$

Using a normal PaCO₂ of 40 mmHg and a normal HCO₃ of 24 mEq/L, the above equation can be rewritten as follows:

$$Expected PaCO_2 = 40 + [0.7 \times (current HCO_3 - 24)]$$

For a metabolic alkalosis with a plasma HCO₃ of 40 mEq/L, the ΔHCO_3 is $40 - 24 = 16$ mEq/L, the $\Delta PaCO_2$ is $0.7 \times 16 = 11$ mm Hg, and the expected PaCO₂ is $40 + 11 = 51$ mm Hg.

25.2.1.1.3 Respiratory Acidosis

25.2.1.1.4 Respiratory Alkalosis

25.2.2 Liver Function Test

25.2.2.1 General Liver Function

1. Excretion Functions (Test bilirubin, urobilinogen)
2. Synthetic Functions (Serum protein, albumin, globulin, prothrombin, ammonia)
3. General Metabolic Functions
4. Storage Site
5. Hematopoiesis
6. Catabolism of Steroid Hormones
7. Clearance of Exogenous Substances (Bromosulphthalein)

liver has a large amount of anatomical and functional reserve and capacity for rapid regeneration, functional deficiency becomes apparent if there is an extensive liver damage. Test for hepatic Injury (ALT, AST, GGT, ALP, 5'-nucleotidase)

Indication of Liver Function Test:

1. Screen for liver disease;
2. Identify the nature of liver disease (hepatocellular, cholestatic, or infiltrative);
3. Assess severity and prognosis of liver disease; and
4. Follow up the course of liver disease

Limitation Liver Function Test:

1. Do not necessarily assess liver function
2. Lack sensitivity (i.e. may be normal in some liver diseases like cirrhosis)
3. Lack specificity (i.e. may be abnormal in non-liver disorders e.g. serum albumin is low in nephrotic syndrome and in cirrhosis)

Non hepatic Cause of Abnormal Liver Function Test: Increased serum bilirubin: – Hemolysis – Ineffective erythropoiesis – Resorption of a large hematoma

Increased aminotransferases: – Muscle injury – Alcohol abuse – Myocardial infarction

Increased serum alkaline phosphatase: – Pregnancy – Bone disease

Low serum albumin: – Poor nutritional status – Proteinuria – Malabsorption – Severe illness causing protein catabolism

25.2.2.1.1 Excretion Function

25.2.2.1.1.1 Bilirubin metabolism

Bilirubin is mostly (85%) from breakdown of HB of Old red cells in reticuloendothelial cells, mainly in Spleen. Normal serum bilirubin is less than 1 mg/dl. Common method to measure bilirubin are Diazo methods, which measure total bilirubin and conjugated bilirubin (thus named as direct bilirubin), while the unconjugated bilirubin were calculated from subtraction of total bilirubin - conjugated bilirubin. Other derived from premature destruction of red cell precursor in bone marrow. The Steps are following:

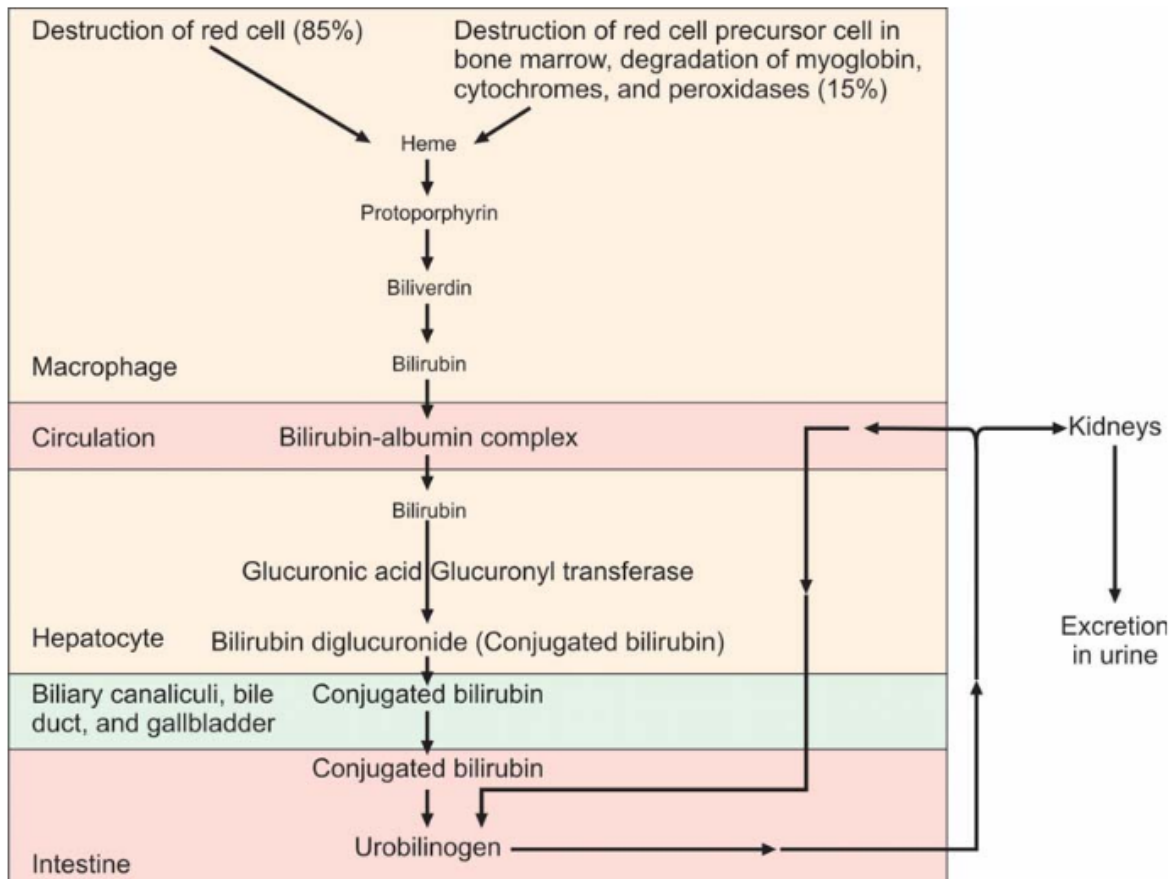


Figure 25.1: Normal Bilirubin Metabolism

1. HB, degraded within macropahes to form heme and globin. Globin consist of amino acids, which are recycled. While Heme (iron + protoporphyrin) release iron, which is stored as ferritin. The protoporphyrin then converted to biliverdin, then to bilirubin.(Kawthalkar 2018)
2. Released bilirubin into the circulation and bind albumin, this called unconjugated bilirubin. It is insoluble to water, but lipid soluble.

3. Billirubin-albumin complex reaches the liver whee it is taken up by the hepatocytes. Albumin released back to the circulation.
4. Bilirubin then conjugated with glucoronic acid to form bilirubin monoglucoronid and diglucoronid (conjugated bilirubin). This Process is mediated by the ezyme glucoronyl transferase. Conjugated bilirubin is more soluble in water.
5. Conjugated bilirubin is secreted from the hepatocytes into biliary canaliculi, from where it passes into the bile duct and gallbladder along with bile.
6. When bilirubin reaches the large instensites, it is converted to urobilinogen by the baterial enzyme.
7. Most urobilinogen is excreted in feses as urobilin, thus coloring the feses yellow. A part of urobilinogen is absorbed into the circlation from where it reaches the liver, taken up and reexcreted into the bile. Small amaount of urobilinogen in cleared in urine.

Jaundice or icterus referes to yellow discoloration of skin, screa and mucous membranes due to increased level of serum bilirubin. Jaundice will clinicaly evident when serum bilirubin level exceeds **2.0 mg/dL**.

1. According to type of dominant bilirubin:

A. Predominant unconjugated hyperbilirubinemia

If > 85% of total is unconjugated bilirubin or indirect bilirubin. The cause are hemolysis, ineffective erythropoeisis, resorption large hematome, gilbert syndrome.

B. Predominant conjugated hyperbilirubinemia

If > 50% of total is conjugated or direct bilirubin. Cause are hepatitis, cirrhosis, cholestasis, drugs (anabolic steroid, oral contraceptives), toxin, dubin johnson syndrome, rotor syndrome

C. Mixed (Conjugated and Unconjugated)

if conjugated bilirubin is 20-50% of total, it may results from viral or alcoholic hepatitis.

2. According to site of disease

A. Prehepatic There is excessive formation of bilirubin exceeding the capacity of the liver to conjugate it for excretion. The dominant bilirubin is unconjugated. Bilirubin is absent in urine, since of is water insoluble. Urobilinogen is increased in urine and feses. Jaudice is usually mild (total bilirubin usually < 5.0 mg/dl, conjugated is < 15% of total). The cuase is usually hematological problem.

B. Hepatic

Both Unconjugated and Conjuated are increased. mainly increased unconjunjugated 1. defective uptake 2. defective conjugation 3. physiologic jaundice (newborn)

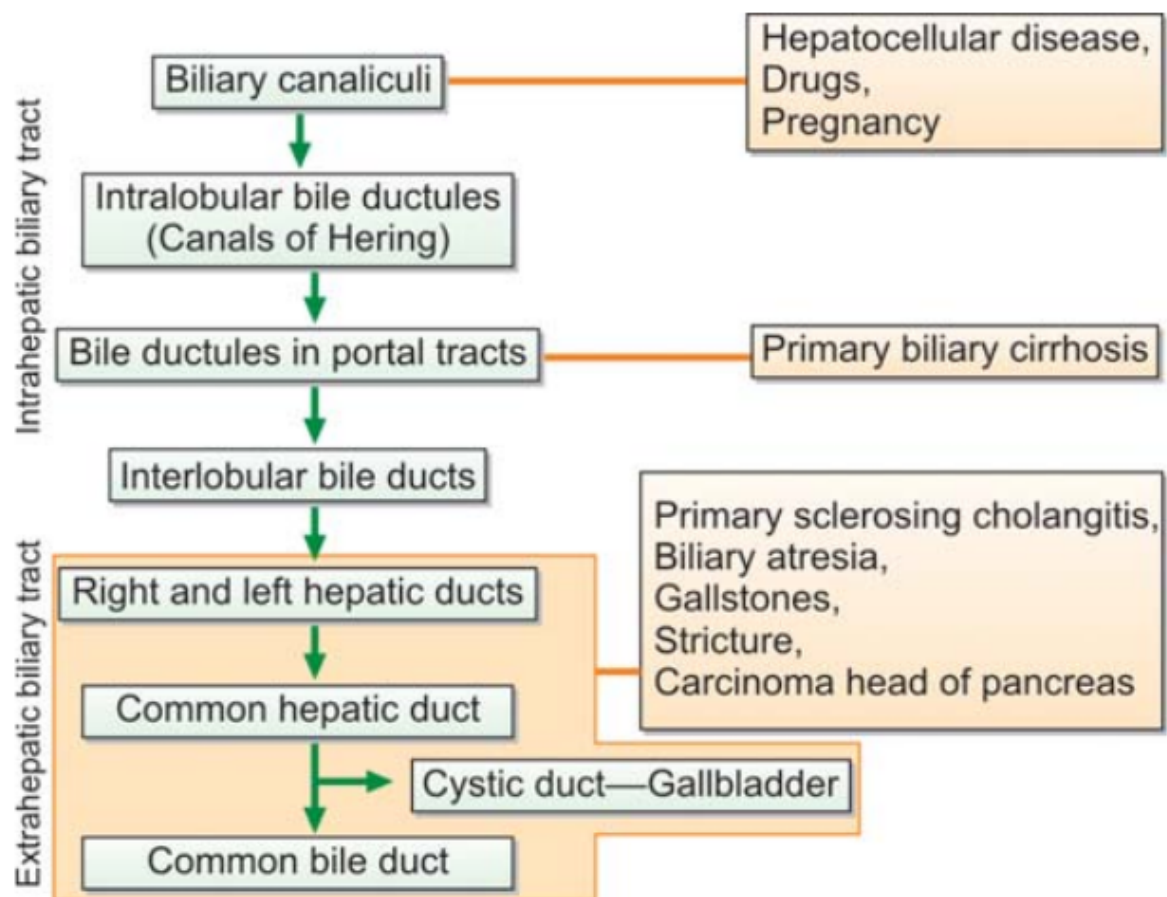


Figure 25.2: Sites of Cholestasis

mainly Conjugated bilirubin 1. Hepatocellular disease: Liver enzyme increase. Bilirubin is around 4.0 s.d 8.0 mg/dl. Conjugated bilirubin is 20-50% of total bilirubin. In hepatocellular injury, both conjugated and unconjugated bilirubins are increased. Unconjugated bilirubin is increased due to reduced ability of liver cells to conjugate bilirubin. Conjugated bilirubin is raised from cholestasis due to hepatocyte swelling.

2. Intrahepatic Cholestasis: In intrahepatic cholestasis, there may be (1) impairment of secretion of bilirubin from hepatocytes into the biliary canaliculi; (2) obstruction of bile flow in canaliculi by swollen hepatocytes; or (3) damage to intrahepatic canaliculi.

Drugs commonly associated with cholestatic injury are oral contraceptives, anabolic steroids, oral anti-diabetics, phenothiazines, and erythromycin.

Primary biliary cirrhosis, there is autoimmune destruction of intrahepatic bile ducts. It predominantly occurs in middle-aged, females and is characterized by chronic elevation of alkaline phosphatase and positive anti-mitochondrial antibody in serum.

Primary sclerosing cholangitis is an autoimmune disorder occurring in young to middle-aged men in whom there is inflammation and destruction of both intrahepatic and extra-hepatic bile ducts. Associated inflammatory bowel disease is often present. Serum alkaline phosphatase is elevated and many patients have circulating perinuclear antineutrophil cytoplasmic antibodies.

C. Post Hepatic

Obstruction of extrahepatic biliary tract prevents flow of bile into the duodenum. This causes "regurgitation" of conjugated bilirubin into the circulation. (Biliary canaliculi distend and rupture due to backpressure of bile and conjugated bilirubin escapes into the sinusoids). Conjugated bilirubin is usually >50% of total in posthepatic jaundice. Urinary and fecal urobilinogen are decreased, faeces are clay-colored, and bilirubin (being conjugated and water-soluble) appears in urine. Cause of Posthepatic Jaundice:

- Carcinoma of head of pancreas
- Carcinoma of ampulla of Vater
- Secondaries in porta hepatis
- Gallstones in or stricture of common bile duct

25.2.2.1.1.2 Clinical Algorithm of Jaundice

Normal serum bilirubin is less than 1 mg/dl. conjugated bilirubin is 10% or less, while unconjugated bilirubin is 90% or more. This is because conjugated bilirubin is rapidly secreted into the bile after its formation and removed through the gut. Conjugated bilirubin is composed of bilirubin glucuronide, bilirubin diglucuronide, and delta (Δ) bilirubin. Delta bilirubin represents bilirubin covalently bound to albumin in circulation. Normally, bilirubin is absent or present in very small amount. In cholestasis, proportion of Δ -bilirubin increases. Owing to its longer half-life, it is cleared slowly from circulation. Conjugated bilirubin is weakly bound to albumin,

Parameter	Prehepatic jaundice	Hepatocellular jaundice	Posthepatic jaundice
1. Basic mechanism of raised bilirubin	Hemolysis leading to excess production	Deficient uptake, conjugation, or excretion by hepatocytes	Deficient excretion due to obstruction of biliary tract
2. Type of serum bilirubin increased	Mainly unconjugated	Unconjugated + Conjugated	Mainly conjugated (>50%)
3. Urine bilirubin	Absent	Present	Present
4. Urine urobilinogen	Increased	Variable	Decreased
5. Prototype	Hemolytic anemia	Viral hepatitis	Common duct stone
6. Prothrombin time	Normal	Abnormal that is not corrected with vitamin K	Abnormal that is corrected with vit K
7. Additional features	Features of haemolysis on blood smear (reticulocytosis, low haptoglobin, low hemoglobin)	Marked rise of serum ALT and AST	Marked rise of serum ALP (>3 times normal)

Figure 25.3: Three types of Jaundice Differential Diagnosis

is water-soluble, and can be excreted in urine. Unconjugated bilirubin is tightly bound to albumin and is water-insoluble.

25.2.2.1.2 Synthetic Function

Markers of hepatic synthetic function are serum albumin and prothrombin time (PT). Hypoalbuminemia and a prolonged PT indicate a severe functional impairment of liver.

25.2.2.1.2.1 Protein Synthetic

Liver is the sole organ to synthesize plasma protein, except gamma globulin which synthesized by Plasma Cell. Normal total serum protein in adults is about 5.5 - 8.0 gm / dl, albumin 3.5 - 5.0 which comprise the 60% of serum protein.

Test for protein in liver disease include total serum protein, serum albumin, calculation of albumin/globulin ratio (normal > 1.5) and serum electrophoresis. The common laboratory method to measure total protein is biuret and refractometer method.

Total serum protein level is affected by both and gamma globulins. In cirrhosis, decrease in albumin level is often compensated by increase in the level of gamma globulins; therefore, estimation of total serum proteins is of limited value in cirrhosis. Estimation of serum albumin and serum protein electrophoresis are more helpful.

Albumin is synthesized exclusively in liver and constitutes about 60% of total proteins in serum; therefore its estimation is an important investigation in liver disease. Half-life of albumin is about 20 days and therefore fall in its level in response to decreased synthesis is not immediately apparent. Therefore, in acute liver disease (e.g. viral hepatitis), there is little change in albumin

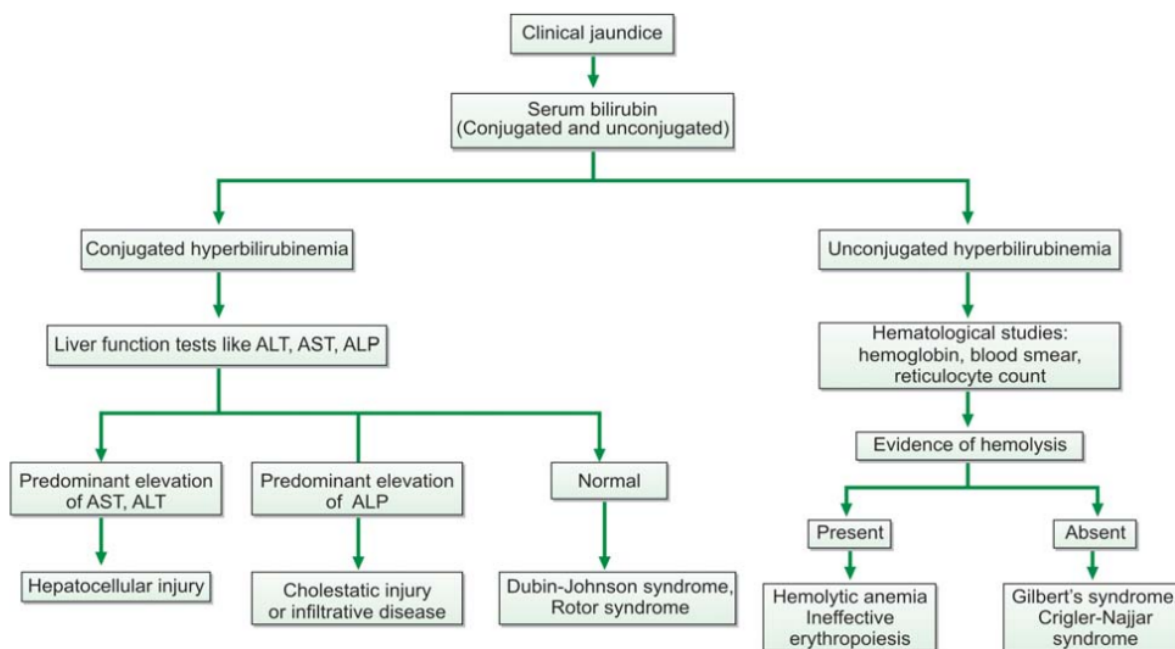


Figure 25.4: Jaundice Algorithm

level. Serum albumin level is low in chronic liver disease (cirrhosis) and correlates with synthetic capacity of hepatocytes; therefore, it is helpful in following progression of cirrhosis. Also, fall in serum albumin level correlates with severity of ascites. In cirrhosis and in chronic active hepatitis, serum gamma globulins are increased due to inflammation. Low albumin and raised gamma globulins in serum cause reversal of albumin/globulin ratio. Serum albumin is estimated by bromocresol green method

Causes of decreased serum albumin: • Decreased intake: malnutrition. • Decreased absorption: malabsorption syndromes. • Decreased synthesis: liver disease, chronic infections. • Increased catabolism: thyrotoxicosis, fever, malignancy, infections. • Increased loss: nephrotic syndrome, severe burns, protein-losing enteropathies, ascites • Increased blood volume: pregnancy, congestive cardiac failure.

Serum Protein Elektrophoresis: 1. In cirrhosis, albumin may be reduced and there may be polyclonal increase of IgG and IgA, with - bridging. (IgA migrates between and regions which obscures the demarcation between and peaks). 2. In primary biliary cirrhosis, there is polyclonal increase of IgM. 3. In 1-antitrypsin deficiency (associated with cirrhosis) 1-globulin band is reduced. 4. In chronic active hepatitis, IgG is elevated.

25.2.2.1.2.2 Prothorombin Time (PT)

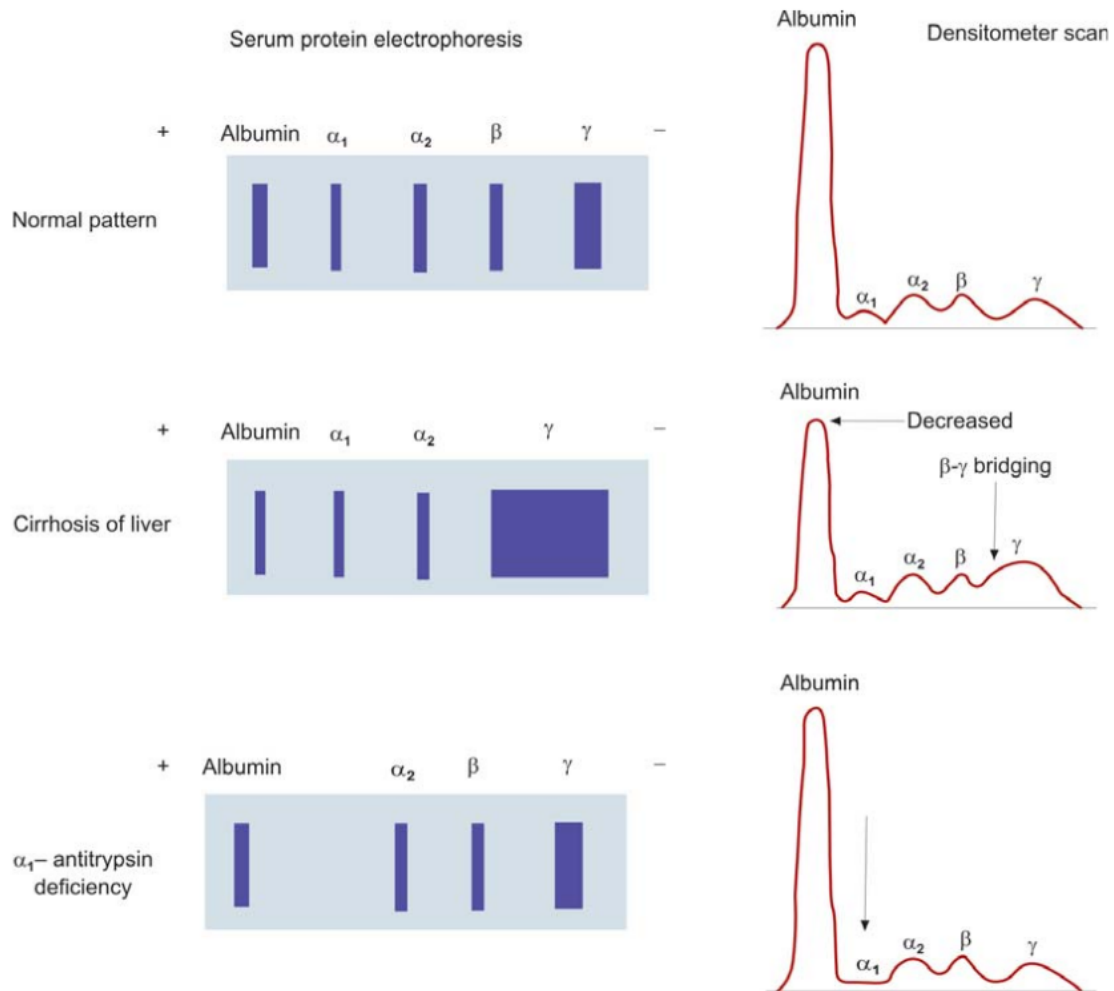


Figure 25.5: Serum Protein Electrophoresis patterns and densitometer scans in Normal, Cirrhosis and Alpha 1 antitrypsin deficient individuals

Most of the coagulation proteins are synthesized in the liver. Vitamin K is required for the synthesis of factors II, VII, IX, X by the hepatocytes, thus it is called vitamin K dependent factors. Synthesis of the factors is deficient in hepatocellular disease. In addition, in obstructive jaundice, vitamin K (a fat soluble vitamin) cannot be absorbed due to the absence of bile in the intestine.

PT measures three out of four vitamin K-dependent factors (II, VII, and X) and is prolonged in hepatocellular disease and in obstructive jaundice. Intramuscular injection of vitamin K corrects prolonged PT in obstructive jaundice but not in hepatocellular jaundice. To distinguish between a prolonged PT due to hepatocellular disease from that due to cholestasis with fat malabsorption, PT is repeated after administration of vitamin K. Reduction of prolonged PT occurs in cholestatic liver disease, but not in hepatocellular disease. In acute fulminant liver failure, marked prolongation of PT is an unfavourable prognostic sign.

25.2.2.1.2.3 Blood Ammonia

Blood ammonia is mainly derived from GI tract. In the gut, bacterial enzymes act on nitrogen-containing foods to produce ammonia, which is carried to the liver via portal vein. In the liver, ammonia is converted to non-toxic urea in urea cycle.

Increased blood ammonia seen in: - Fulminant hepatic failure - cirrhosis - Reye's Syndrome - "Shunting" of portal blood to systemic circulation - Gastrointestinal hemorrhage (there is increased production of ammonia from blood proteins by bacterial enzymes). In hepatic disease, gastrointestinal hemorrhage is associated with increased risk of hepatic encephalopathy. - Inherited deficiencies of ure cycle enzymes.

If the ammonia is increased, it might help indicate a hepatic encephalopathy.

25.2.2.1.3 Test Hepatocellular Injury

Serum enzyme changes in liver disease result from hepatocyte damage and do not indicate hepatic functional capacity.

- Serum aspartate aminotransferase or AST (formerly called serum glutamic-oxaloacetic transaminase or SGOT)
- Serum alanine aminotransferase or ALT (formerly called serum glutamic-pyruvic transaminase or SGPT)
- Serum alkaline phosphatase or ALP
- -Glutamyl transferase or GGT (also called as -glutamyl transpeptidase)
- 5'-nucleotidase (5'-NT)

25.2.2.1.3.1 Serum Aminotransferase (Transaminitis)

Both AST and ALT are sensitive marker of hepatocellular injury. ALT is cytosolic enzyme, AST is cytosolic and mitochondria. Normally, aminotransferases are present in serum at a low level. When necrosis or death of cells containing these enzymes occurs, aminotransferases are released into the blood and their concentration in blood increases. This level correlates with extent of tissue damage.

“Most marked elevations of ALT and AST (>15 times normal) are seen in acute viral hepatitis, toxin-induced hepatocellular damage (e.g. carbon tetrachloride), and centrilobular necrosis due to ischemia (congestive cardiac failure).”

Moderate elevation (5-15X normal) occurs due to chronic hepatitis, autoimmune hepatitis, alcoholic hepatitis, acute biliary tract obstruction, and drug induced hepatitis. Mild elevation (1-3 x normal) seen in cirrhosis, non-alcoholic steatosis and cholestasis.

Normal ratio AST/ALT is 0.7 to 1.4. Increased ratio (>2.0) is highly suggestive of alcoholic hepatitis, while ratio <1.0 is seen in acute viral hepatitis. ALT and AST are elevated in acute viral hepatitis even before the appearance of jaundice. Persistence of elevated ALT and AST beyond 6 months in a case of hepatitis indicates development of chronic hepatitis. Measurement of ALP is helpful in differentiation of hepatocellular jaundice from cholestatic jaundice.

25.2.2.1.3.2 Serum Alkaline Phosphatase (ALP)

Alkaline phosphatase is distributed widely in various tissues like liver, bones, intestine, kidney, and placenta. In the liver, ALP, GGT, and 5'-NT are located normally on canalicular surface of hepatocytes. In cholestasis, accumulated bile acids dissolve canalicular side of hepatocyte membrane and enzymes are released in blood. Therefore, diseases that affect mainly hepatocyte secretion have elevated levels of ALP.

ALP is increased in most cases of cholestatic type of jaundice. While hepatocellular injury is characterized by marked elevation of ALT and AST, cholestasis is characterized by marked increase (more than 3 times normal) of ALP. Since there are many other sources of ALP apart from liver, simultaneous measurement of serum GGT and serum 5'-NT may be used to ascertain whether increase of ALP is of hepatic origin. Main cause of increased ALP due to hepatobiliary disease are: Bile duct obstruction, primary biliary cirrhosis, primary sclerosing cholangitis, infiltrative disease of liver.

Disease of bones also could cause increased ALP. ALP present within osteoblasts, thus ALP increase when there is an increase osteoblastic activity. Pregnancy also increase ALP due to increased secretion from placenta.

25.2.2.1.3.3 Serum Gamma-Glutamyl Transferase (GGT)

GGT presents in liver, pancreas, kidney, and prostate. The estimation of this enzyme is particularly useful in following: alcoholism (acute alcoholic hepatitis); Cholestasis (elevation of ALP and GGT significantly points toward liver disease); recovery from acute hepatitis (GGT is the last enzyme to return to normal, thus its normalization indicative favourable outcome)

25.2.2.2 Interpretation of Liver Function Test

Liver disease could broadly classified into two: hepatocellular and cholestasis.

1. Typical LFT profile in hepatocellular disease

- Marked elevation of AST and ALT (usually >500 IU)
- Mild increase of ALP (<3 times normal)
- Hyperbilirubinemia, if present, is of both conjugated and unconjugated type
- Chronic hepatocellular injury, aminotransferases are moderately elevated, and serum albumin is reduced.

If the LFT showed a hepatocellular disease, then next investigation is to detect underlying cause:

- Viral serology (viral hepatitis: IgM anti-hepatitis A antibody, hepatitis surface B antigen (HBsAg), antihepatitis C antibody),
- search for injurious drugs, toxins or alcohol,
- autoantibodies like antinuclear antibodies and anti-smooth muscle antibodies (autoimmune hepatitis), and serum ceruloplasmin (Wilson disease).

If cause is not detected in the presence of persistent elevation of aminotransferases, liver biopsy is performed (that may reveal chronic viral hepatitis, autoimmune disorders, Wilson disease, haemochromatosis, and infiltrative diseases)

2. Typical LFT profile in cholestatic jaundice

- Marked elevation of ALP (>3 times normal)
- Elevation of GGT and 5'-NT
- Mild or no increase of ALT and AST (usually <200 IU)
- Elevation of conjugated bilirubin
- The pattern of elevated ALP but normal serum bilirubin is seen in infiltrative diseases

In patients with cholestatic or infiltrative pattern of injury, imaging studies should be done for diagnosis of obstruction. If there is no evidence of obstruction, liver biopsy is usually done

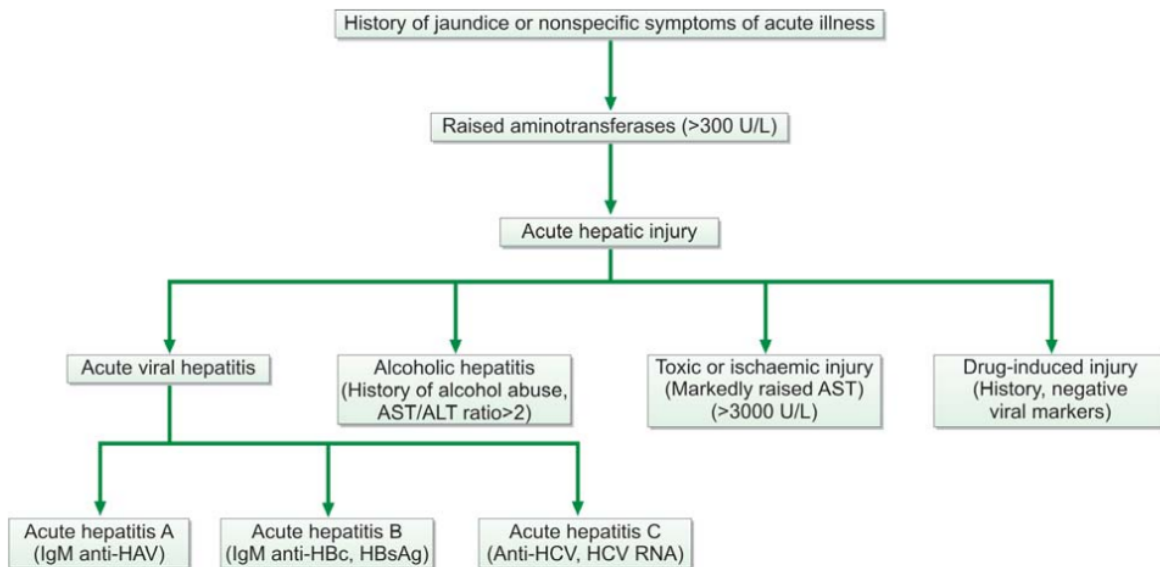


Figure 25.6: History of Jaundice of nonspecific acute illness in the setting of increased aminotransferase

Finally, Severity of liver disease is assessed by serum bilirubin, serum albumin, and prothrombin time. Child-Turcotte-Pugh classification is commonly used to assess severity of cirrhosis and is based on both clinical and laboratory parameters

25.2.2.2.1 Normal Values

Serum alanine aminotransferase (ALT, SGPT): 5-42 U/L

Serum aspartate aminotransferase (AST, SGOT): 5-40 U/L

Serum alkaline phosphatase (ALP):

- Children: 25-350 U/L
- Adult males: 25-120 U/L
- Adult females: 25-90 U/L

AST/ALT ratio: 0.7-1.4

Serum bilirubin:

- Total: 0.3-1.0 mg/dl
- Direct (Conjugated): 0-0.2 mg/dl

Serum proteins, total: 5.5-8.0 gm/dl

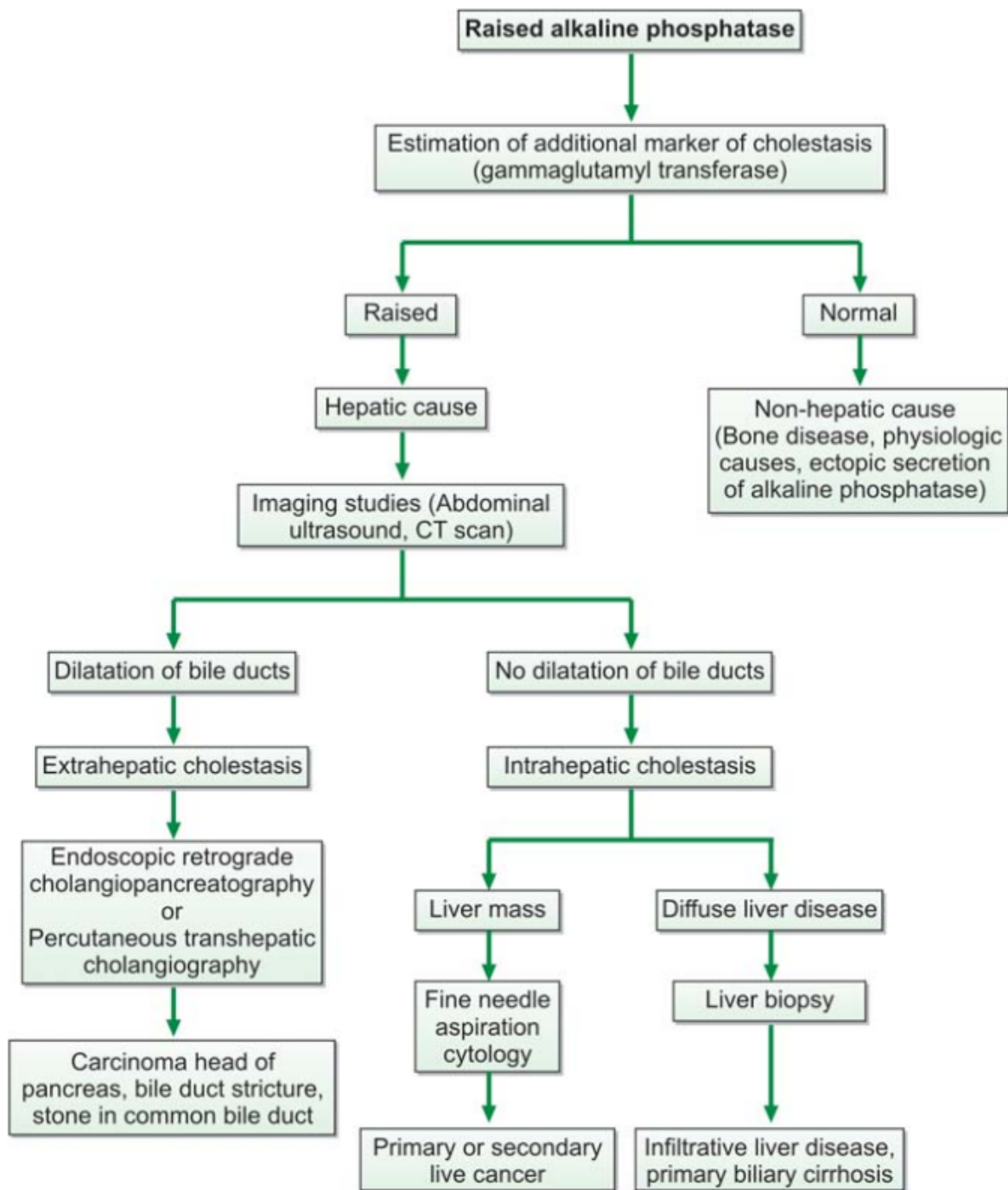


Figure 25.7: Evaluation of Raised ALP in the suspicion of cholestatic injury

Parameter	Score 1	Score 2	Score 3
Serum bilirubin (mg/dl)	<2	2-3	>3
Serum albumin (g/dl)	≥ 3.5	2.8-3.5	<2.8
Prothrombin time (difference in seconds)	<4	4-6	>6
Ascites	None	Slight or controlled by diuretics	Moderate despite diuretics
Encephalopathy (grade)	0	1-2	3-4

Figure 25.8: Child-Turcotte-Pugh score to assess severity of cirrhosis

Serum albumin: 3.5-5.0 gm/dl

Serum globulins: 1.8-3.5 gm/dl

Albumin/Globulin (A/G) ratio: >1.5

Prothrombin time: 11-15 seconds

Plasma ammonia: 9-33 mol/L

Serum gamma-glutamyl transferase:

- Males: Up to 40 U/L
- Females: Up to 25 U/L

Serum protein electrophoresis:

Albumin 52-65%

1 globulin: 2.5-5%

2 globulin: 7-13%

globulin: 8-14%

globulin: 12-22%

25.2.3 Disorder of Lipids

25.2.4 Cardiac markers

25.2.5 Coagulation Study

25.2.6 Renal Function Test

Kidney performs following functions: 1. maintenance of extracellular fluid volume & composition
2. Excretion of metabolic waste 3. Regulation of blood pressure 4. synthesis of erythropoietin

5. Production of Vitamin D3

Kidney functions affected by:

1. Diffuse renal disease
2. Pre-renal conditions
3. Post-renal conditions

Classification of renal Function Test

A. Evaluation of Glomerular Function

B Evaluation of Tubular Function

25.2.6.1 A. Test to Evaluate Glomerular Function

The best test to evaluate overall kidney functions is estimation of GFR. GFR varies with age, sex and BSA. Normal GFR in young adults is 120-130 ml/min per 1.73 m² of body surface area. GFR declines progressively with age (due to arteriosclerosis of glomeruli). After 40 years of age, there is a steady and progressive fall in the GFR at the rate of 1 ml/minute/year because of reduction in the number of glomeruli due to arteriosclerosis.

Kidney 2022 divide Chronic Kidney Disease into :

1. Stage 1; Kidney damage with Normal to increased GFR ($\text{GFR} \geq 90 \text{ ml / min / } 1.73 \text{ m}^2$)
2. Stage 2; Kidney damage with Mildly reduced GFR ($\text{GFR } 60 - 89 \text{ ml / min / } 1.73 \text{ m}^2$)
3. Stage 3; Moderately reduced GFR ($\text{GFR } 30 - 59 \text{ ml / min / } 1.73 \text{ m}^2$)
4. Stage 4; Severely reduced GFR ($\text{GFR } 15 - 29 \text{ ml / min / } 1.73 \text{ m}^2$)
5. Stage 5; Kidney Failure ($\text{GFR} \leq 15 \text{ ml / min / } 1.73 \text{ m}^2$)

Kidney damage refers to presence of pathological abnormalities in blood or urine test or imaging studies. Symptoms usually present at GFR less than 60, which reflect loss of 50% Kidney Function.

GFR, glomerular filtration rate refers to the rate in ML/min at which a substance is cleared from the circulation by the glomeruli. The ability of the glomeruli to filter a substance from the blood is assessed by clearance studies. If a substance is not bound to protein in plasma, is completely filtered by the glomeruli, and is neither secreted nor reabsorbed by the tubules, then its clearance rate is equal to the glomerular filtration rate (GFR).

Clearance of a substance refers to the volume of plasma, which is completely cleared of that substance per minute; it is calculated from the following formula:

$$clearance = \frac{UV}{P}$$

U: concentration of substance in urine (mg/dl)

V: volume of urine excreted in ml/min

P: concentration of substance in plasma (mg/dl)

all clearance value are adjusted to standard body surface area: 1.73 m².

Clearance tests are cumbersome to perform, expensive, and not readily available. One major problem with clearance studies is incomplete urine collection.

The Particular “substance” are molecules which having the following properties: 1. It should be physiologically inert and preferably endogenous,

2. It should be freely filtered by glomeruli and should be neither reabsorbed nor secreted by renal tubules,

3 It should not bind to plasma proteins and should not be metabolized by kidneys

4 It should be excreted only by the kidneys

Example of substance used to measure GFR are:

1. Exogenous: Inulin, Radiolabelled ethylenediamine tetraacetic acid (51Cr- EDTA), 125I-iothalamate
2. Endogenous: Creatinine, Urea, Cystatin C

25.2.6.1.1 Inulin Clearance

Inulin is inert polysaccharide derived from the plant. It is called inert as it is freely filtered by the glomerulus, and is neither reabsorbed nor secreted by the tubules thus Inulin is an ideal agent for measuring GFR.

Protocol: A bolus dose of inulin (25 ml of 10% solution IV) is administered followed by constant intravenous infusion (500 ml of 1.5% solution at the rate of 4 ml/min). Timed urine samples are collected and blood samples are obtained at the midpoint of timed urine collection. Average inulin clearance for males is 125 ml/min/1.73 m² and for females is 110 ml/min/1.73 m².

25.2.6.1.2 Cystatin C Clearance

This is a cysteine protease inhibitor of MW 13,000, which is produced at a constant rate by all the nucleated cells. It is not bound to protein, is freely filtered by glomeruli and is not returned to circulation after filtration. It is a more sensitive and specific marker of impaired renal function than plasma creatinine. Its level is not affected by sex, diet, or muscle mass. It is thought that cystatin C is a superior marker for estimation of GFR than creatinine clearance.

25.2.6.1.3 Creatinine Clearance

Creatinine is being produced constantly from creatine in muscle. It is completely filtered by glomeruli and is not reabsorbed by tubules; however, a small amount is secreted by tubules.

A 24-hour urine sample is preferred. After getting up in the morning, the first voided urine is discarded. Subsequently all the urine passed is collected in the container provided. After getting up in the next morning, the first voided urine is also collected and the container is sent to the laboratory.

A blood sample for estimation of plasma creatinine is obtained at midpoint of urine collection.

Creatinine clearance is calculated from:

1. concentration of creatinine in urine in mg/ml (U),
2. volume of urine excreted in ml/min (V) (this is calculated by the formula: volume of urine collected/collection time in minutes e.g. volume of urine collected in 24 hours ÷ 1440)
3. concentration of creatinine in plasma in mg/dl (P).
4. Creatinine clearance in ml/min per 1.73 m² is then derived from the formula UV/P .

Because of secretion of creatinine by renal tubules, the above formula overestimates GFR by about 10%. In advanced renal failure, secretion of creatinine by tubules is increased and thus overestimation of GFR is even more. Overestimation of GFR due to tubular secretion of creatinine is somewhat balanced by slight overestimation of serum creatinine by Jaffe's reaction. To provide values closer to the actual GFR, cimetidine (which blocks secretion by renal tubules) can be administered before commencing urine collection

Creatinine clearance has few weakness:

1. Tubular secretion of creatinin that icrease as the kidney disease advance
2. Collection of urine often incomplete
3. Creatinin level is affected by protein intake and muscle mass
4. Creatinine level also affected by drugs, ie Cimetidine, Trimetophrim (which block tubular secretion of creatinine)

Tips: BUN and serum creatinine, by themselves, are not sensitive indicators of early renal impairment since values may be normal e.g. if baseline values of serum creatinine is 0.5 mg/dl, then 50% reduction in kidney function would increase it to 1.0 mg/dl. Thus clearance tests are more helpful in early cases. If biochemical tests are normal and renal function impairment is suspected, then creatinine clearance test should be carried out. If biochemical tests are abnormal, then clearance tests need not be done.

25.2.6.1.4 Urea Clearance

Urea is filtered by the glomeruli, but about 40% of the filtered amount is reabsorbed by the tubules. The reabsorption depends on the rate of urine flow. Thus it underestimates GFR, depends on the urine flow rate, and is not a sensitive indicator of GFR.

25.2.6.1.5 Estimation of Creatinine Clearance from serum creatinin by equation

Estimation of GFR from Serum creatinin depend on age, sex, body weight and serum creatinine value (and in some formula includes ethnicity). This is called as estimated GFR (eGFR).

1. Modified Cockcroft Gault Equation (if SC Mg/Dl)

$$CreatineClearance(ml/min) = \frac{(140 - age) \times (Weight)}{72 \times SC}$$

In females, the value obtained from above equation is multiplied by 0.85 to get the result.

2. Modification of Diet in Renal Disease (MDRD) equation for estimated eGFR:

$$eGFR(mL/min/1.73m^2) = 175 \times (s.creatinine in mg/dL)^{-1.154} \times (Age in year)^{-0.203} \times (0.742 if female) \times$$

This is applicable for adults between 18 and 70 years of age. It is more accurate than measured creatinine clearance from 24-hour urine collection and Cockcroft-Gault equation. This is the most commonly used equation.

3. CKD-EPI (chronic kidney disease-epidemiology) equation: This is a recently introduced more accurate equation.

$$eGFR(mL/min/1.73m^2) = 141 \times \min(Scr/k, 1)^a \times \max(Scr/k, 1)^{-1.209} \times (0.993)^{Age} \times 1.018(if female) \times 1.159($$

where, Scr = Standardized serum creatinine (mg/dL) K = 0.7 (female) or 0.9 (male) a = -0.329 (female) or -0.411 (male), min = minimum of Scr/k or 1, max = maximum of Scr/k or 1, Age = years

4. In Children, Schwartz formula is used for estimation of GFR:

$$EstimatedGFR = \frac{K \times Height(cm)}{Sc}$$

$k = 0.33$ in preterm babies for the first year of life; $k=0.45$ for full term infants; for infants and children up to 12 years of age k is 0.55.

Disadvantages of estimated GFR: 1. GFR estimated by Cockcroft-Gault and MDRD formulas tends to underestimate renal function at high levels of GFR (>60 mL/minute/1.73 m²);

2. Values may be misleading in acute kidney injury, pregnancy, decreased muscle mass (e.g. paraplegia, muscle wasting disorders, critically ill, patients with cancer), increased muscle mass in athletes and body builders, unusual diets, and extremes of age or weight.

25.2.6.1.5.1 Blood Urea Nitrogen

Urea is produced in the liver from amino acids. Amino acids are utilized to produce energy, synthesize proteins, and are catabolized to ammonia. Urea is produced in the liver from ammonia in the Krebs urea cycle. Ammonia is toxic, and hence, is converted to urea, which is then excreted in urine.

The concentration of blood urea is usually expressed as blood urea nitrogen. This is because older methods estimated only the nitrogen in urea. Molecular weight of urea is 60, and 28 g of nitrogen are present in a gram mole of urea. As the relationship between urea and BUN is 60/28, BUN can be converted to urea by multiplying BUN by 2.14, i.e. the real concentration of urea is $BUN \times (60/28)$. Urea is completely filtered by the glomeruli, and about 30–40% of the filtered amount is passively reabsorbed in the renal tubules depending on the person's state of hydration.

Blood level of urea is affected by a number of nonrenal factors (e.g. dehydration, hypoperfusion of kidneys, high protein diet, protein catabolism, upper gastrointestinal hemorrhage, liver function, steroid administration, etc.), and therefore, utility of BUN as an indicator of renal function is limited. Also considerable destruction of renal parenchyma is required before elevation of blood urea can occur.

The term azotemia refers to the increase in the blood level of urea; uremia is the clinical syndrome resulting from this increase. Reference range for BUN in adults is 7–18 mg/dL. In adults >60 years, level is 8–21 mg/dL. Laboratory method to analyze BUN are Diacetyl monoxime urea and urease-Berthol reaction.

25.2.6.1.5.2 Serum Creatinine

Creatinine is a nitrogenous waste product formed in muscle from creatine phosphate. Endogenous production of creatinine is proportional to muscle mass and body weight. Exogenous creatinine (from ingestion of meat) has little effect on daily creatinine excretion.

Serum creatinine is a more specific and more sensitive indicator of renal function as compared to BUN because: - It is produced from muscles at a constant rate and its level in blood is not affected by diet, protein catabolism, or other exogenous factors - It is not reabsorbed, and very little is secreted by tubules.

Significant kidney reserve, increase of serum creatinine level (from 1.0 mg/dl to 2.0 mg/dl) in blood does not occur until about 50% of kidney function is lost. Therefore, serum creatinine is not a sensitive indicator of early renal impairment (**SC_gfr?**). Laboratory report showing serum creatinine “within normal range” does not necessarily mean that the level is normal; the level should be correlated with body weight, age and sex of the individual. If renal function is absent, serum creatinine rises by 1.0 to 1.5 mg/dL/day.

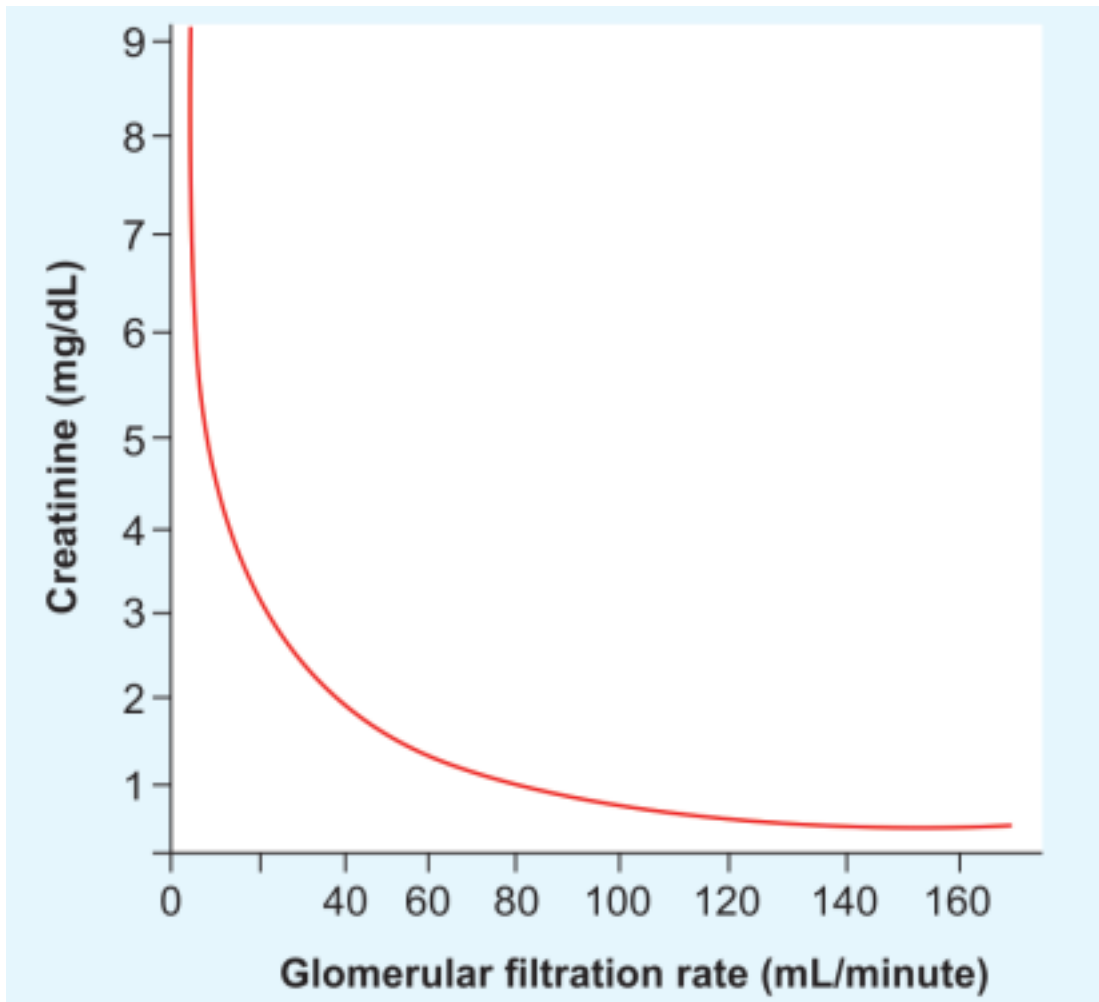


Figure 25.9: Relationship between glomerular filtration rate and serum creatinine. Significant increase of serum creatinine does not occur till a considerable fall in GFR

Causes of decreased serum creatinine level:

1. Female sex.
2. Vegetarian diet.
3. Malnutrition, muscle wasting.
4. Increasing age (reduction in muscle mass)

Methods to estimate serum creatinine is Jaffe reaction and Enzymatic Methods. Reference range Adult males: 0.7–1.3 mg/dL. Adult females: 0.6–1.1 mg/dL.

25.2.6.1.5.3 BUN/Serum Creatinine Ratio

BUN/creatinine ratio to discriminate prerenal and postrenal azotemia from renal azotemia. Normal ratio is 12:1 to 20:1.

Causes of increased BUN/creatinine ratio (>20:1): 1. Increased BUN with normal serum creatinine:

- Pre-renal azotemia (reduced renal perfusion)
- High protein diet
 - Increased protein catabolism
 - Gastrointestinal hemorrhage.
2. Increase of both BUN and serum creatinine with disproportionately greater increase of BUN:

Postrenal azotemia (Obstruction to the outflow of urine) Obstruction to the urine outflow causes diffusion of urinary urea back into the blood from tubules because of backpressure.

3. Causes of decreased BUN/creatinine ratio (<10:1):
- Acute tubular necrosis
 - Low protein diet, starvation
 - Severe liver disease.

25.2.6.2 B. Test To Evaluate Tubular Functions

Renal tubules efficiently reabsorb 99% of the glomerular filtrate to conserve the essential substances. Abnormality tubular function includes:

1. GLycosuria
2. Generalized Aminoaciduria
3. TUbular Proteinuria
4. Urinary concentration of sodium
5. Fractional excretion of sodium.

25.2.7 Urine Analysis

25.2.7.1 Macroscopy

25.2.7.2 Microscopy

25.2.7.3 Chemical

25.2.7.3.1 Albuminuria

Normally, a very small amount of albumin is excreted in urine. The earliest evidence of glomerular damage in diabetes mellitus is occurrence of microalbuminuria (albuminuria in the range of 30 to 300 mg/24 hours). An albuminuria >300 mg/24 hour is termed clinical or overt and indicates significant glomerular damage. Urine protein to creatinine ratio on a random urine sample has been shown to correlate well with estimation of 24-hour urine protein.

26 Clinical Microbiology

27 Ear, Nose & Throat

28 Ophthalmology

29 Rehabilitation & Physical Medicine

30 Disaster Medicine, Wilderness medicine, Transport Medicine

31 Pharmacology

32 Common Medical Procedures for Internist

32.1 Diagnostics

32.2 Therapeutics

32.2.1 Blood Transfusion

Part III

Bioinformatics & Genomic Medicine

33 Genomic Medicine

34 Linux & Bioinformatic Programs

Part IV

Statistics and R language

Dataframe Operation

Checking the columns

```
[1] "Field.Name"  "Area"        "Slope"       "Vegetation"  "Soil.pH"
[6] "Damp"        "Worm.density"
```

SUmmary

Field.Name	Area	Slope	Vegetation	Soil.pH
Length :20	Min. :0.800	Min. : 0.00	Length :20	Min. :3.500
N.unique :20	1st Qu.:2.175	1st Qu.: 0.75	N.unique : 5	1st Qu.:4.100
N.blank : 0	Median :3.000	Median : 2.00	N.blank : 0	Median :4.600
Min.nchar: 7	Mean :2.990	Mean : 3.50	Min.nchar: 5	Mean :4.555
Max.nchar:17	3rd Qu.:3.725	3rd Qu.: 5.25	Max.nchar: 9	3rd Qu.:5.000
	Max. :5.100	Max. :11.00		Max. :5.700

Damp	Worm.density
Mode :logical	Min. :0.00
FALSE:14	1st Qu.:2.00
TRUE :6	Median :4.00
	Mean :4.35
	3rd Qu.:6.25
	Max. :9.00

The use of aggregates function

Arable	Grassland	Meadow	Orchard	Scrub
5.333333	2.444444	6.333333	9.000000	5.250000

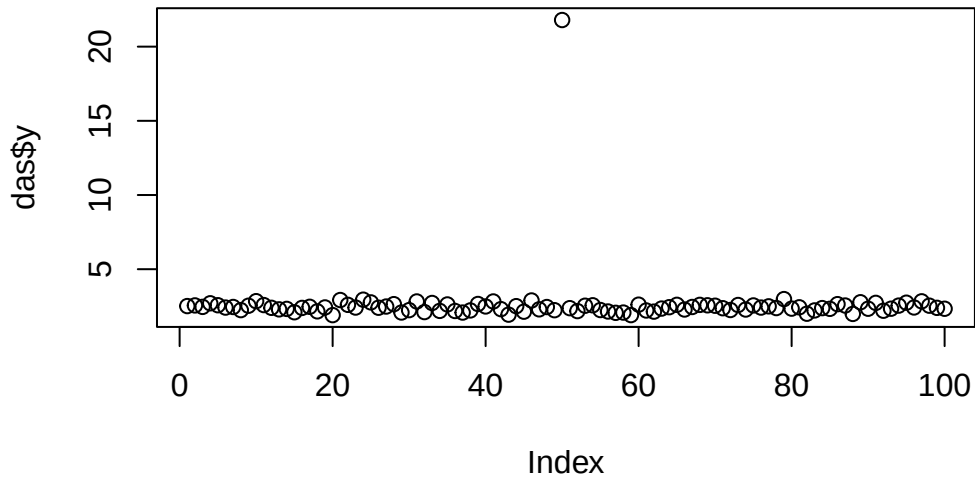
Community	Area	Slope	Soil.pH	Worm.density
1 Arable	3.866667	1.333333	4.833333	5.333333
2 Grassland	2.911111	3.666667	4.100000	2.444444
3 Meadow	3.466667	1.666667	4.933333	6.333333
4 Orchard	1.900000	0.000000	5.700000	9.000000
5 Scrub	2.425000	7.000000	4.800000	5.250000

Multiple classification

	moisture	community	Area	Slope	Soil.pH	Worm.density
1	FALSE	Arable	3.866667	1.333333	4.833333	5.333333
2	FALSE	Grassland	3.087500	3.625000	3.987500	1.875000
3	TRUE	Grassland	1.500000	4.000000	5.000000	7.000000
4	TRUE	Meadow	3.466667	1.666667	4.933333	6.333333
5	FALSE	Orchard	1.900000	0.000000	5.700000	9.000000
6	FALSE	Scrub	3.350000	5.000000	4.700000	7.000000
7	TRUE	Scrub	1.500000	9.000000	4.900000	3.500000

Get to Know the Data

Check if any unusual data points



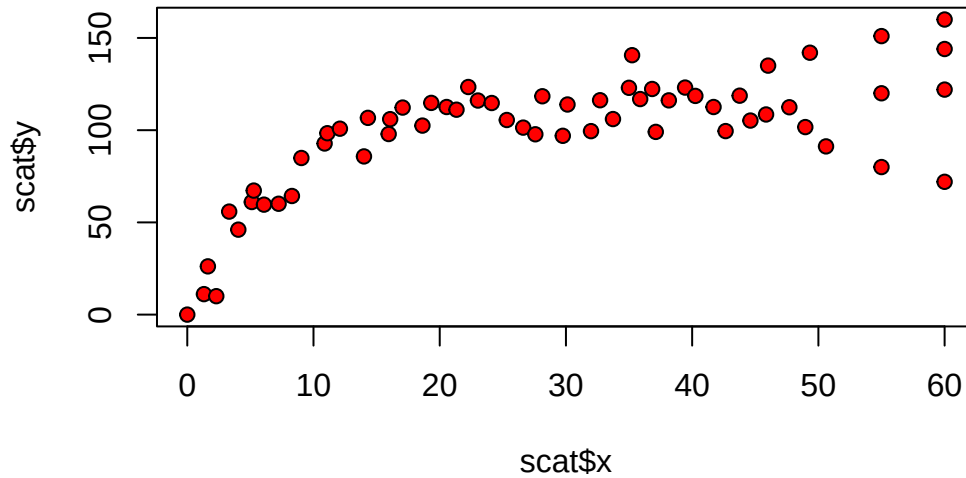
To find which data is the outlier in the above scatter plot

```
[1] 50
```

So it is the the 50 th data point.

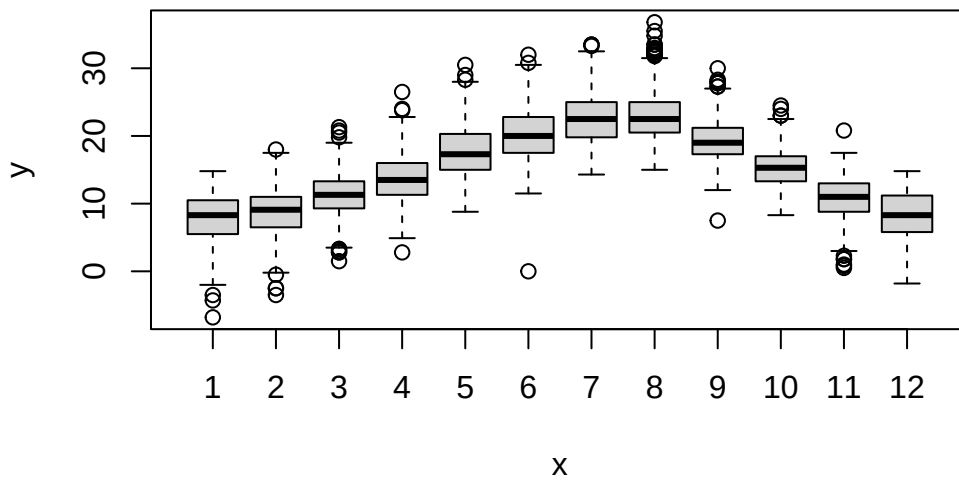
Find Relathionsip

Numeric vs Numeric Variables



From plot above we can see : 1. The relationship between response and explanatory variables is curved 2. Degree of scatter from left to right increases (non heterogenous), thus heteroscedasticity.

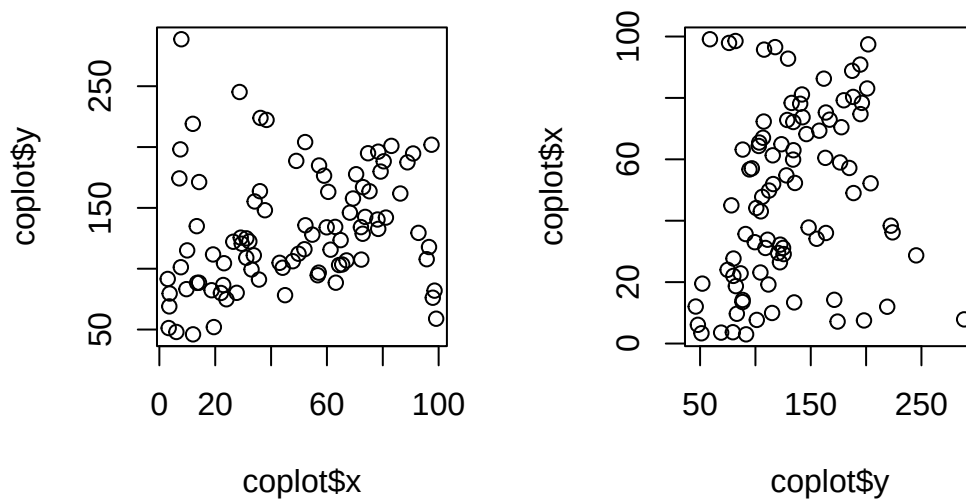
Categorical vs Numeric Variables



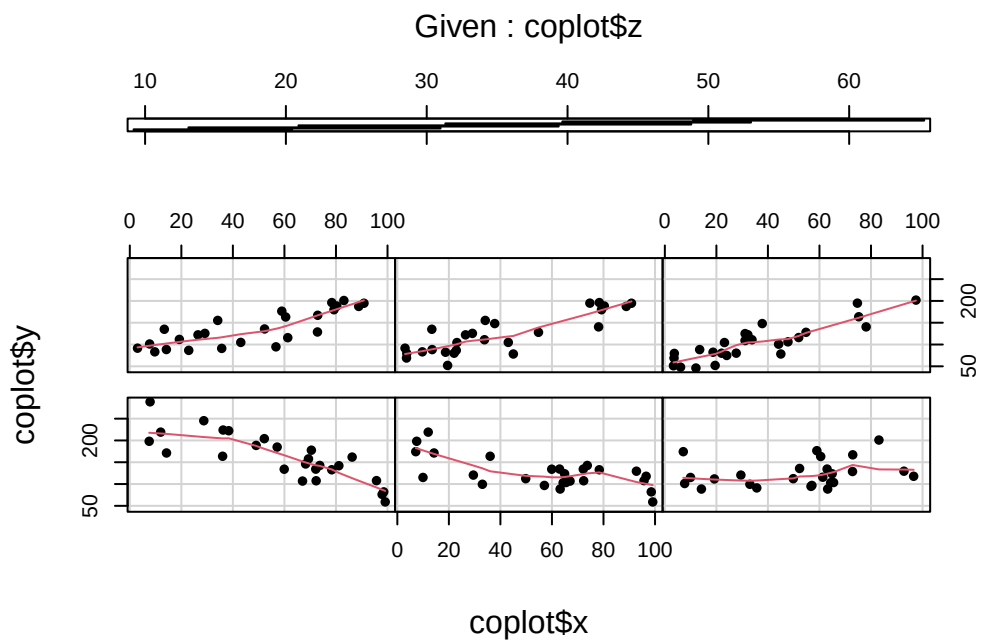
We see an overall pattern of temperature in a year. We see an outlier at month-6, it turns out a missing data that inputted as zero.

making Conditional Plot (COPLLOT)

Alter graphics parameter to specify two sets of axis on the same row then make a simple scatter plot

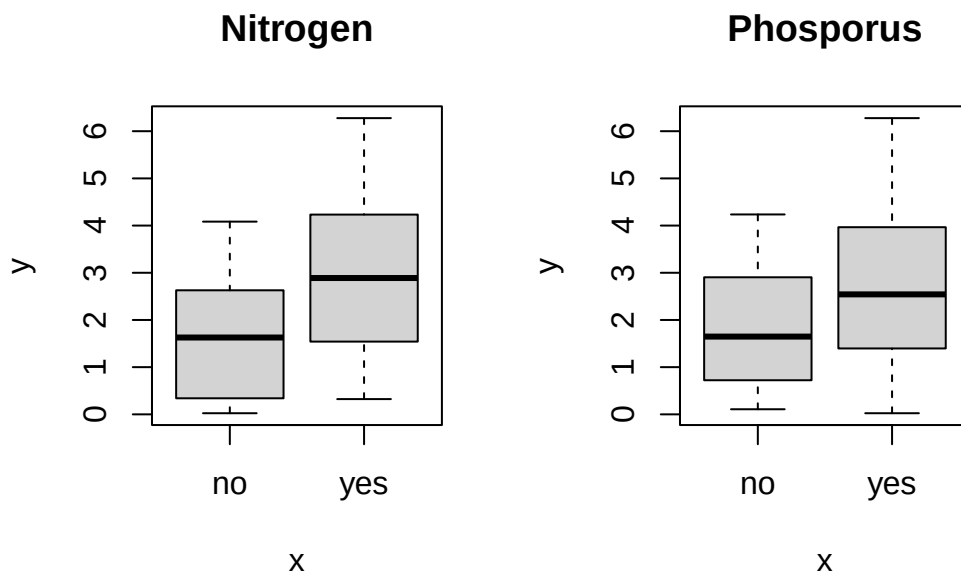


To look for interaction between explanatory variables and response variables, we employ coplot. It plot Y against X conditional on the value of Z. Z is numeric variables, by default the coplot split Z into six graphs, with the lowest value appear in the bottom left-hand.



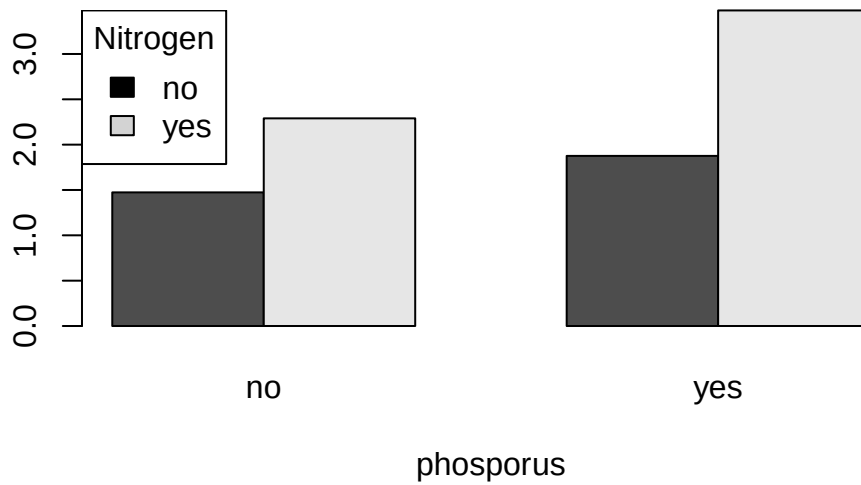
The relationship of X and Y change according to Z, as increasing Z the trend comes positive.

Interaction in Categorical Data



Finding mean for each groups

	no	yes
no	1.47384	1.875928
yes	2.28999	3.480184



Nitrogen only, gives (effect sizes) 2.29/1.47 (1.55) while nitrogen + phosphorus gives 3.48/1.87 (1.86) increase in yield. Thus effect size of nitrogen increase depends on phosphorus. This is called statistical interaction.

Central Tendency

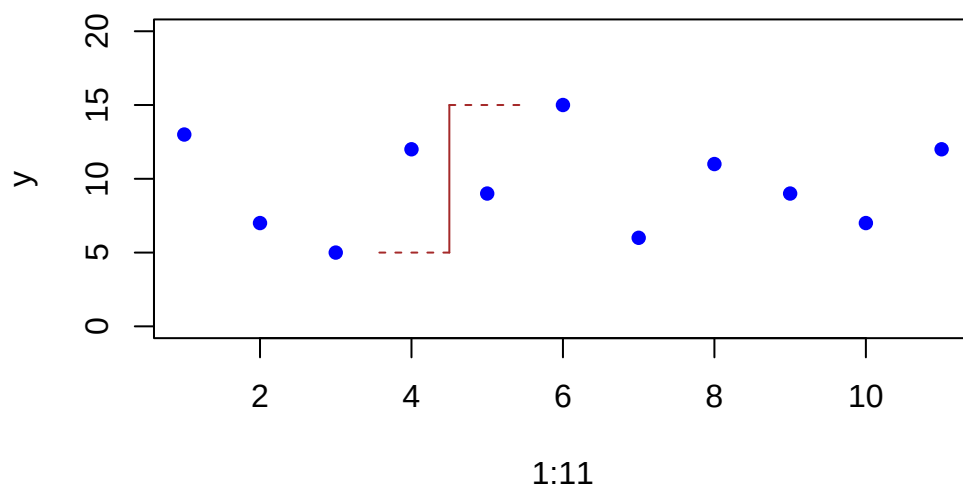
Variance

A measure of variability is perhaps the most important quantity in statistical analysis. The greater variability ~ the greater the uncertainty, the harder to distinguish competing hypothesis.

Illustration of Variance

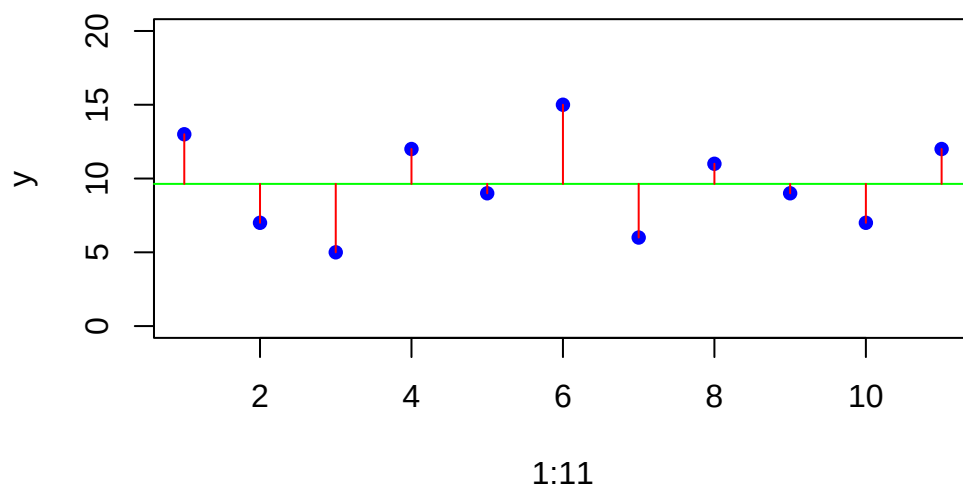
Just learn how to make the graph

Finding Range



[1] 5 15

Illustration of Mean Value



When Variance are different, dont compare the means.

Single-Samples

- what is the mean value?
- is it different from current expectation or theory?
- what is the estimate?

Two-Samples

Regression

Analysis of Variance

Analysis of Covariance (ANCOVA)

Ancova involves a combination of regression and analysis of variance. The response variable is a continous, and there is at least one continous explanatory variable & at least one categorical variable.

Principle of parsimony If the simpler model does not explain significantly less of the variation in the response, then the simpler model preferred. Test explanatory power: Anova or AIC

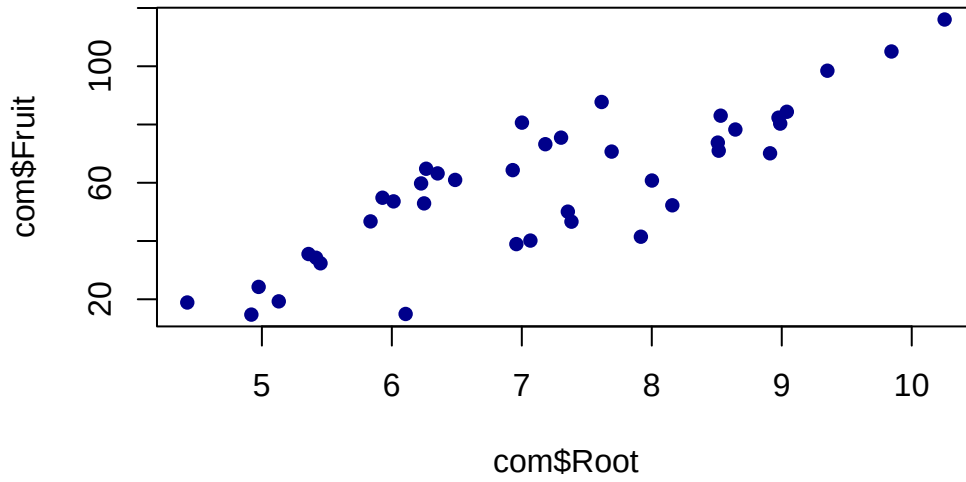
- anova: retain the more complicated model,
- AIC: prefer model with lower value.

Illustration: response variable: fruit explanatory variable: root size and grazing

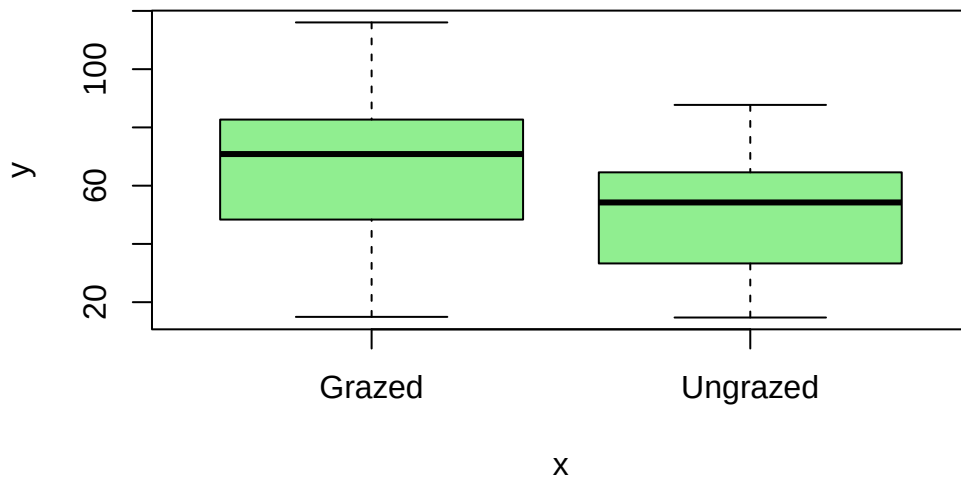
```
[1] "Root"      "Fruit"      "Grazing"
```

```
'data.frame':  40 obs. of  3 variables:
 $ Root      : num  6.22 6.49 4.92 5.13 5.42 ...
 $ Fruit     : num  59.8 61 14.7 19.3 34.2 ...
 $ Grazing: chr   "Ungrazed" "Ungrazed" "Ungrazed" "Ungrazed" ...
```

First lets see the data



Bigger roots produced more seeds



Grazed Ungrazed

67.9405 50.8805

It seems that the grazed plants produced more fruits. Lets do the ANOVA

```
      Df Sum Sq Mean Sq F value Pr(>F)
com$Grazing 1    2910   2910.4    5.309 0.0268 *
Residuals  38   20833    548.2
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

It is indeed significant from ANOVA.

Lets do the ANCOVA, we use the most complex model, two intercept and two slopes for the grazed and ungrazed (use asterisk*). in ANCOVA, order of the explanatory variables matter.

ROOT first

```
      Df Sum Sq Mean Sq F value    Pr(>F)
com$Root      1  16795   16795 359.968 < 2e-16 ***
com$Grazing    1   5264    5264 112.832 1.21e-12 ***
com$Root:com$Grazing 1     5      5  0.103    0.75
Residuals     36   1680      47
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Grazing first

```
      Df Sum Sq Mean Sq F value    Pr(>F)
com$Grazing    1   2910    2910  62.380 2.26e-09 ***
com$Root      1  19149   19149 410.420 < 2e-16 ***
com$Grazing:com$Root 1     5      5  0.103    0.75
Residuals     36   1680      47
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both produces similar error sum of square (1680) and interaction sum of squares (5). The regression sum of square where higher when root fitted after grazing (19149) then before grazing (16795) due to non-orthogonal data.

The SSRdiff, representing differences in slope between grazed and ungrazed treatments appear insignificant, thus we can remove it. Then we fit difference intercepts for grazed and ungrazed plants but fit the same slope to both graphs (we use + sign in formula):

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
com\$Grazing	1	2910	2910	63.93	1.4e-09 ***
com\$Root	1	19149	19149	420.62	< 2e-16 ***
Residuals	37	1684	46		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Does the simpler model have significantly lower explanatory power? we use anova:

Analysis of Variance Table

Model 1: com\$Fruit ~ com\$Grazing * com\$Root

Model 2: com\$Fruit ~ com\$Grazing + com\$Root

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	36	1679.7				
2	37	1684.5	-1	-4.8122	0.1031	0.75

the simpler model does not produce lower explanatory power ($p = 0,75$), thus we can adopt it.

if we see the linear model of model2

Call:

lm(formula = com\$Fruit ~ com\$Grazing + com\$Root)

Residuals:

	Min	1Q	Median	3Q	Max
	-17.1920	-2.8224	0.3223	3.9144	17.3290

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-127.829	9.664	-13.23	1.35e-15 ***
com\$GrazingUngrazed	36.103	3.357	10.75	6.11e-13 ***
com\$Root	23.560	1.149	20.51	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.747 on 37 degrees of freedom

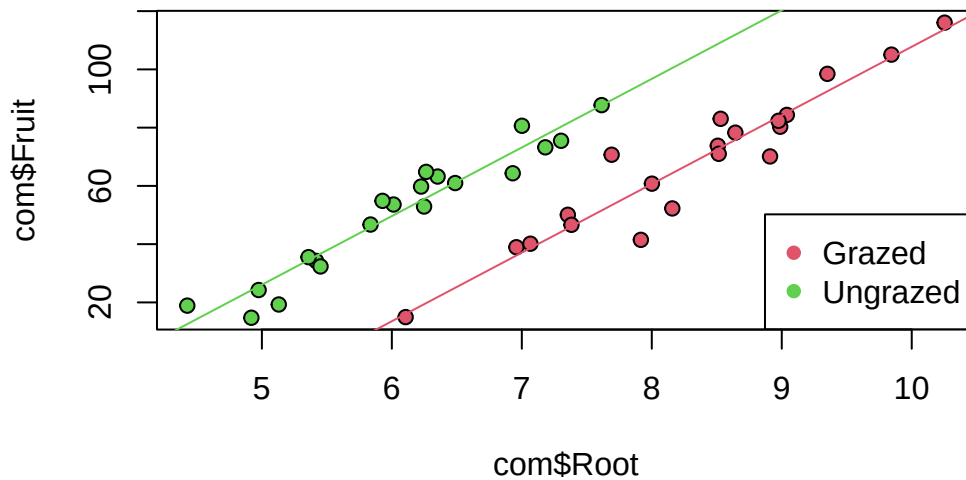
Multiple R-squared: 0.9291, Adjusted R-squared: 0.9252

F-statistic: 242.3 on 2 and 37 DF, p-value: < 2.2e-16

The model has high explanatory variables, accounting for 93% variation (multiple r squared). The intercept (-127.8) is the intercept for the graph of fruit production against plant rootstock size for the grazing variable which the factor level whose come first (in this case grazed).

```
[1] "Grazed"    "Ungrazed"
```

The $\text{comGrazingUngrazed}$ is the difference in intercept for the ungrazed plants ($-127.8 + 36.1 = -91.726$). The comRoot is the slope, the gradient of fruit production against initial rootstock size. It is same for both grazed or ungrazed, if it is difference it will showed in the fourth row the difference between slopes.



This showed that, the Grazed plants had relatively larger rootstock size then ungrazed plants, thus explain why in the boxplot of fruit ~ grazing showed that grazed plants produce more fruits while it actually reduce fruit production.

Multiple Regression

Contrast

Other Response Variables

Count Data

Proportion Data

Binary Response Variable

Death and Failure

35 Basic R and Base Plot

36 Tidyverse & ggplot2

Part V

Mata Kuliah Dasar Umum

Mata kuliah disampaikan kepada seluruh PPDS pada bulan pertama semester satu.

Materi terdiri dari 5 bidang:

1. Filsafat Ilmu
2. Biologi Molekuler²
3. Statistik Dasar
4. Metode Penelitian
5. Evidence-based Medicine

37 Biologi Molekular

37.1 Cell Structure & Organel

Tujuan: memahami teknik teknik biologi molekuler untuk kepentingan klinis.

Sel: bagian terkecil fungsional penyusun organisme

Prokaryotic - ukuran diameter 0.2 - 2.0 mikrometer - no nuclear membrane - DNA sirkuler - etc

Eukaryotic - ukuran diameter 10 - 100 mikrometer - ada nuclear membrane - DNA dalam kromosom - etc

37.1.1 Organel Sel

1. Inti Sel

terdapat kromosom, mengandung DNA, protein histone dan non histone.

Central Dogma: DNA ditranskripsi menjadi mRNA DNA direplikasi menjadi kopi DNA lain

mRNA akan di translasi menjadi Protein,

Jenis RNA: mRNA, tRNA, rRNA, miRNA, siRNA

2. Membran sel

Terdiri dari Fosfolipid bilayer, protein integral & perifer,

3. Sitoplasma

4. Ribosom subunit kecil dan besar bersatu menjadi kompleks. Ribosom bisa menempel pada RE kasar.

5. retikulum endoplasma

6. mitokondria

7. Badan Golgi

8. Lisosom

9. Peroxisom

37.2 Genome Organization

Genome => chromosome => Chromatin => Gene

Gen vs DNA

- Gen = unit fungsional dan struktural hereditas
- DNA = Struktur template RNA

Ukuran panjang DNA nukleus => 3200 million base pair ukuran panjang DNA mitokondria
=> 16.6 kilo base pair

PhantomJS not found. You can install it with `webshot::install_phantomjs()`. If it is installed

pol 1 mRNA pol 2 lain lain pol 3 rRNA

37.2.1 Protein coding genes

- Genes memiliki exon dan intron, ukuran bervariasi
- ada genes yang tidak memiliki intron => gene coding histone
- gene yang berukuran besar, pengkode dystropin, 99% mengandung intron
- Gen yang sering di transkripsi biasanya memiliki intron yang pendek

RNA Coding Sequence DNA => Enhancer – Promoter —————

terminator

transkripsi

pre-RNA => 5'cap – leader – exon-intron-exon-intron-exon-intron-exon-intron—trailer— poly
A tail 3' cap dan poly A tail ditambah post transkripsi

mature mRNA 5'cap – leader – exon—trailer— poly A tail 3'

translasi di sitoplasma

mRNA => monosistronik = 1 untai RNA menyandi 1 gen => manusia, polisistronik di
prokaryotic

37.2.2 Codon

1 kodon terdiri dari 3 basa. terdapat 64 kodon, termasuk 3 stop kodon Lihat tabel kodon

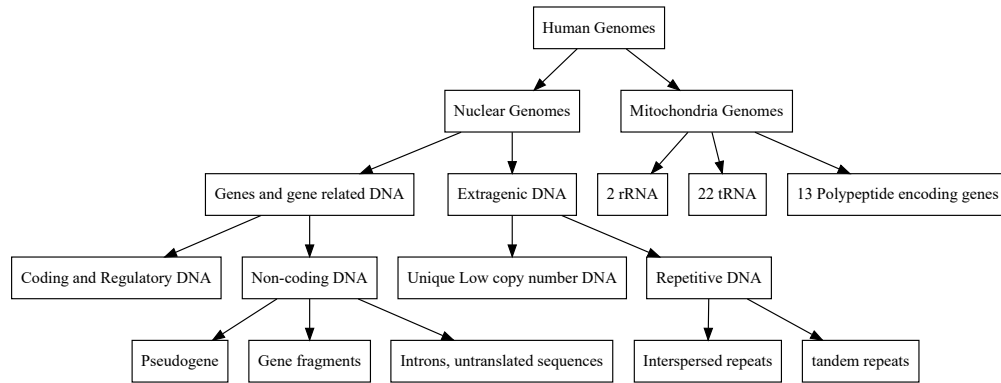


Figure 37.1: Human Genomes

37.3 DNA & RNA Structure

Central dogma.....

37.4 Signal Transduction

Komponen membran Sel:

- Phospholipid bilayer bersifat amphipathic yaitu ada sisi hidrofobik dan sisi hidrofilik. Komponen terdiri dari
 - Phosphate Head
 - Glycerol Backbone
 - Fatty Acid Tails
- Protein:
 - Integral protein, melewati 2 lapis membran
 - Peripheral protein, menempel hanya pada 1 lapis membran

Sifat fosfolipid semipermeable, yakni permeable terhadap zat selektif. - Small non polar, mudah melewati membran (CO₂, O₂) - Small polar, bisa melewati membran tapi lambat (ethanol) - Large Non-polar - Large Polar - Charged

Komponen sel signalling

1. Signal
 2. Reseptor Sensing the signal. Terjadi perubahan struktur reseptor saat berinteraksi dgn signal
 - 2 tipe reseptor
 - Intraseluler/cytosolic receptor eg. steroid hormone
 - ekstraseluler/transmembrane/cell surface membrane receptor eg. polypeptide hormone (insulin, glukagon).
- Tipe transmembrane reseptor:
- * Ion channel linked receptor
 - * G Protein Linked Receptor, aktivasi protein meningkatkan konsentrasi second messenger, terjadi proses amplifikasi signal.
 - * Enzyme Linked Receptor, signal mengaktivasi aktivitas enzim yang juga menjadi reseptor. contoh : tyrosine kinase, phospholipase C memotong PIP₂ menjadi IP₃ dan DAG

to continued.....

37.5 Cell Cycle

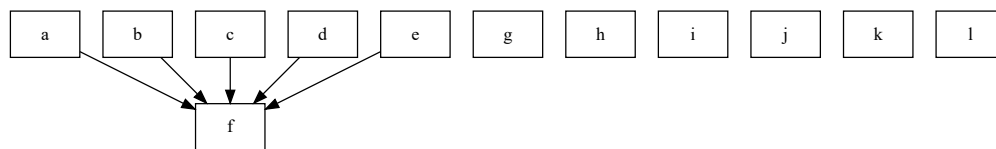
Siklus sel terdiri dari 2 tahap besar yaitu replikasi DNA dan segregasi sel. Siklus sell terdiri dari 4 fase: G1, S, G2 dan M. Proses siklus sel bukan hanya pembelahan, tetapi juga terkait dengan pertumbuhan sel dan proses lainnya.

Proses melibatkan poerubahan chromatin dari closed conformation (heterochromatin) yang sulit mengalami transkripsi (represed) menjadi Euchromatin (open chromatin) atau mudah transkripsi (activation)

Terdapat beberapa checkpoint siklus sel. checkpoints ukuran sel (protein dll, massa ribosomal), DNA damage. DNA damage bisa akibat:

- faktor intrinsik
- faktor ekstrinsik
- Kontrol dari protein: p53 chk1

Faktor lain berpengaruh terhadap siklus sel - mutasi faktor checkpoint - epigenetic - aktivitas histon - smallDNA atau smallRNA, panjang 10-25 nucleotida, berfungsi mengenali transkrip untuk memotong mRNA yang sesuai.



37.5.1 G1

Fase Gap 1, menentukan apakah pasien akan masuk ke tahap pembelahan. Jika ada kondisi yang tidak optimal, maka sel tidak akan masuk ke pembelahan. Durasi paling panjang ada pada G1 (12-15 hrs)

Ada peran dari fosforilasi protein retinoblastoma (RB) oleh protein CDKs (cyclin dependent kinases).

Cyclin berikatan dengan CDKS, mengaktifkan CDK, CDK lalu melakukan fosforilasi enzim enzim terkait. CDK adalah kelompok serine/threonine protein kinases.

CDK 4/6 dan cyclin D dari unphosphorylated pRB menjadi hypofosforilasi pRb, oleh CDK4/6-cyclinD CDK 2 - cyclin E (post Replication point)

Perain penting protein p53 dan Mdm2 pada G1. P53 penting juga pada fase G2 jika ada yang terlewat pada fase G1. p53 berfungsi posttranslational modification, termasuk fosforilasi.

37.5.2 S

durasi 6-8 jam

CDK 2 dan cyclin A dari hypo menjadi hyperforilasi pRb, oleh CDK2-cyclinA

37.5.3 G2

Durasi 3-5 jam

CDC2 - cyclin A hyperforilasi pRb

37.5.4 M

Durasi 1 jam CDC2 - cyclin B hyperforilasi pRb

post M, defosforilasi pRB

37.6 Mutastion & Free Radical & Antioksidan

Mutasi basa: substitusi, delesi, insensi.

37.6.1 Substitusi biasanya terjadi 1 basa saja, sehingga sering juga disebut point mutation.

Substitusi bisa transversi menjadi transisi. Transisi: purin > purin Transversi: purin > pirimidin atau sebaliknya.

Hasil preotein: Silent mutation: asam amino sama Missense mutation: berubah kode asam amino Nonsense mutation: berubah menjadi stop kodon stop codon mutation: merubah stop kodon menjadi kodon asam amino.

37.6.2 Delesi dan Insersi

Delesi dan insersi mengubah reading frame, sehingga disebut dengan frameshift mutation.

37.6.3

37.6.4 Molecular effect of Mutation

Within coding region - Silent (biasanya mutatasi pada basa ketiga di kodon, sering tidak mengganti asam amino) - Missense - Non-sense (premature stop codon) - Frameshift

Within non-coding region - SPlicing mutation - Regulatory Sequence mutation

37.6.5 Clinical Effect of Mutation

- Acceptable
- Partially acceptable
- Unacceptable (fatal/letal)

bisa menimbulkan kanker, inborn error metabolism.

Kanker muncul: - altered protein

37.6.6 DNA Repair

37.6.6.1 SPontaneous Mutation

- Mismatch Repair DNA kita secara natural ada metilasi, jika belum termetilasi setelah replikasi maka akan di excisi oleh endonuclease.

37.6.6.2 Metagen

- Photoreactivation Timin Dimer, dipotong pada dimernya lalu dilakukan excision repair.

Tipe Repair - Base excision repor - Nucleotide excision repair.

37.6.7 Free Radical

Molekul yang memiliki elektron yang tidak berpasangan di orbit terluarnya. Radikal bebas tidak stabil, sehingga mencuri elektron dari molekul lain, dan menciptakan radikal bebas lain.

Oksidant: elektron acceptor, $\text{Fe}^{3+} + e \Rightarrow \text{Fe}^{2+}$

Sifat: 1. Highly reactive, 2. Unstable 3. Very short half-life 4. Generate new radicals by chain reaction 5. Cause damage to biomolecules, cells

37.6.7.1 Reactive Oxygen Species (ROS)

- H_2O_2

ada juga Reactive nitrogen species

Sulit diperiksa secara langsung. Pemeriksaan secara tidak langsung dengan pemeriksaan produk dari hasil paparan radikal bebas.

Sumber radikal bebas: - Mitokondria pada proses pembentukan ATP,

37.6.7.2 Sumber Radikal Bebas

- Endogen
- EKsogen

37.7 DNA Replication

Replikasi DNA terjadi pada fase S pada siklus sel bertujuan membuat copy DNA untuk diwarikan ke sel keturunanya.

37.8 Mutation & DNA Repair

37.9 Gene Expression

Jumlah gene sel manusia ada sekitar 20 rb - 30 rb. Setiap sel somatik sama jumlahnya, yang berbeda adalah aktivasinya pada masing-masing tipe sel.

Proses transkripsi: ada inisiasi, elongasi dan terminasi. Proses translasi: iniasi, elongasi, terminasi

37.9.1 Transkripsi

37.9.1.1 Inisiasi

RNA Pol 1, RNA pol 2, RNA pol 3.

Promoter mengenali sekuens DNA spesifik yang disebut responsive element dalam proses inisiasi transkripsi. Ada kompleks TATAA Box. Kompleks promotor menarik aparatus faktor faktor yang membentu proses transkripsi. Selain promotor adalah enhancer. Enhancer terletak lebih jauh dari gene terkait, baik upstream atau downstream yang membentuk sebuah DNA loop berikatan dengan kompleks inisiasi.

37.9.1.2 Elongasi

37.9.1.3 Terminasi

Terminasi terjadi karena pemotongan strand RNA pada sekuens tertentu. RNA kemudian ditambah poly A tail untuk meningkatkan stabilitas. Makin panjang Poly A tail, makin stabil.

37.9.1.4 Post transcriptional Step in Gene Expression.

1. mature RNA megalami splicing
2. Penambahan 5' end cap, menjadi 7 mentilguanosin cap
3. Penambahan 3' poli adenin tail

Terjadi regulasi paska transkripsi melibatkan small/micro RNA.

37.9.2 Translasi

37.9.3 Regulasi Transkripsi

Regulasi mRNA tergantung 1. kestabilan mRNA 2. pasangan mirna (RNAi) 3. keadaan sel 4. Metilasi DNA (pada sitosin)

Transcription control of genes transcriptional regulation occurs at two interconnected levels - Histon complex - **histon COde** -

37.10 Control of Gene Expression

37.11 Genetic Variation

Microevolution is a change in allele frequencies in a population over generations. Three mechanism:

- Natural Selection
- Genetic Drift
- Gene flow

only natural selection causes adaptive evolution. Hardy - Winberg Equilibrium: genetic variation will remain constant from one generation to the next in the absence of disturbing factors.

37.12 Genotyping

Process determining the genetic constitution of an individual and comparing to the other individual or reference sequence. Perbedaan interspecies <1%.

Genotyping pada manusia untuk mengetahui susceptibilitas atas penyakit dan response terhadap pengobatan.

37.12.1 Polimorfisme vs Mutasi

Perubahan basa DNA, mutasi perubahan sekuens dari normal. Polimorfisme adalah perubahan basa yang merupakan variasi normal. Cut-off point antara mutasi dan polimorfisme adalah 1%, artinya variasi alele setidaknya memiliki frekuensi minimal 1% di populasi.

Polimorfisme bisa berupa silent point mutation.

37.12.1.1 SNP

Variation when a single nucleotide in genome is altered. SNP do not always affect the way protein functions.

- Linked SNP/Indicative SNP : do not reside within genes, do not affect protein function, but correspond to certain characteristics
- Causative SNP: affect the way protein functions, whether:
 - Coding SNP: located in coding region, changes the amino acid sequence
 - Non-coding SNP: within gene regulatory sequences: change timing and expression of gene.

37.13 Stem Cell

Sel yang belum terspesialisasi.

- Embryonic Stem Cells (inner cells mass)

37.14 Molecular Biology Technique

DNA universal

37.15 Application of Molecular Biology in Medicine

38 Metodologi Penelitian

38.1 Overview of Clinical Research

clinical research is medical research that involves people, to learn more about disease and improve health care for people in the future.

include: epidemiology, behavioral, health services, clinical trials.

clinical research, not just clinical trial, it is any research design that studies humans or any materials taken from humans. it may even include animal studies that the results apply to humans.

type of clinical research:

- observational studies => studies that aim to identify and analyze patterns in medical data.
- clinical trials/interventional studies

Research Question: FINER - Feasible - interesting - novel - ethical - relevant

38.2 Association & Causation

Sebuah hubungan belum tentu menunjukkan sebab-akibat. Banyak peneliti menghindari dogmatic statement: "cause" or "produces"

Causation: occurs when a change in one variable produces a change in another variable

- Contributory Cause, no single specific cause of a disease. Most of disease caused by multifactorial etiologies. A risk factor may be a contributory cause.

Three criteria: 1. Association may be reflected by effect size 2. Prior Association, does the cause (independent var) precede the effects (dependent var) (case control or cohort), 3. Altering the cause alters the effects

Some other features of contributory causation:

1. Consistency: the results of study (effect size) should be consistent regardless the design, the population used.

2. Dose response relationship
 3. Biologic Plausibility: should make biologic sense.
 4. Experimental confirmation: phenomenon found in clinical studies have to be consistent with studies in animal models.
- Necessary cause, dibuktikan sebagai satu satunya penyebab. Sebagai contoh: fulfilled by Koch's Postulates: the organism is always found with the disease.

Association: can be observed between many variable, but causation only be determined from data collected from a controlled experiment.

- Spurious association, association not actually present in population 3 mekanisme spurious causation
 - Chance, due to random error and sampling variation (Individual chance to be a sample not equal), karakteristik sampel terlalu heterogen. Minimize random error: adequate sample size and low type 1 and type 2 error. Effect of random error can be judged from the width of the 95% confidence interval and the consistency of the results with previous evidence.
 - Design, conduct of the study, and analyze of the results,
 - * selection bias, pemilihan sampel yg tidak sesuai => randomized and blind intervention
 - * performance bias, pemberian intervensi tidak setara antar sampel => randomized and blind intervention
 - * Detection bias, deteksi efek atau cause tidak sama => randomized and blind intervention
 - * Transfer bias, loss to follow up
 - Confounding, occurs when factors which related to both independent and dependent variables. Contoh Kopi menyebabkan AMI, tetapi ada faktor merokok dimana orang ngopi juga suka merokok. Ternyata merokok yang berhubungan dgn AMI, tetapi merokok tidak berhubungan dengan AMI. Kontrol dengan by design atau by analysis.
- Real Association, true biological phenomena. However not due to a cause effect relationship

Critical Appraisal about Harm/Etiology/Causation. follow checklist.

PR: 1 jenis penelitian causation atau association yang menjadi ide kalian nanti penelitian.

38.3 Research Measurement

38.4 Variables and Relationship Between Variables

Variable: independent, dependent, confounding variables

other variable: mediator, moderator, external, random

relationships among variables

- chances
- bias
- confounding
- causal

38.5 Hypothesis and Hypothesis Testing & Effect Size, Confidence Interval, and P Value

Hypothesis:

a statement about relationship between two or more variables that suggest an answer to the research question, is declarative statement that predicts an expected outcome.

Research hypothesis: kalimat deklaratif yang mencoba memberikan jawaban sementara atas pertanyaan penelitian.

Statistical hypothesis: biasanya tidak ditulis di proposal.

Research hypothesis nanti akan di uji apakah hypothesis didukung oleh data data.

Jika hypothesis yang kita uji tidak sesuai dengan data, maka bukan berarti ilmu itu salah

38.6 Sample Size and Power

Sample size pentingnya untuk bisa mewakili populasi dan dapat inferensi ke populasi itu kembali, serta:

- untuk mendeteksi treatment effects jika memang ada di populasi (mendapatkan hasil statistik signifikan dan secara klinis penting)
- Presisi estimasi
- avoid wasting resources
- avoid misleading conclusion

Two many subject proves everything, too few subject proves nothing

Kapan ketika perhitungan sample size tidak diperlukan? - Penelitian kualitatif - Pilot studi, biasanya 10 -20 subjek untuk menentukan sample size studi yang lebih besar.

Sample size sesuai dengan design, analysis plan dan outcome. Sample size harus sesuai untuk memenuhi statistical significance dan clinically important, tanpa membuang sumber daya terlalu banyak.

Sample size berhubungan power. Power = makin kuat penelitian mampu mendeteksi sebuah efek di populasi, maka semakin banyak sample size dibutuhkan.

Definition of Power: the probability of concluding the new treatment is effective if it truly is effective. Biasanya 80% type I error = the probability of concluding that the new treatment is effective if it truly is not effective. (positive palsu, α) type II error = the probability of concluding the new treatment not effective if it is truly effective (negative palsu, $1 - \text{power}$ atau β)

Elemen menentukan sample size: Δ = **effect size** or difference under null hypothesis (perbedaan klinis yang dianggap penting) diambil dari studi sebelumnya dan tinjauan pustaka. Jika effect size besar, maka sampel di butuhkan lebih kecil, begitu kebalikannya α = type I error, 0,05 β = type 2 error, 0,2 σ = standard deviation, makin heterogen populasi maka sample size makin besar. r = ratio of number of patients in the two groups, usually $r = 1$

Hubungan sampel size dengan hasil presisi, dimana hasil confidence interval akan lebih sempit (lebih presisi) jika sample size besar. Jika hipotesis 2 arah, sample lebih besar, biasanya lebih disarankan hipotesis 2 arah. Pertimbangan tambahan sampel jika kemungkinan loss to follow up terutama pada kohort prospektif dan clinical trial. Jika cross sectional, tidak perlu. Jika tidak ada studi sebelumnya sebagai patokan sampel, maka tetap harus memperhitungkan sampel dengan cara pilot study.

Jika tidak bisa memenuhi jumlah sampel sesuai perhitungan sebelum penelitian (apriori), kita bisa menentukan apakah hasil studi kita underpower dengan memperhitungkan hasil tes post-hoc.

Strategi meminimalkan sample size - pakai variabel kontinu - paired measurement.

rule of thumb: 10 - 50 per variabel, atau 30 sampel. Pakai rule of thumb jika tidak ada data yang bisa mendukung perhitungan jumlah sampel.

Effect size tidak boleh mengutip dari penelitian lain secara *eksplisit* di proposal.

38.7 Research Design

Jenis penelitian digunakan untuk membuktikan tujuan penelitian.

Sebainya tidak perlu dilakukan analisis untuk mendapatkan nilai P untuk membandingkan karakteristik sampel antar 2 kelompok sebagai usaha untuk menunjukan 2 kelompok tersebut komparabel, jika kita tidak menghipotesiskan seperti itu.

Desain:

38.7.1 Cross sectional

Pengukuran variabel bebas dan tergantung terjadi bersamaan. Baik untuk penyakit yang durasi sakitnya lama, dan kejadian cukup banyak.

38.7.2 Kasus-Kontrol

Identifikasi sampel dengan variabel tergantung dulu, kemudian dicari kontrolnya, lalu dilihat secara retrospektif mencari variabel bebasnya. Lalu dihitung rasio odds. Odds: Perbandingan terjadinya efek dibandingkan tidak terjadinya efek pada suatu kelompok Rasio Odds: perbandingan odds antar 2 kelompok.

Matching case control:

Variabel bebas bisa banyak

38.7.3 kohort

Mengidentifikasi variabel bebas, lalu diobsersevari untuk melihat variabel tergantung. Baik untuk kasus yang jarang, dan perjalananya penyakit jarang.

Variabel tergantung bisa banyak

38.7.4 Nested Case-Cohort Study atau Nested Control Study

Umumnya penelitian yang sampelnya bisa disimpan di Lab dan pemeriksaannya mahal. Controlnya: di matching => case cohort study Controlnya: di Random => case control study

38.7.5 Uji Diagnostik

Selalu tabel 2 x 2.

38.8 Cross Sectional & Case-Control Study

38.8.1 Cross-sectional Study

Cross sectional study assessing prevalence (old and new patient).

1. Formulate appropriate research question and hypothesis
2. Identify independent and dependent variables
3. Determine research subjects
4. Carry out measurements
5. Do analysis

Tabel 2x2 harus risk factor di kiri, effect di atas.

38.8.2 Case-Control Study

38.9 Cohort Study

38.10 Experiments

38.11 Diagnostic Test

38.12 Data Management and Analysis

38.13 Research Proposal

38.14 Ethical Issue

38.15 Novelty Aspect

39 Filsafat Ilmu

39.1 Pengantar Filsafat Ilmu

Filsafat ilmu: untuk mencari tujuan hidup.

Ontology : study of existence, becoming and reality, mengetahui segala sesuatu yang ada sebagaimana adanya. **Mencari tahu tentang hal terkait.** **Epistemology** : Study of knowledge, bagaimana kita bisa mendapatkan pengetahuan yang hakiki itu atau sesuatu yang ada sebagai mana adanya. **Mengetahui semua hal untuk mencapai sesuatu/how to achieve it.**

Deduktif: Mencari ke arah spesifik

Induktif: Mencari teori (ke arah umum)

Axiology : Study of Value, ilmu pengetahuan yang didapat itu bisa diterapkan untuk kemahasiswaan umat atau justru sebaliknya. **bagaimana cara menggunakan ilmunya**

pengetahuan $\rightarrow$ 5 panca indra, pengetahuan bisa menjadi ilmu jika melalui ilmiah dan dapat diturunkan/semua bisa mempelajari dengan hasil yang sama.

Filsafat ilmu sebagai kumpulan teori digunakan memahami dan mereaksi dunia pemikiran.

- 39.2 Ilmu Sebagai Produk Berpikir**
- 39.3 Ilmu dan Teknologi Kedokteran**
- 39.4 Ontologi, Epistemologi, dan Aksiologi Kedokteran**
- 39.5 Intelektualitas dan Aktivitas Intelektual**
- 39.6 Pengetahuan dan Ilmu Pengetahuan**
- 39.7 Teori Kebenaran Ilmiah**
- 39.8 Perkembangan Ilmu Kedokteran**
- 39.9 Aksiologi, Sikap Ilmiah dan Etika Ilmiah**
- 39.10 Complementary & Alternative Medicine**
- 39.11 Komunikasi Ilmiah**
- 39.12 Bioetika Penelitian**
- 39.13 Publikasi dan Hak Paten**
- 39.14 Filsafat Ketuhanan dan Kemanusiaan**
- 39.15 Metode Penalaran Ilmiah**

40 Statistik Kedokteran Dasar

40.1 Dasar-dasar Penulisan Ilmiah Kedokteran dan Plagiarisme

Metode sains: bisa diulang dengan hasil yang sama. observasi, measurement, verification and evaluation (self correction)

40.1.1 Methods of acquiring knowledge

- Method of Tenacity
- Method of Authority
- Method of Intuition
- Method of Science

40.1.2 Benefits of Conducting Research

- Expanding frontiers of knowledge
- New inventions and discoveries
- Solving problems affecting the society
- Increasing efficiency and reducing costs
- Research strives to make life easier
- Luxury and comfort
- Infotainment
- Economic growth

40.1.3 Qualities of a researcher

- Scientific attitude
- Research aptitude
- Persistence
- Self Motivation
- Courage to ask questions
- Skepticism and receptivity
- Objectivity

- Industriousness
- Honesty and truthfulness
- Open-mindedness
- Above-average intelligence
- Knowledge
- Imagination
- Self-confidence
- Search for perfection
- Team spirit

40.1.4 Bioethical

- Autonomy - consent to treatment
- Beneficence
- Justice
- Utility - balancing benefit
- Inform Consent
- Confidentiality

40.1.5 Induction and Deduction in Research

- Deductive reasoning is the process of reasoning from a general assumption to a specific application. Contoh: General proposition: all crows are black Specific proposition: this bird is a crow, therefore it is black
- Induction normally involves generalization from the behavior of a few samples to that of a population contoh: all jasmine flowers are scented, if a situation or condition is true in all observed cases, then the situation or condition must be true in all cases.
- Hypothetico-deductive method or inductive-deductive method.

40.1.6 Scientific Method

It should be unprejudice

- Make observations or gather information
- Develop hypothesis
- Predicts results
- Design an experiment to falsify the hypothesis
- Conduct the experiment and collect data
- Evaluation and conclusion
- Acceptance, modification or rejection of the hypothesis

40.1.7 First Draft

- Why did you start the experiment? (introduction)
- What did you do? How was it done (The materials and methods)
- What did you find? What was learned? (the results)
- What do the results mean? How do they relate to what is already known? (the discussion)

40.1.8 Structure of a Research Paper

- Title
- Running Title
- Author
- Institution
- Abstract
- Keywords
- Introduction
- Materials and Methods
- Results
- Discussion
- Conclusion
- Acknowledgements
- References
- Tables and Figures

40.1.9 Seven Categories of plagiarism

- direct plagiarism
- mosaic plagiarism
- paraphrase plagiarism
- insufficient acknowledgement
- complete plagiarism
- self plagiarism
- accidental plagiarism

40.1.10 Characteristics of Good Research

- Research is based on the work of others
- Research is a blend of logic and imagination
- Research tries to identify and avoid bias
- Repeatability

- Research must be generalizable to other settings
- Research is systematic
- Research generates new questions
- Research is an apolitical activity

40.2 Kiat-kiat Publikasi Majalah Terakreditasi dengan OJS

40.2.1 Kendala Publikasi di Indonesia:

- Indonesia lack of language access. Why we dont make one in Bahasa Indonesia?
- Lack of Cultural backgroud dalam menulis.

40.2.2 Kenapa publikasi penting?

- Rekognisi atas hasil penelitian anda baik pengakuan regional, nasional dan internasional
- H Index
- Jaminan kolaborasi di tingkat regional, nasional, dan internasional

40.2.3 Jurnal

Jurnal biasanya di ranking berdasarkan dimana jurnal tersebut di indeksasi (misal scopus, medline, ISI, Sinta, dll)

40.2.3.1 Kriteria Jurnal International

- Tulisan memenuhi kaidah ilmiah dan etika keilmuan
- Memiliki ISSN
- Ditulis dengan bahasa resmi PBB (inggris, france, arab, rusia, and china)
- Memiliki terbitas versi online
- Dikelola secara profesional
- Editorial board pakar di bidangnya, dan berasal dari berbagai negara
- artikel ilmiah diterbitkan dalam satu isu berasal dari penulis berbagai negara
- memuat karya ilmiah dari penulis yang berasal dari berbagai negara dalam setiap penerbitannya
- Terindeks database international berputasi.

Publikasi:

- Submit ke jurnal dgn OJS
- Jika ditolak, edit ulang dan perbaiki.

Perlu diketahui: - Penulisan naskah/manuskrip ilmiah merupakan suatu tantangan, seni dan kemampuan mengelaborasi hasil penelitian. - Ikuti kaidah IMRAD. Introduction biasanya sudah menggambarkan M,R,D nya.

Author vs Behavior

- Author
 - Want to publish more, write in detail
 - peer review essential
 - other journal functions crucial
 - wider dissemination
- Reader
 - want to read less. (judul yg baik)
 - Want integrated system
 - Browsing is crucial
 - Quality information is important
 - Want to read less

Unsur Utama dalam Publikasi:

- Isue Etik + pernyataan tidak ada conflict of interest + Pencantuman sumber dana + Image manipulation guidelines
- Gaya Selingkung dan Bahasa
 - Sesuai author guidelines
- Struktur naskah/manuskrip ilmiah
 - Title: judul clear, precise, including keywords. do not use abbreviations and jargon
 - Authors Listing: only include those who have made an intellectual contribution to the research, or those who will publicly defend the data and conclusion, and who have approved the final version. Gift (atasan, dekan, kepala lab), Guest (pemberi dana), and Ghost author (ini tidak boleh, atas dasar sembarangan mengikutkan namanya). Ghost writer, penulis lain yg menawarkan jasa (secara etika diijinkan).
 - Abstract: ditulis selesai IMRAD. Tidak boleh sebelum lengkap dan menyeluruh menulis IMRAD. Bisa terjadi inkonsistensi dgn discussion.
 - Introduction: clearly state the problem being investigated, background that explains the problem, reasons for conducting the research. Summarize relevant research to provide context. State how your work differs from published work. Identify the questions you are answering.

- Methods: Jika RCT sebaiknya diregister, agar SOP penelitian bisa dibaca orang lain dan terrekognisi. Provide the readers enough details so they can understand and replicate your research. Explain how you studied the problem identify procedures, order the chronologically where possible. Include the frequency of observations, what types of data were recorded.
- Results: present the finding objectively, follow logical sequence, show figures/table and provide brief description.
- Discussion: Describe your results in the context of what was already known, how the result relate, do not extend your conclusion beyond what is directly supported by your results, outline the next step of the study.
- References: acknowledge sources, citation follow journal guideline.

- Submit artikel/pemilihan jurnal
- proses publikasi/peer review

Dulu hierarki tertinggi evidence adalah: Validitas Pada Penelitian Randomized, controlled, clinical trial.

Validitas Internal

- Ada kontrol
- Ada Randomisasi

Validitas External

- loss to follow up < 20%?

Saat ini sudah bergeser pada Systematic Review dan Metanalysis.

40.3 Effect Size, COnfidence Interval

40.4 Uji Hipotesis

Mengacu pada distribusi populasi:

Asumsi dasar H_0 ; $\mu_1 = \mu_2$ (hipotesis nol, tidak ada perbedaan) H_1 : $\mu_1 \neq \mu_2$ (hipotesis alternatif, ada perbedaan)

H_1 diterima bila terjadi kesalahan kecil: kesalahan tipe 1, false positive : dihitung peluang kesalahan akibat variasi sampel $\Rightarrow$ nilai P batas penolakan (α) 0,05; 0,01

- beta: kesalahan tipe 2, false negative: ditetapkan
- metafor: seperti persidangan

Jenis Hipotesis: - superiority testing: A lebih baik daripada B - non-inferiority: membuktikan perbedaan dalam rentang yang tak berbeda. (contoh penelitian obat biosimilar)

makna p value - chance play a role in sample value if H_0 accepted

significant level (alpha) if $p < 0.05$ or 0.01 as cut off dicari: power(1-beta) di set di awal 80% atau 90% ditentukan the answer yes or no tetapi tidak bisa melihat seberapa.... sehingga saat ini tidak hanya boleh melihat P value, karena harus melihat effect size dengan confidence interval.

menghitung confidence interval :

40.4.1 Clinician and Statistician

Nilai P tidak memberikan informasi besar perbedaan hasil (effect size) beda yang kecil saja memberikan signifikansi statistik bila N besar

beda yang besar bisa tidak memberikan signifikansi statistik bila N kecil (error type 2)

Maka perlu ukuran sampel yang sesuai, dan perlu memaknai sebuah statistical significance dengan makna klinis (clinical importance)

40.5 Korelasi Linier dan multipel

Dasar-dasar dan aplikasi biostatistika klinik

- mengestimasi kuatnya hubungan atau besarnya perbedaan (besarnya efek)
- melakukan uji kemaknaan statistik (inferensial)
- mengontrol pengaruh variable pengganggu terhadap besar efek dan inferensi

hubungan semu dapat diperoleh dari analisis bivariat, lalu multivariat untuk mengekslusi hubungan semu.

Korelasi = antara independen dan dependennya memiliki hubungan garis lurus. Korelasi ada 2, linier dan non linier. Korelasi ada fleksibilitas variabel independen dan dependennya.

Jenis analisis statistik: - Tentukan pertama parametrik atau non-parametrik - Skala variabel: kontinyu (variabel skala dan interval), ordinal, nominal, diskrit - Uji normalitas: normal/non normal (tes kolmogorov smirnov atau shapiro-wilk) - Uji kesamaan variansi: sama/tidak sama (uji levene) - Uji kolinearitas: hubungan antar variabel independen (penting untuk uji partial korelasi, regresi linier berganda)

variabel: - jika bisa dipecah menjadi 5 atau lebih, bisa dikatakan kontinyu. - jika kontinyu dipecah menjadi beberapa kategori : bisa ordinal - jika yes no= nominal - diskrit = tidak bisa dibedakan antara kategori?

Pemilihan uji statistik: 1. tentukan variabel dependen (umumnya satu variable dependen)
2. tentukan jumlah variabel independen - bila tidak ada variabel independen, statistik yang dipakai adalah analisis univariat (deskriptif) - bila ada satu variabel independen, statistik yang dipakai adalah bivariat - bila ada lebih dari satu variable independen, statistik yang dipakai adalah analisis multivariat.

Sentral limit theorem: jika lebih dari 30 bisa berdistribusi normal.

PhantomJS not found. You can install it with `webshot::install_phantomjs()`. If it is installed

var_ind	var_dep	ana_biv
kontinyu (normal)	kontinyu(normal)	korelasi pearson(rho) ; regresi linier sederhana (slope,
kontinyu(non-normal)	kontinyu (non-normal)	korelasi speramn (rho)

40.6 Regresi Linier Sederhana

.. lanjutan korelasi linier dan regresi linier sederhana dan multipel

40.6.0.1 Memilih analisis bivariat

40.6.0.1.1 Regresi Linier

asumsi regresi linier: - distribusi normal - tidak ada kolinearity - variansi asumsi sama

Regresi nilai: intercept (a), slope/tangent (bx) dan error (e) Intercept: nilai Y saat X = 0
slope: perubahan rerata variabel dependen untuk setiap unit perubahan variabel independen

$Y_i = a + bX + e_i$ (a atau b0 dan b1 adalah effect size) dan p value

40.6.0.1.2 Korelasi Linier

Mengukur seberapa konsisten perubahan variabel indepen diikuti oleh perubahan variabel dependen (hubungan kovariansi) koefisien korelasi (r), menunjukkan kekuatan hubungan antara variabel dependen dan variable independen.

ada 2 nilai: (rentang nilai -1 sd +1) - koefisien korelasi (rho), Ada kecenderungan hubungan x dengan y, tetapi tidak menunjukkan seberapa kuat. - koefisien determinasi (r atau R^2), seberapa persen pengaruh independen mempengaruhi dependen (effect size) - tambahan p value

contoh R^2 adalah R^2 45% pada konsumsi garam dan tekanan diastolik adalah variasi tekanan darah dapat dijelaskan sebesar 45% oleh konsumsi garam, sisanya 55% dijelaskan oleh factor lain. Nilai ini yang dimaknai oleh klinisi serta signifikansi korelasi (p) (kemungkinan hasil korelasi akibat by chance).

tambah interpretasi regresi: a (intercept) = jika seorang tidak konsumsi garam berapa tekanan darahnya b (slope) = kenaikan tekanan darah setiap unit peningkatan konsumsi garam.

Perbedaan korelasi dan regresi adalah korelasi belum bisa menentukan ind dan dependen, sedangkan regresi harus sudah ditentukan mana independen dan dependent.

Visualisasi dengan scatter plot.

40.6.0.1.3 Regresi dan korelasi Linier Berganda

Multipel variabel independen melihat pengaruhnya terhadap variabel dependen.

Jika menggunakan korelasi maka nilainya rho partial (korelasi partial), jika regresi multipel (regresi berganda) maka slopenya multipel ($a + b_0 + b_1 + b_x \dots + e$)

Nisal: peneliti ingin mengetahui antara LFG (ml/menit) dengan estimasi data IMT (kg/m^2), umur (tahun), dan TDD (mmhg), semua variabel independen berskala kontinyu.

$$\text{lfg} = 72,74 + 3,11 (\text{IMT}) - 1,15 (\text{umur}) - 0,23 (\text{tdd}).$$

$R^2 = 0,35$, maka 3 variabel independen berperan 35% menjelaskan variabel dependen.

Uji hipotesis: hipotesis nol omnibus = kesalahan tipe 1 (α) merupakan hipotesis nol, artinya keseluruhan var indepen tidak dapat mengestimasi var dependen. Menguji $B=0$.

Koefisien regresi (β) pada persamaan regresi $\Rightarrow$ hubungan antara variabel masing masing independen dan variabel dependen.

40.6.0.1.3.1 Asas Kolinearitas

Perhatikan nilai tolerance dan VIF (variance inflation factor). Nilai VIP besar lebih dari 10 atau tolerance $< 0,10$ maka dianggap terjadi multikolinearitas antara variabel dependen dalam model.

40.7 Distribusi normal

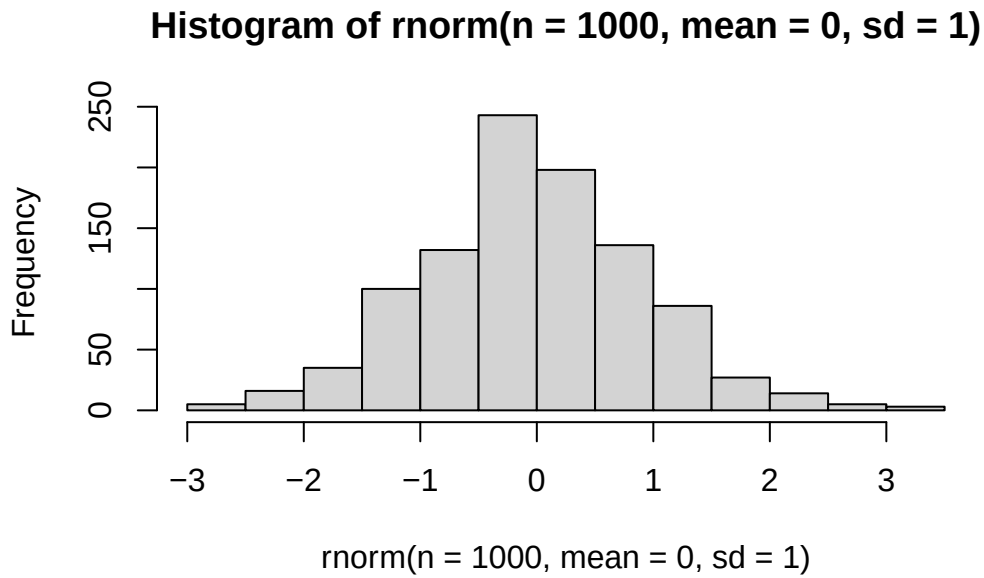
Two problems: Important differences are often obscured and overgeneralize.

Generalisasi: - Sampel cukup dan representatif. - Random sampling. - Ada pembanding.

Cara membuat distribusi normal: 1. di log 2. di akar kuadratkan 3. Dibagi akar kuadratkan

Jika tetap tidak berdistribusi normal, lanjutkan dengan analisis non-parametrik. Non-parametrik bukan berarti tidak bisa dianalisis. Tetap bermakna.

Normal distribution: bell shaped, symmetrical distribution with mean, median and mode all coincidence at its peak.



Insert Zscore Formula:

40.8 Uji T berpasangan dan tak berpasangan

40.9 One Way ANOVA

40.10 Ancova dan Repeated Measurements ANOVA

Variabel Y = Kuantitatif Variabel X = Kuantitatif/Kualitatif

Jika prediktornya adalah kontinuous dan categorical variable, dan response variabel kontinuous, maka menggunakan ANCOVA.

Asumsi: 1. distribusi normal 2. Tidak ada outliers 3. Variansi homogen, 4. Linearity antar DV dan RV 5. Independence of errors. 6. Absence of multicollinearity 7.

40.11 Tabel 2 x 2, Odd ratio, Prevalence Ratio dan Relative Risk

40.12 Prinsip Analisis Data Biner

40.13 Regresi Logistik Sederhana

40.14 Regresi Logistik Berganda

41 Evidence-based Medicine

41.1 Introduction to EBM and Critical Appraisal

Adanya gap antara hasil penelitian dengan praktek sehari hari

ODDS Ratio: rasio terjadinya outcome pada kelompok dgn faktor risiko dengan tidak terjadinya outcome pada kelompok tersebut.

How EBM help: membantu informasi baru, memberikan data empiris dan pasien bisa berpartisipasi dalam decision, reduce litigation, keep up to date.

Interpretasi OR: < 1 , misal 0,6 : risiko 60% kali dibandingkan kontrol, mengurangi risiko 40% (berapa dari 1) Interpretasi OR: > 1 , misal 1,5 : risiko 150% atau 1,5 kali lebih besar pada klpk perlakuan, menambah risiko 50%

High sensitivity, if negative, rule out (SnNout) High spesifisity, if positive, rule in (SpPin)

STeps in EBM

1. Formulate an answerable question
2. Track down the best evidence of outcomes available
3. Critically appraise the evidence
4. Apply the evidence

3 prinsip EBM: 1. Valid

2. Penting (penting secara klinis dan bermakna secara statistik: hasil, lihat PR, OR, RR,ARR, NNT, mean difference) 3. Aplikabel - Feasible? - kebutuhan lain yg dibutuhkan untuk memanfaatkan evidence? - Alternatif ada? - kemiripan sampel dan pasien kita? - benefit outweigh risk? - Bagaimana tanggapan pasien?

f method untuk estimasi risiko pasien. NNT pada trial dibagi NNT pada pasien kita.

5. Evaluate our performance
 - evaluasi informal dan formal.

41.2 Clinical Question berdasarkan PICO

memecahkan masalah klinis dengan pendekatan...

41.2.1 Step 1

proses EBM dimulai dengan skenario klinis yang memerlukan jawaban terbaik berdasarkan bukti ilmiah yang terbaik. “mengubah suatu masalah klinis menjadi masalah research”

- Background question: general knowledge (a question root & disorder and other aspect of health care.), dicari dari literatur
- Foreground question: pertanyaan pengetahuan spesifik untuk menentukan pilihan terapi atau tindakan. Penting menggunakan PICO

P: patient, population, problem **I** : intervensi, exposure **C**: control, comparator **O**: outcome, efficacy

Clinical problem = belum sistematis, clinical question= sesuai komponen PICO, dan answerable question.

Contoh: kejang pada pasien 15 tahun, dengan demam. pertanyaan klinis yang mungkin muncul:
- apa penyebab demam? - penyebab kejang? - obat terbaik?

lalu masing masing pertanyaan diubah menjadi komponen PICO. umpama, pada pasien anak, apakah pemeriksaan lumbar punksi lebih baik dari pemeriksaan DL dalam diagnosis DL??

Contoh 2: anak, 8 tahun, dirawat di PICU dengan severe GBS dan menggunakan ventilator. Pengetahuan dasar ada terapi prednison, plasma exchange and other. apakah IVIG therapy deliver outcome better than other therapy? => Clinical Problem.

PICO: P: child with severe GBS I/C: prednisone, IVIG, atau plasma exchange O: cepat kesembuhan

good, clear and focused

41.2.2 Prinsip Dasar Searching

boolean logic: AND, OR, NOT

Tugas mencari permasalahan klinis, ubah menjadi pertanyaan research, lalu search dengan Boolean logic di Pubmed : <https://wiryadana.github.io/picolo/>

41.3 Effect Size, Hypothesis, and Confidence Interval

besarnya efek: suatu perbedaan klinis disebut juga point estimate. Contoh: Rasio prevalens, risiko relative, rasio odds, beda rerata.

ada 4 kemungkinan, - secara statistik signifikan, secara klinis signifikan - secara statistik signifikan, secara klinis tidak signifikan - secara statistik tidak signifikan, secara klinis signifikan - secara statistik tidak signifikan, secara klinis tidak signifikan.

Sebagai klinisi kita mengutamakan clinically important dengan desain yang valid.

Hipotesis positif = adanya hubungan, perbedaan terkecil yang secara klinis yang dianggap penting. hipotesis negatif = tidak ada hubungan, perbedaan terbesar yang secara klinis tidak dianggap penting (non-inferiority)

Contoh: sebuah jurnal melaporkan kesembuhan obat A 80%, obat B 55%. Jika melakukan penelitian ulang, maka tidak dibenarkan menggunakan angka 25% sebagai effect size, ada 2 alasan.

1. alasan konseptual, kenapa harus diulang jika sudah ada nilai 25%
2. alasan teknis, jika efek size ternyata 23, maka bisa secara statistik tidak signifikan, padahal masih bermakna klinis.

Oleh sebab itu, penentuan efek size adalah judgement peneliti, tidak boleh dari penelitian orang lain.

Untuk generalisasi atau inferensi ke populasi asal sampel diambil ada 2 cara, yaitu nilai P dan IK.

Variabel adalah karakteristik subjek penelitian yang berubah antar subjek. Var perancu adalah variabel yang berhubungan dengan var bebas dan tergantung, tetapi bukan variabel perantara.

Tipe data: Data adalah hasil pengukuran. Pengukuran adalah proses kuantifikasi hasil observasi yang memperhatikan referensi tertentu dan dinyatakan dalam unit baku atau dianggap baku.

1. skala kategorik
2. skala numerik

skala kategorik a. skala nominal (tidak ada derajat yang superior inferior) b. skala ordinal (ada derajat tapi tidak bisa dikuantifikasi secara matematika)

Skala numerik a. skala interval: nilai nol tidak absolut (arbitrary) b. skala ratio: nilai nol absolut

lainnya Kontinyu: ada desimal diskreet: tidak ada desimal contoh jumlah keluarga.

bebas vs tergantung (uji bivariat) Nominal vs nominal : chi kuadrat, uji fischer nominal (dikotom) vs numerik: uji T nominal (>2 nilai) vs numerik Anova Numerik vs numerik: korelasi/regresi

Jika tidak ada sebab akibat, pakai korelasi. Jika ada sebab (var bebas) dan akibat (tergantung) pakai regresi

Multivariat bebas vs tergantung numerik vs numerik : regresi multipel nominal, numerik vs nominal : regresi logistik

Data berpasangan: adalah data yang berasal dari orang/subjek yang sama atau subjek yang berbeda yang telah dilakukan matching. (ada before dan after)

Uji parametrik: a. koef varians: <30% b. uji K-S atau S-W

Uji homogenitas varians Uji levans

Uji non-parametrik

Interpretasi nilai P: 0,035 besarnya kemungkinan untuk memperoleh hasil tersebut disebabkan oleh faktor peluang bila H_0 benar adalah sebesar 3,5%. Artinya 96,5% lainnya adalah akibat faktor obat.

bila sampel kecil, nilai P bisa bermakna tetapi interval kepercayaan melewati nilai batas tidak signifikan.

41.4 Study About Risk Factor/Case Control Study

Start from dependent variable to independent variable. Harus jelas bagaimana kita memilih kasus dan kontrolnya.

41.5 Study About Diagnosis

EBM: integration of best research evidence with clinical expertise and patients values.

Sensitivitas dan spesifitas relatif stabil terhadap prevalensi penyakit PPV dan NPV harus memperhatikan prevalensi penyakit di tempat kita.

41.6 Study about Therapy

41.7 Study about Prognosis

41.8 Summary Capita Seleкта

42 Tugas MKDU

42.1 Biologi Molekuler

42.2 Statistik

42.3 Metode Penelitian

42.3.0.1 Tugas Metpen-1 - Association & Causation

42.3.0.1.1 Tugas/Pertanyaan/Instruksi

Mencari jurnal yang bisa menjadi acuan penelitian saat PPDS, jelaskan apakah jurnal tersebut association atau causation.

42.3.0.1.2 Jawaban/Hasil

Nama : Kadek Adit Wiryadana Prodi : Ilmu Penyakit Dalam Nim : 2371041012

42.3.0.1.2.1 Judul Jurnal:

Importance of Venous Congestion for Worsening of Renal Function in Advanced Decompensated Heart Failure

42.3.0.1.2.2 Tautan/Link:

akses: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2856960/>

42.3.0.1.2.3 Screenshot:



NIH Public Access

Author Manuscript

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Importance of Venous Congestion for Worsening of Renal Function in Advanced Decompensated Heart Failure

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Abstract

Background—Reduced cardiac output is traditionally believed to be the main determinant of worsening renal function (WRF) in advanced decompensated heart failure (ADHF).

Objective—To determine if venous congestion, rather than impairment of cardiac output, is primarily associated with the development of WRF in ADHF.

Methods—A total of 145 consecutive patients admitted with ADHF treated with intensive medical therapy guided by pulmonary artery catheter were studied. WRF was defined as an increase of serum creatinine ≥ 0.3 mg/dl during hospitalization.

Results—In the study cohort (age 57 ± 14 years, cardiac index 1.9 ± 0.6 l/kg.m², LVEF $20 \pm 8\%$, serum creatinine 1.7 ± 0.9 mg/dl), 58 patients (40%) developed WRF. Patients who developed WRF had a higher central venous pressure on admission (CVP, 18 ± 7 versus 12 ± 6 mmHg, $p < 0.001$) and after intensive medical therapy (11 ± 8 versus 8 ± 5 mmHg, $p = 0.04$). The development of WRF occurred less frequently in patients that achieved a CVP < 8 mmHg ($p = 0.01$). Furthermore, the ability of CVP to stratify risk for development of WRF was apparent across the spectrum of systemic blood pressure, pulmonary capillary wedge pressure, cardiac index, and estimated glomerular filtration rates.

Conclusions—Venous congestion is the most important hemodynamic factor driving WRF in decompensated patients with advanced heart failure.

Keywords

worsening renal function; venous congestion; cardiac index; decompensated heart failure

Figure 42.1: screenshot Jurnal

42.3.0.1.2.4 Alasan Association/causation:

Penelitian Mullens et al diatas meneliti hubungan antara kongesti vena (CVP/central venous pressure) terhadap perburukan fungsi ginjal pada pasien dewasa dengan ADHF. Pertanyaan penelitian ini adalah apakah kongesti vena (diukur dengan kateter arteri pulmoner) merupakan faktor yang berasosiasi dengan perburukan fungsi ginjal pada pasien ADHF. Tujuan lainnya adalah menjawab apakah asosiasi tersebut lebih bermakna dibandingkan asosiasi penurunan cardiac output (diukur dengan cardiac index) terhadap penurunan fungsi ginjal, dimana secara umum teori yang diterima adalah faktor utama perburukan fungsi ginjal pada ADHF adalah penurunan cardiac output.

Penelitian ini cenderung mendukung **Causation** antara kongesti vena dengan penurunan fungsi ginjal. Hal ini karena variabel independen berkontribusi terhadap penurunan fungsi ginjal (ditemukan asosiasi melalui perhitungan statistik). CVP ditemukan lebih tinggi baik saat admisi dan setelah mendapatkan terapi ADHF secara intensif pada pasien yang mengalami perburukan fungsi ginjal. Secara prospektif, ketika pasien mendapatkan terapi ADHF dan CVP berhasil diturunkan <8 mmHg, pasien yang kemudian mengalami perburukan fungsi ginjal lebih jarang ditemukan (mendukung prior association). **Causation yang ditemukan merupakan contributory cause.** Terdapat beberapa variabel yang bisa menjadi confounding karena tidak dilakukan pemeriksaan diantaranya regional renal perfusion, kelainan struktural ginjal, serta analisis data secara multivariabel tidak dilakukan

42.3.0.1.2.5 Ide Penelitian Selanjutnya:

Hubungan diameter dan collapsibility index inferior vena cava terhadap perburukan fungsi ginjal pada pasien acute decompensated heart failure.

42.4 Filsafat Ilmu Kedokteran

42.4.0.1 Tugas Filsum-1 - Pengenalan filsafat ilmu

42.4.0.1.1 Tugas/Pertanyaan/Instruksi

Epistemology

42.4.0.1.2 Jawaban/Hasil

42.5 Evidence-based Medicine

42.5.0.1 Tugas EBM-1 - Introduction to EBM & Critical Appraisal

42.5.0.1.1 Tugas/Pertanyaan/Instruksi

Instruksi : Membuat pertanyaan klinis berdasarkan prinsip PICO !

Skenario : Kita menghadapi anak dg defisiensi growth hormone (GH), apakah pemberian terapi GH eksogen dapat memacu pertumbuhan anak tersebut?

42.5.0.1.2 Jawaban/Hasil

42.5.0.1.2.1 PICO

P (patient/population) : Anak usia 2 - 14 tahun dengan isolated growth hormone deficiency (GH)

I (intervention) : Terapi GH eksogen dengan dosis awal 25 mcg/kgBB/hari

C (comparison/control): plasebo

O (outcome) : laju peningkatan tinggi badan per tahun (Δ height velocity) >3 cm dalam 12 bulan terapi.

42.5.0.1.2.2 Pertanyaan Klinis

Pada pasien anak usia 2 -14 tahun dengan isolated growth hormone deficiency, apakah pemberian growth hormone eksogen sebagai terapi dapat meningkatkan Δ height velocity >3 cm/tahun pada 12 bulan terapi dibandingkan anak yang diberikan terapi plasebo.

42.5.0.1.2.3 Contekan

<https://academic.oup.com/jcem/article/93/2/352/2597982> <https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2265.2012.04420.x> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6979443/>

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